

Integrating Wildfire Risk in Forest Road Network Planning

A Linear Optimization Approach to Select a Cost-Minimizing and Fire Response Enhancing Road Network in Timber Forests

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Dissertation to obtain the Master of Science Degree in

Data Science in Agriculture, Food, Forest and Environment

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(Versão provisória)

Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Acknowledgments

First of all, I express my sincere gratitude to my supervisors, Dr. Marta Guerreiro Duarte Mesquita de Oliveira and Dr. Susete Maria Gonçalves Marques, for providing the idea for this thesis and the relevant data, and for their invaluable guidance, patience, and support throughout this work. Their understanding of my absences and their guidance in shaping the work I returned with were truly indispensable.

My appreciation is extended to all the professors teaching in the [Green Data Science](#) Master's program, a visionary initiative that leverages data science to tackle environmental challenges and promote sustainable resource management through evidence-based decision-making. The barrier is often not only the lack of evidence but the willingness to act on it at the political level, but the program is a crucial step in the right direction – preparing students to extract actionable insights from data and communicate them effectively to stakeholders, helping future decisions move toward better outcomes.

This journey would not have been the same without the support of my colleagues and friends, particularly Aziza Ben Tanfous and Joana Esteves, whose own thesis battles paralleled mine, and my friends in Germany who had no idea what I was talking about, but patiently listened to me thinking aloud on weird data doing weird things challenges.

Finally, my heartfelt thanks go to my family for their constant support and unwavering encouragement. I am grateful to my mother for always lending an ear in times of need, to my father for carefully reading my thesis and providing valuable feedback, and to my siblings for asking not too many questions about my thesis, with a special thanks to my grandparents, whose love, wisdom, acceptance and encouragement have been a constant source of strength and support, which brightened even the most stressful days.

To all of you, I express my deepest gratitude and appreciation.

Abstract

Wildfires are an increasingly severe threat to Mediterranean forests, where climate, vegetation, and topography amplify fire risk. Forest road networks play a dual role in timber forests: they are indispensable for harvesting operations and can provide access for firefighting crews. Yet, current forest road planning approaches focus almost exclusively on minimizing road construction and maintenance costs for timber extraction, without accounting for wildfire resilience.

This master's dissertation develops and applies a linear optimization model that integrates wildfire risk considerations into multi-period forest road network planning. The presented integer linear program minimizes total road construction and maintenance costs across five ten-year periods, while ensuring both mechanized harvesting access and ground-based fire vehicle access to forest stands with low fire resistance. Two road width options are incorporated: narrow roads sufficient for timber extraction and wider roads allowing fire response operations. Timber access needs are derived from the stands' harvesting schedules, while fire-vehicle access needs are based on stand-level wildfire-resistance indicators: stands falling within the lowest quantile of resistance values are flagged for fire-vehicle access, providing an adjustable threshold that allows users to specify the desired extent of protection.

The model is applied to the Paiva forest intervention zone, a fire-prone landscape in northern Portugal. A data preprocessing and model implementation workflow was developed in Python using Jupyter Notebooks, presenting a reusable and adaptable framework for future case studies. The preprocessing results – particularly the graph of potential roads that follow the ownership boundaries of stands and connect to the preexisting public road network – can be reused for other projects in Paiva, and the preprocessing performed to generate this graph can be applied to other case study areas. Model results show that the framework generates minimum-cost timber road networks within a known optimality gap while ensuring firefighting access to the least fire resistant areas – providing quantitative and actionable guidance for multi-purpose forest road network planning.

Keywords

Forest Road Network Planning; Wildfire Resilience; Operations Research; Decision Support; Integer Linear Programming

Resumo

Os incêndios florestais representam uma ameaça crescente para as florestas mediterrânicas, onde clima, vegetação e topografia intensificam o risco de fogo. As redes viárias florestais desempenham um papel duplo: suportam as operações de exploração madeireira e garantem acesso terrestre para veículos de combate a incêndios. No entanto, o planeamento viário continua maioritariamente centrado na minimização dos custos de construção e manutenção associados à extração de madeira, sem integrar a resiliência ao incêndio.

Esta dissertação desenvolve um modelo de otimização inteiro linear que integra o risco de incêndio no planeamento multiperíodo da rede viária florestal. O modelo minimiza os custos totais de construção e manutenção ao longo de cinco períodos de exploração, assegurando simultaneamente o acesso mecanizado à exploração e o acesso de veículos de combate a incêndios aos povoamentos com baixa resistência ao fogo. Consideram-se duas larguras de via: estradas estreitas adequadas à extração de madeira e estradas largas capazes de suportar operações de combate a incêndios. As necessidades de desenvolvimento resultam dos planos de corte, enquanto os requisitos de desenvolvimento para combate a incêndios resultam dos valores-limite para resistência ao fogo que podem ser definidos no modelo.

A abordagem é aplicada à Zona de Intervenção Florestal do Paiva, uma paisagem com elevada propensão a incêndios no norte de Portugal. O processamento de dados e a implementação do modelo foram realizados em Python com Jupyter Notebooks, oferecendo uma estrutura replicável para estudos futuros. Os resultados do pré-processamento dos dados, em particular o grafo criado para uma rede de estradas florestais ligada à rede rodoviária pública existente, com todas as estradas potenciais ao longo dos limites de propriedade existentes, também podem ser reutilizados para outros projetos. Os resultados calculados após a aplicação mostram que o modelo gera redes de custo mínimo dentro de um intervalo conhecido de optimalidade, garantindo simultaneamente o acesso aos povoamentos menos resistentes ao fogo. Assim, fornece orientações quantitativas para o planeamento de redes viárias florestais com múltiplos propósitos em regiões suscetíveis a incêndios.

Palavras-chave

Planeamento de Redes Viárias Florestais; Resiliência face a Incêndios Florestais; Investigação Operacional; Apoio à Decisão; Programação Linear Inteira

Resumo Alargado

Propósito e Problema de Investigação

Os incêndios florestais constituem uma ameaça crescente para as florestas mediterrânicas, particularmente em regiões como o norte de Portugal, onde clima, topografia e composição de espécies intensificam o risco de fogo. Embora as práticas de gestão florestal integrada incluam medidas preventivas como a redução de combustíveis e a restauração ecológica, o planeamento das redes viárias florestais raramente incorpora requisitos de acesso para veículos de combate a incêndios. Nas florestas de produção, não existe, até onde é do conhecimento do autor, um enquadramento abrangente que integre a minimização multiperíodo dos custos económicos com as necessidades de acessibilidade para exploração madeireira e veículos de combate a incêndios, nem ferramentas de apoio à decisão que permitam o planeamento dual das infraestruturas viárias.

Esta dissertação de mestrado propõe um modelo de optimização linear inteira que identifica, ao longo de cinco períodos de dez anos, os troços de estrada a construir ou a manter, de modo a recomendar uma rede viária economicamente eficiente que assegure simultaneamente a extração de madeira e o acesso de veículos de combate a incêndios às zonas menos resistentes ao fogo. O estudo de caso centra-se na Zona de Intervenção Florestal (ZIF) Paiva, situada no distrito de Aveiro, abrangendo aproximadamente 7.620 hectares e gerida colaborativamente por 185 proprietários privados associados à Associação Florestal do Vale do Sousa. A área é caracterizada por povoamentos pequenos e dispersos, predominantemente de *Eucalyptus globulus* e *Pinus pinaster*, e apresenta elevada vulnerabilidade a incêndios — cerca de 46% da zona queimou entre 2014 e 2023. Apenas os povoamentos inacessíveis (315) foram considerados na modelação, refletindo as zonas onde a intervenção viária era necessária.

Contexto e Motivação

O aumento da frequência e intensidade de incêndios extremos, impulsionado pela acumulação de combustível, alterações climáticas, prolongamento das estações secas e aumento da temperatura, tem levado a consequências sociais, económicas e ambientais graves, incluindo perda de vidas, emissões de gases de efeito estufa e degradação dos ecossistemas (Turco et al. 2019; Ruffault et al. 2020). Apesar da eficácia do combate aéreo, a supressão terrestre

continua a ser essencial, sendo as estradas florestais um ativo estratégico para a intervenção rápida e segura (Stefanović et al. 2016; Stergiadou 2014).

Em florestas de produção, o desenvolvimento de estradas é inevitável para a extração de madeira. Assim, estradas planeadas de forma dual, que suportem simultaneamente a exploração florestal e o combate a incêndios, permitem acrescentar uma função crítica a uma infraestrutura já necessária, potenciando a sua utilidade num contexto em que a presença de estradas está associada a ignições antropogénicas. Estudos prévios demonstram que a densidade e a distribuição de estradas não só influenciam a ocorrência de ignições antropogénicas, como também a eficácia do combate a incêndios, contribuindo para a redução da área queimada quando a resposta é rápida (Narayanaraj et al. 2012; Pinto et al. 2020).

Assim, a integração de considerações de resiliência a incêndios no planeamento viário pode ser considerada um elemento-chave para a gestão sustentável de florestas mediterrânicas propensas a incêndios.

Desenho do Estudo e Metodologia

O modelo desenvolvido é um Integer Linear Programming (ILP) determinístico, com variáveis de decisão binárias para construção e manutenção de troços e variáveis inteiras de fluxo que asseguram conectividade entre povoamentos e a rede pública existente. Duas larguras de estrada foram consideradas: estradas estreitas adequadas à extração de madeira e estradas largas capazes de suportar operações de combate a incêndios. O horizonte temporal compreende cinco períodos de dez anos, correspondendo ao planeamento multi-período dos cortes e à necessidade de manutenção ou construção de troços viários em cada período.

O trabalho iniciou-se com o processamento e análise de dados geoespaciais, incluindo: a rede pública existente, os limites dos povoamentos, os calendários de exploração madeireira e estimativas de custo por quilómetro para construção, manutenção e alargamento de estradas. Seguidamente, a área de estudo foi segmentada espacialmente para decompor o problema em subproblemas mais manejáveis, e foi gerada uma estrutura em grafo representando os troços de estrada potenciais. Este pré-processamento, implementado em Python e Jupyter Notebooks, utilizou bibliotecas de geoprocessamento e foi seguido da utilização do solver CPLEX através da sua API, permitindo uma integração sistemática e reproduzível dos dados com o modelo de optimização.

As necessidades de acesso para a exploração madeireira foram derivadas do calendário de cortes, enquanto os requisitos de acesso para o combate a incêndios se basearam nos in-

dicadores de resistência ao fogo ao nível dos povoamentos, em conjunto com um parâmetro de apetite ao risco definido pelo utilizador, que permite seleccionar os povoamentos incluídos no quantil inferior do indicador de resistência ao fogo — por exemplo, o 5.º percentil ou qualquer outro quantil definido pelo utilizador — que devem ser servidos por estradas alargadas, acessíveis a veículos de combate a incêndios. O modelo minimiza os custos totais de construção, manutenção e alargamento ao longo de todos os períodos, assegurando simultaneamente a acessibilidade mecanizada para exploração e a capacidade de intervenção dos veículos de combate a incêndios nas áreas mais vulneráveis.

Resultados e Contribuições

O trabalho apresenta contribuições de dois tipos: metodológicas, relacionadas com o desenvolvimento do modelo e do pipeline de dados, e aplicadas, consistentes nos resultados operacionais obtidos com a implementação na ZIF Paiva.

O estudo introduz um modelo ILP determinístico inovador que integra explicitamente a acessibilidade para veículos de combate a incêndios na planificação económica de redes viárias florestais, combinando objetivos de logística madeireira e gestão do risco de incêndios. O pipeline de pré-processamento de dados permite criar grafos reutilizáveis a partir dos limites dos povoamentos e das estradas existentes, estabelecendo uma base sistemática e replicável para o modelo. O grafo resultante pode ser utilizado como base para outros trabalhos ou análises na mesma área de estudo que necessitem de uma representação estruturada da rede viária, enquanto a metodologia subjacente pode ser aplicada a diferentes áreas de estudo para gerar grafos similares adaptados a novos contextos. Este enquadramento integra minimização de custos, acessibilidade para exploração madeireira e acessibilidade para combate a incêndios ao longo de múltiplos períodos, incluindo restrições adaptadas ao fogo com limiares ajustáveis pelos gestores florestais. O workflow completo, implementado em Python, é reproduzível e fornece suporte operacional robusto para a tomada de decisões.

A aplicação do modelo à ZIF Paiva demonstrou a sua capacidade de gerar redes viárias de custo mínimo que cumprem simultaneamente os critérios de acesso para exploração madeireira e para combate a incêndios. Para cada cenário de calendário de cortes, o modelo identificou os troços a construir, manter ou alargar, distinguindo entre estradas estreitas e largas, garantindo conectividade realista com a rede pública e evitando percursos por outros povoamentos. Mesmo sem comprovar a optimalidade global devido a limitações computacionais, foram obtidas soluções viáveis com gaps de optimalidade conhecidos, permitindo aos gestores avaliar a proximidade das soluções aos custos mínimos. Estes resultados oferecem

orientações para o planeamento multipropósito de redes viárias em florestas mediterrânicas, equilibrando eficiência económica, produção madeireira e resiliência face a incêndios, e podem ser generalizados a outras regiões suscetíveis a incêndios.

Conclusão

Em síntese, a dissertação demonstra que a integração de requisitos de combate a incêndios no planeamento de redes viárias florestais multiperíodo é viável. O modelo desenvolvido oferece uma abordagem rigorosa para decidir quais troços construir ou manter, garantindo simultaneamente a acessibilidade para exploração e a resiliência ao fogo. Esta abordagem contribui para a mitigação de riscos em paisagens mediterrânicas propensas a incêndios, fornece uma ferramenta prática para gestores florestais e abre caminhos para trabalhos futuros.

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Acronyms

AFVS Associação Florestal de Vale do Sousa

bwd backward

CAOF Comissão de Acompanhamento para as Operações Florestais

CPLEX IBM® ILOG® CPLEX® Optimization Studio

CRS Coordinate Reference System

EWE Extreme Wildfire Event

fwd forward

FFPR Forest Fire Prevention Road

GIS Geographic Information System

ILP Integer Linear Programming

LB Lower Bound

MIP Mixed Integer Programming

MILP Mixed Integer Linear Programming

UB Upper Bound

ZIF Zona de Intervenção Florestal

Glossary

digraph

A digraph is a graph in which edges have orientations. It is written as a pair $G = (V, A)$, where V is a set of elements called *vertices*, *nodes*, or *points*, and A is a set consisting of ordered pairs of vertices (v_i, v_j) called *arcs*, *directed edges*, *arrows*, or *directed lines*. 49

FIRE-RES

4-year project (2021-2025) dedicated to developing an integrated forest management strategy to address the challenges posed by Extreme Wildfire Events (EWEs) in Europe. It is led by the [Forest Science and Technology Centre of Catalonia](#), Spain, and funded under the [European Union's H2020 research and innovation program](#). See also appendix A xx, 9, 86

graph

Mathematical structure used to model pairwise relationships between objects. It consists of a set of vertices (also called nodes) and a set of edges that connect pairs of vertices. Graphs can be directed (edges have a direction) or undirected (edges have no direction), and they are fundamental in fields such as computer science, network analysis, and combinatorics. xix, 35, 47

IBM® ILOG® CPLEX® Optimization Studio

[IBM® ILOG® CPLEX® Optimization Studio](#) is a commercial decision optimization software developed by IBM for building and solving complex optimization models. Analytical decision support toolkit for rapid development and deployment of optimization models using mathematical and constraint programming. It combines an integrated development environment with the powerful Optimization Programming Language and high-performance CPLEX and CP Optimizer solvers. xvii, 64

Integer Linear Programming

In ILP, a variant of Linear Programming, the key requirement is that some or all decision variables must take integer values. This makes ILP suitable for problems involving discrete choices

(e.g., scheduling, routing, allocation, or selecting which road segments to build to obtain a minimum-cost road network, as in this work)..... v, xvii, 12

Linear Programming

Mathematical optimization method used to achieve the best outcome in a mathematical model, where the objective and constraints are expressed by linear relationships. By systematically selecting values for the decision variables using techniques such as branch-and-bound or the simplex method, the goal is to find the values that optimize the objective function while satisfying all the constraints. xix

Living Lab

name for case study regions within the FIRE-RES project 9, 86

nonindustrial private forest

Privately owned forests that are not managed for industrial purposes but for other objectives such as recreation, conservation, or personal use. 9

QGIS

Quantum Geographic Information System, an open-source GIS software for spatial data visualization and analysis..... viii, 18, 19

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Introduction

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1.1 Background and Motivation

FIRE occurrences play a significant role in the dynamics of forest ecosystems, particularly in the Mediterranean region. Although they can have beneficial ecological effects (Keeley et al. 2011; Pereira et al. 2021; Davim et al. 2022), each year, some extreme wildfires occur globally that exceed suppression capabilities, cause extensive damage, and frequently result in fatalities (Tedim et al. 2018; Schwörer et al. 2024). Large wildfires release large quantities of greenhouse gases and air pollutants, thereby deteriorating air quality and compromising ecosystems, their provisioning services, and public health (Wang et al. 2023; Bolan et al. 2025; Carreras-Sospedra et al. 2024). They lead to widespread vegetation loss, which accelerates soil erosion, disrupts water and nutrient cycles, and can increase the risks of floods and landslides (Shakesby et al. 2006; Francos et al. 2024; Roces-Díaz et al. 2022; Sutanto et al. 2024; Culler et al. 2023). Hence, preventing wildfires from escalating into such extreme events is a major environmental and societal concern.

However, both the frequency and intensity of the most extreme wildfire events worldwide have increased in recent decades (Cunningham et al. 2024). A growing consensus among scientists, fire managers, and agencies recognizes that wildland areas are increasingly exposed to challenging fire weather, longer fire seasons, and more extensive fires — all exacerbated by climate change (San-Miguel-Ayanz et al. 2017; Jolly et al. 2015; Bowman et al. 2017; Robinne et al. 2018; Ellis et al. 2022). In 2017, Portugal experienced an unprecedented wildfire season, with approximately 500,000 hectares burned and more than 120 fatalities reported (Turco et al. 2019). Climate change is expected to further intensify fire-conducive conditions, thereby increasing the risk of large wildfires (Dupuy et al. 2020; Turco et al. 2018). Projections for Europe suggest that, by 2090, the annual area burned could rise by 120–270% compared to the 2000–2010 average (Robinne et al. 2018; FAO 2020). Specifically, in the Mediterranean Basin, the frequency of heat-induced fire-weather types — closely associated with large wildfire activity — is projected to increase by 14% under the RCP4.5 scenario and by 30% under the RCP8.5 scenario by the end of the century, significantly raising the likelihood of extreme wildfire events (Ruffault et al. 2020).

As wildfires are projected to become more frequent and intense, particularly in fire-prone regions such as the Mediterranean basin, it is increasingly important to adopt a holistic approach to wildfire management. This includes moving away from zero-burn policies¹ and prioritizing

¹The suppression of low-intensity fires can lead to fuel accumulation, increasing the severity of subsequent wildfires (Mateus et al. 2014; Collins et al. 2013; Kreider et al. 2024).

preventive measures and fuel management, such as prescribed burning and agro-silvo-pastoral practices (Pais et al. 2023; Oliveras Menor et al. 2025; Mauri et al. 2023). Within the broader context of integrated forest management, it is both reasonable and necessary to incorporate wildfire response considerations into forest road network planning. As Stefanović et al. (2016) noted, forest roads are the backbone of ground-based firefighting access:

"The importance of forest roads in fire protection is great, because they represent the basic infrastructure necessary for effective fire prevention and suppression."

Developed forest road networks facilitate faster fire response and are associated with reduced fire size (burned area) – but they also increase the risk of anthropogenic wildfire ignitions due to improved public access (Viegas et al. 2022; Thompson et al. 2021; Narayanaraj et al. 2012; Pinto et al. 2020). In a study on the influence of forest roads on the spatial patterns of human- and lightning-caused wildfire ignitions, Narayanaraj et al. (2012) found that although most fires occurred in roaded areas, they accounted for only a small proportion of the total burned area, whereas a few large fires, typically lightning-induced in roadless areas, accounted for most of the burned area. Similarly, in an analysis on the impact of several landscape features on the occurrence and size of forest fires, Pinto et al. (2020) observed that while higher road density and more firebreaks were associated with increased fire occurrence, these features also correlated with smaller fire sizes. The authors suggested a link to the improved fire suppression: Shorter travel distances between ignition points and fire brigade locations can reduce response time and enable earlier control, whereas sparse road networks lead to a longer period of uncontrolled spread, resulting in larger burned areas.

In areas without road access and beyond the operational reach of fire vehicle pumps, fire suppression at the front line requires either aerial deployment, ground crews advancing on foot², or a strategic “awaiting”³ of the fireline until it reaches accessible terrain — at which point roads can act as natural control lines and support corridors. Where such infrastructure is lacking or distant, this strategy allows fires to spread unhindered until ground-based firefighting becomes feasible — significantly increasing the risk of large-scale ecological and economic loss. While aerial suppression can effectively support containment efforts — and enjoys broad public and media appeal — it has notable limitations, including high costs, limited availability, and operational constraints (Peña et al. 2022; Thompson et al. 2013) such as dependence on

²This reliance on manual methods limits the amount and type of equipment deployable and increases physical risks for personnel.

³This does not mean inaction by fire responders; it refers to indirect tactics such as preparing containment zones, conducting backburns, creating defensible space, staging resources, and evacuating at-risk areas.

weather conditions and is not a sufficient substitute for accessible, strategically located roads. Yocom et al. (2019) identified distance to roads as the most influential variable in predicting fire perimeters, noting that approximately 26% of observed perimeters aligned with existing roads. This suggests that responders routinely utilize roads as containment boundaries in areas with adequate road infrastructure.

"The first thought as forest engineers is to use constructions that already exist (road network, water springs, etc) so as to become useful for forest protection and fire fighting." (Stergiadou 2014)

Building on this idea, the importance of incorporating wildfire protection and response considerations into the early stages of forest infrastructure planning becomes evident. In fire-prone landscapes, road infrastructure is not only essential for access, but also serves as a strategic asset for wildfire control. When road networks are planned and developed with dual purposes in mind, they can significantly enhance the effectiveness of suppression efforts. Roads designed to support wildfire response enable fire crews to intervene earlier and more safely, reducing the risk of fires escalating beyond containment capacity and causing extensive ecological or economic damage. Given the increasing frequency and severity of wildfires, the question is no longer whether roads can serve multiple purposes — but why they are not routinely designed to do so.

1.2 Integrating Wildfire Risk in Forest Road Network Planning – An Overview of Related Work

Integrating wildfire risk into forest road network planning represents a small, yet existing field of research. The crucial role of forest roads in wildfire management is well documented: there is broad agreement that early suppression efforts are essential for containing wildfire spread and that forest roads provide the critical access needed for both prevention and emergency response (Viegas et al. 2022; Thompson et al. 2021; Majlingova 2012; Stefanović et al. 2016; Stergiadou 2014; Laschi et al. 2019). To the best of the author's knowledge, no existing method directly addresses the selection of road segments for construction from a predefined set of alternatives while explicitly accounting for wildfire risk. Nonetheless, a variety of related research efforts — ranging from Geographic Information System (GIS)-based spatial analyses to multi-criteria decision-making tools and spatial decision support systems — incorporates

wildfire considerations in forest road network planning or offers valuable conceptual foundations. These approaches, while not directly solving the constrained selection problem, inform how spatial wildfire vulnerability and operational objectives can be jointly considered. Some of these contributions are presented below.

Liampas et al. (2019) observed that while forest opening up was initially driven by the need for timber transportation and connectivity between forest settlements, it is increasingly acknowledged as a vital preventive measure for forest protection, particularly within the context of multipurpose forestry in the Greek Mediterranean mountains. In response to this shift in perspective, they proposed a method for forest road planning that employs least cost path analysis and multi-criteria analysis within a GIS environment. Although wildfire risk was not explicitly integrated into their model, the criteria and indicators used in the multi-criteria analysis were informed by a previous work developing a forest fire risk zone map based on satellite imagery and GIS data (Dong et al. 2005).

Psilovikos et al. (2009) examined the role of forest roads in enhancing wildfire suppression effectiveness in the suburban forests of Thessaloniki, Greece. Focusing on two major forest roads that had been the subject of environmental and functional debates, the authors assessed their contribution to fire protection in terms of spatial coverage and critical response time. Specifically, they evaluated whether fire-fighting vehicles traveling along these roads could cover the entire forest area and reach ignition points within the crucial 15-minute window considered essential for successful early fire suppression. Their findings emphasized the significance of strategically placed roads in achieving rapid response and effective protection while also addressing the need to balance technical efficiency with environmental compatibility, and offers practical and theoretical guidance for improving forest road networks as an integral component of wildfire defense planning.

Kravanja et al. (2007) examined the role of Forest Fire Prevention Roads (FFPRs) as a proactive fire protection measure in the Kras Forest Management Region in Slovenia, an area characterized by high wildfire risk due to climatic, geological, and anthropogenic factors. They found that although FFPRs are critical for enabling access for emergency interventions, the existing and even the planned FFPR network in the Kras I Forest Management Unit falls short of providing sufficient openness to support modern fire suppression techniques. Their work highlights the importance of aligning road density and layout with fire-prone zones to ensure effective protection, underscoring the spatial mismatch that can exist between planned infrastructure and operational firefighting needs.

Akay et al. (2017) delineated “fire-access zones” in Turkey by combining terrain slope, road

position, and firetruck hose-reach capacities. Their GIS analysis highlighted areas beyond current access that warrant new road construction. Eastaugh et al. (2012) introduced a GIS metric comparing reference fire intensities at roads and fuelbreaks to forest averages. They found that roads generally coincide with average-risk zones, whereas fuelbreaks often intersect higher-intensity areas, guiding targeted siting and maintenance. Zhang et al. (2020) proposed a new firefighting distance criterion to evaluate the actual firefighting coverage of the road network and an approach that aims at mapping the optimal forest road network based on the firefighting distance criterion.

Drosos et al. (2014) developed an optimization model for the W. Nestos forest (Greece) that uses uphill/downhill buffer zones (150 m/250 m, 200 m/400 m, 300 m/500 m) around firetruck positions to quantify protected area given hose-length limits. Their GIS analysis showed that, despite effective coverage by 4×4 patrols with 500 m hoses and early warning systems, expanding road density is economically warranted to ensure comprehensive fire protection and efficient timber operations.

Stefanović et al. (2016) evaluated four forest road network variants in Serbia using a multi-criteria decision-making framework that incorporated fire frequency data, the behavior of a recent large wildfire, and seven criteria weighted through the entropy weight coefficient method. These criteria were selected to reflect trade-offs between construction costs, forest accessibility, and the effectiveness of firefighting operations. Their findings revealed that although one road layout was more expensive from a construction cost perspective, its superior capacity to reduce fire-related tree losses justified the investment as the avoided damage costs would outweigh the initial expenditure.

These contributions are highly relevant; yet there remains a need to further support decision-making in forest road network planning in a more practical manner, like already done in other areas: For instance, Sakellariou et al. (2020) presented a spatial decision support system that allocates fire vehicles based on critical response times and dynamically reroutes them around evolving hazards, which provides a practical decision-making tool for emergency services. By contrast, no comparable tool exists that integrates wildfire resistance considerations into forest road network planning. What is still lacking is a decision support tool targeted at timber forest managers and addressing an earlier, fundamental step: guiding construction and maintenance decisions for dual-use forest roads that serve both timber harvesting and fire vehicle access. Such a tool would assist in prioritizing which roads should be built or upgraded to a wider standard, thereby integrating fire vehicle access to critical low-resilience areas directly into the road planning process.

1.3 Problem Statement, Research Objective and Contributions

Although integrated forest management practices include fuel reduction, ecosystem restoration, and other preventive measures, forest road network design does not typically account for access requirements of ground-based fire suppression resources. For timber forests, no comprehensive framework currently integrates multiperiod economic cost minimization with timber and fire vehicle accessibility requirements, nor offers a decision support tool for dual-purpose road planning. Existing optimization models for forest road network planning focus primarily on minimizing construction and maintenance costs or maximizing timber-harvesting efficiency (e.g., Mesquita et al. 2022). In contrast, this master's dissertation develops a decision support tool for forest managers that, over multiple harvesting periods, identifies road segments for construction and maintenance, thereby recommending a cost-efficient road network that supports timber harvesting when and where needed, while also ensuring fire vehicle access to the least fire-resistant areas.

Problem Statement

Mediterranean-type timber forests, where climate and topography amplify fire risk, require road infrastructure that simultaneously supports timber extraction and wildfire containment. This work addresses a case study in the fire-prone [Vale do Sousa](https://www.valsousa.pt/valsousa/o-vale-do-sousa)⁴ region (described in detail in section 1.4), where private forest owners engage in collaborative planning of timber harvests over ten-year periods. In each period, some forest stands (also called management units) are scheduled for harvesting and therefore require access roads wide enough to accommodate timber extraction operations. These roads may either be newly constructed during that period or maintained if they had been built or maintained in the previous period. At the same time, certain stands are less wildfire-resistant than others, and the least fire-resistant zones require wide access roads to enable effective ground-based firefighting in the event of a nearby fire.

With the goal of ensuring equitable cost distribution among landowners, road construction is limited to stand boundaries. The selected road network should be optimized to minimize total construction and maintenance costs.

⁴<https://www.valsousa.pt/valsousa/o-vale-do-sousa>

Research Objective

The aim of this Master's dissertation is to develop a linear optimization model for selecting road segments along stand boundaries for construction and maintenance across five ten-year periods, which:

- minimizes total economic costs of road construction and maintenance over all periods,
- ensures mechanized harvesting access according to the harvesting schedule,
- provides ground-based fire vehicle access to stands with low fire resistance,
- while taking into account two different road width options (small ones suitable for timber harvest and transport, and wider ones suitable for fire-fighting operations).

Contributions

To the best of the author's knowledge, this work introduces a novel integer linear programming model that integrates fire vehicle accessibility into cost-optimized forest road network planning, bridging timber logistics and wildfire risk management. Building upon traditional models focused solely on timber harvest and transport cost minimization, this approach explicitly incorporates constraints ensuring fire vehicle access to stands with low fire resistance⁵. The model supports multi-period planning, capturing the dynamic interplay between forest growth and fire risk. Its application to the study region Paiva demonstrates how stand-boundary constraints and collaborative management can be aligned with wildfire response goals. In the context of increasing wildfire frequency and intensity in Mediterranean-type ecosystems, this framework offers a practical tool for climate-resilient forest infrastructure planning.

In light of prior research, key contributions include:

- **Exemplary Data Preprocessing Pipeline:** Generating reusable forest road network graphs derived from data on stand boundaries and preexisting public roads.
- **Integrated Modeling Framework:** Combining cost efficiency, multi-period timber harvest accessibility and firefighting accessibility in a single optimization model.

⁵The designation of these high-risk zones is based on the stand-level fire wildfire resistance indicator developed by Ferreira et al. (2015) and implemented in form of a quantile threshold. The level of this threshold is to be adjusted by forest managers to determine the degree of emphasis placed on wildfire protection (i.e., how many areas are classified as high risk and therefore prioritized for wider road access).

- **Fire-Adapted Constraints:** Novel inclusion of fire vehicle access requirements for vulnerable stands, based on stand-level fire resistance indicators.
- **Real-World Application:** Case study in Castelo da Paiva illustrating the integration of timber road planning with wildfire resilience considerations.

1.4 Description of the Study Area

Study area for this research is the Zona de Intervenção Florestal (ZIF)⁶ [Paiva](#)⁷, located in Portugal's northwestern Aveiro district near the Douro River. The ZIF Paiva encompasses approximately 7,620 hectares and is representative of the forest structure in Northern Portugal, characterized by small and scattered stands and a high incidence of wildfires. Its administrative scope includes the entirety of the municipality [Castelo de Paiva](#) with all its parishes. The [Associação Florestal de Vale do Sousa](#) (AFVS)⁸ serves as the managing entity (Ministério da Agricultura, do Desenvolvimento Rural e das Pescas [2008](#)), overseeing the participation of 185 registered members. Paiva lies within the broader region of [Vale do Sousa](#) (see fig. 1.1), which serves as a Living Lab for the Horizon 2020 project FIRE-RES (2021–2025), aimed at increasing wildfire resilience across European landscapes.

Vale do Sousa extends over 14,316 hectares and comprises small-scale properties. The current dominant tree species include [Eucalyptus globulus](#) (blue gum) and [Pinus pinaster](#) (maritime pine) – both of which are fire-prone and managed primarily for timber production. Apart from forestland, the landscape composition features agricultural plots, urban regions, as well as shrubland and abandoned agricultural areas. Stakeholders include nonindustrial private forest owners, forest industry actors, investment funds, and municipal entities - all coordinated under the umbrella of the AFVS and collaboratively managed by over 360 landowners.

The region is highly vulnerable to wildfires: Approximately 46% of the ZIF has burned between 2014-2023, highlighting the urgent need for more strategic and integrated risk-reduction strategies. In response, stakeholders are implementing a landscape-scale joint management plan that seeks to balance wildfire mitigation with sustained forest productivity and ecosystem services. (FIRE-RES [2023](#)) Despite its wildfire exposure, Paiva is a touristically attractive region known for its scenic landscapes and cultural heritage.⁹

⁶For details on the concept of forest intervention zones in Portugal, see <https://www.icnf.pt/florestas/zif>

⁷For details on the forest intervention zone 075/07 — Paiva, see [Portaria n.º 1515/2008](#)

⁸English: Association of Forest Producers of Vale do Sousa

⁹A tourist map of Paiva can be found on the [Castelo de Paiva Turismo facebook page](#).

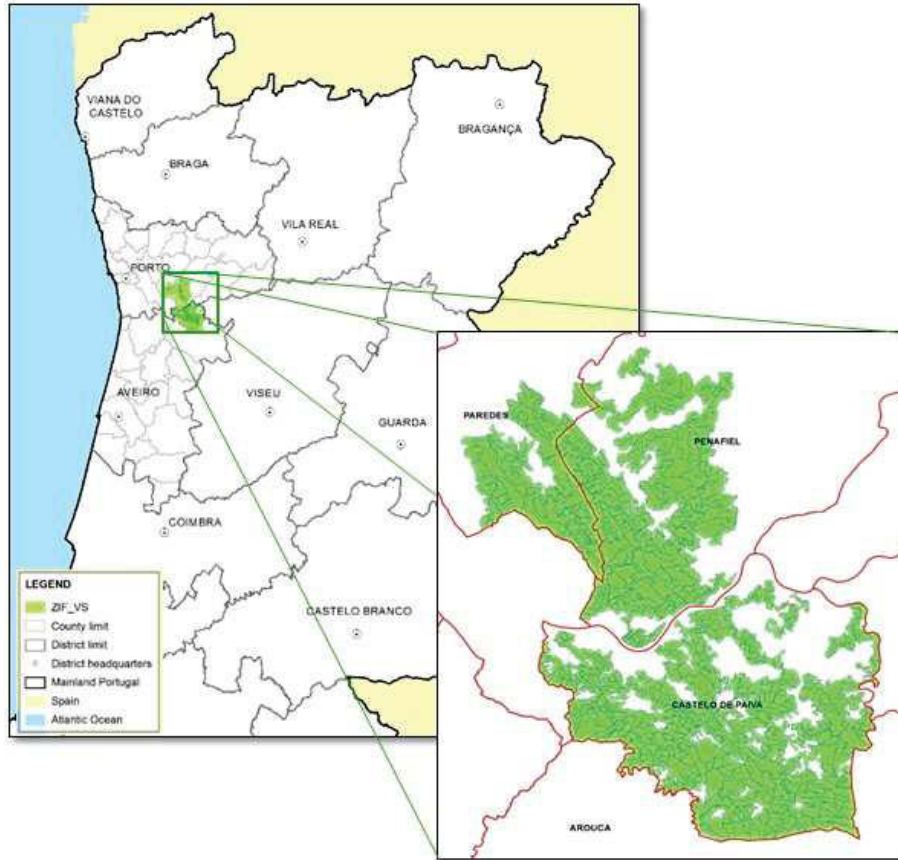


Figure 1.1: Location of ZIF Vale do Sousa. Source: **marto`addressing`2017**

1.5 About the Structure of this Document

This document comes in five chapters: Following the current chapter which introduces the reader to the research context, problem statement and proposed solution, chapter 2 provides a detailed overview of the data used and preprocessed, including characteristics, preprocessing steps and challenges and outlines the implementation details of the preprocessing. In chapter 3 a single-objective integer linear programming model is developed and the solution approach is described, while chapter 4 presents and discusses the resulting solution, i.e. the forest road network design considered optimal under given constraints for the case study area, as well as potential limitations and some indications on potential further research. Chapter 5 concludes with a summary of the main findings and contributions of the work.

2

Data Preprocessing

Contents

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THIS chapter presents the data used in this work and describes the preprocessing steps necessary to transform it into a suitable input for the Integer Linear Programming (ILP) model. Each section within this chapter covers a different dataset, all of which are essential to addressing the problem. These sections are further organized into subsections, beginning with an introduction that presents the data source and a general data description, followed by subsections on preprocessing performed. Each section ends with a brief summary of the most relevant preprocessing results.

As a foundation for this study, we begin by analyzing the geospatial data relevant to the problem. This includes an examination of the existing public road infrastructure (section 2.1) and the forest management units (section 2.2) within the study area. A key aspect of this analysis is assessing the spatial relationships between existing roads and forest stands to determine which stands are already accessible and can therefore be excluded from the optimization model. Additionally, we review the harvesting schedule data (section 2.3), identifying the relevant information for this study and ensuring its compatibility with the subsequent modeling steps. Finally, we estimate the costs associated with constructing, maintaining, and upgrading forest roads on a per-kilometer basis (section 2.4). This structured approach transforms the data provided into the necessary foundation for this project. Then, the problem was decomposed into smaller, more manageable subproblems through spatial segmentation (section 2.5). Finally, the graph structure representing the network potential forest road segments was created (section 2.6).

Before delving into the data and the preprocessing steps undertaken to make it usable, some preliminary remarks regarding the scope and relevance of the data discussed might help the reader's understanding of the presented extracts: In all datasets used for this study, a wide range of attributes is available, as the data were originally collected for various purposes beyond the scope of this work. Many of these attributes pertain to other domain-specific aspects which, while valuable in their original context, are not directly relevant to the problem addressed here. Given the breadth of available information, it is neither practical nor necessary to present and discuss every attribute in detail. Instead, the following sections focus solely on the attributes that are essential for this study — those that contribute directly or indirectly to the analysis and modeling performed.

Regarding the preprocessing presented, only the key steps are outlined here for the sake of brevity and readability. Standard cleaning operations, validation checks, and other routine procedures have been omitted but can be found in the accompanying Jupyter notebooks, along with the full list of libraries used. Data manipulation and geospatial analysis were primarily conducted using the [pandas](#) and [GeoPandas](#) libraries, with visualizations mostly generated through [Matplotlib](#). Supplementary libraries were incorporated as required throughout the process.

2.1 Public Road Infrastructure (Geospatial Data)

The study area is intersected by public roads which serve as a critical component of the transportation infrastructure for timber and ground-based firefighting forces within the region. This data is particularly relevant to our problem for two reasons: First, because some forest stands are already accessible via existing roads, which reduces the complexity of the problem by limiting the need for new road construction; and second, because any future road network must connect to this existing infrastructure to ensure continuity and accessibility.

This section provides a brief description of the road dataset, followed by an overview of the preprocessing steps applied. Key steps performed on this data included the removal of roads marked as "inoperable" in the dataset (see 2.1.2), filling relevant data gaps (see 2.1.3), and distinguishing between two different road networks (see 2.1.4): one consisting of standard timber roads, and the other consisting exclusively of wider roads deemed suitable for fire vehicle access.

2.1.1 Introduction to the Road Dataset

The data on the roads considered in this study originates¹ from a geospatial dataset provided as a shapefile `RVF_CPV_clip.shp`, projected in the [Lisboa Hayford Gauss IGeoE \(ESRI:102164\)](#) coordinate system.² The dataset comprises 311 road segments in the municipality of Castelo de Paiva, each represented as a vector linestring. In total, the road network spans approximately 390 kilometers. Figure 2.1 visualizes the geometries of these segments.

Apart from the spatial geometries, the following attributes were used in this work:

Attribute	Description
Id	Unique identifier for the road
COMPRIM	Length of the road
OPERAC	Operational status of the road
LARGURA	Width of the road
DESIGNACAO	Designation or name of the road

¹According to its provider (Susete Marques, PhD), this dataset was — prior to its inclusion in this work — [clipped](#) from a bigger dataset originally obtained from the Municipal Forest Office.

²A projected coordinate system typically used for mapping in mainland Portugal; see <https://epsg.io/102164>.

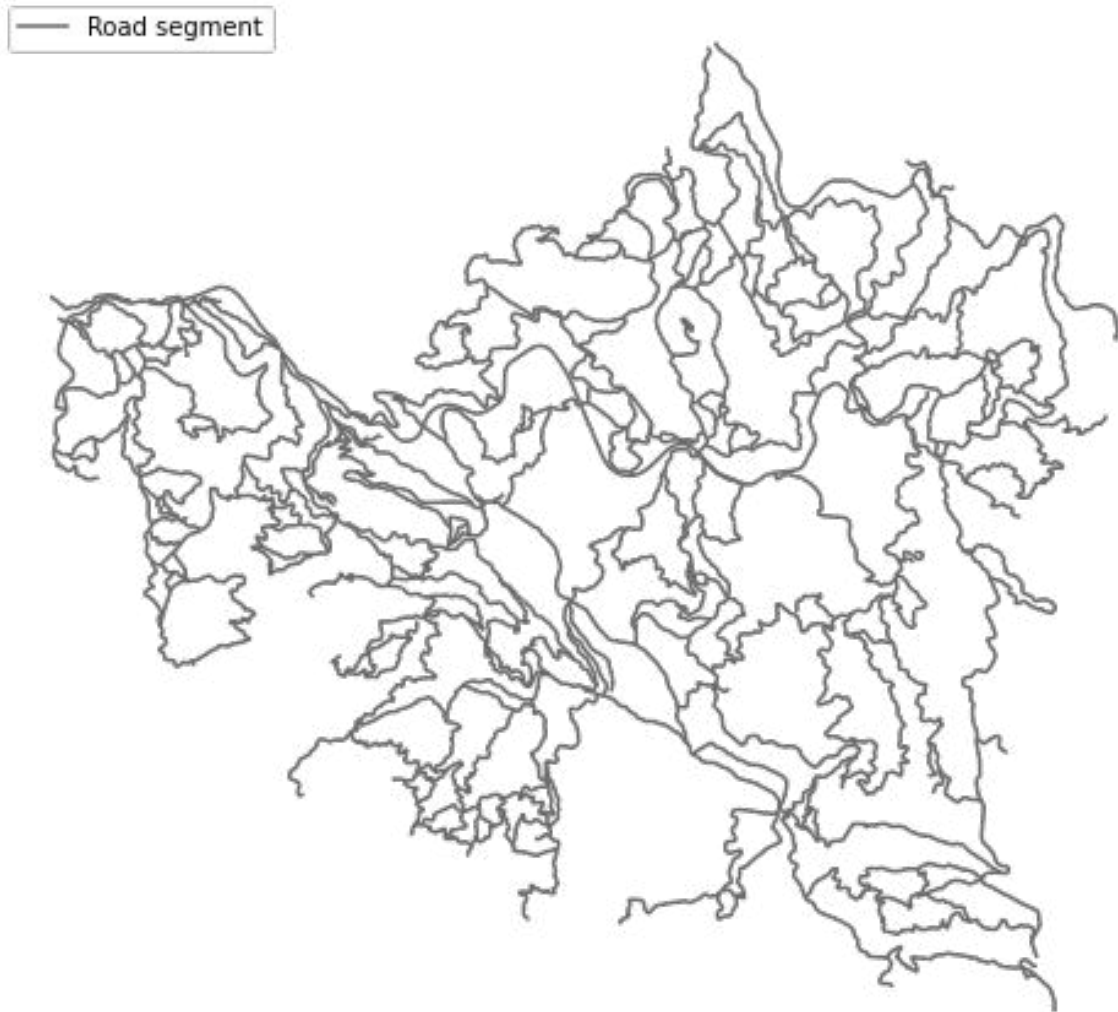


Figure 2.1: Road network in the study area of Paiva, prior to preprocessing

2.1.2 Removal of Inoperable Roads

Some roads in the dataset are marked as inoperable and can not be used. Only operable roads are relevant to this study. An assessment of road operability based on the `OPERAC` column revealed that 13 road segments are inoperable (visualized as red dotted lines in fig. D.1). As these roads do not contribute to the problem under investigation, they were removed from the dataset.

→ After this adjustment, 298 operable road segments remain with a total length of approximately 376 km.

2.1.3 Addressing Data Deficiencies: Road Width Values Reported as Zero

Upon examining the frequency distribution of the road width values (column `LARGURA`, see Table 2.1 and a corresponding plot in fig. D.2), it becomes evident that there are some "holes" in the data. Out of 298 operable road segments, 70 segments have a reported width of 0 meters, which is clearly implausible. Expected road widths typically fall between 3 meters and 10 meters, and they should always be greater than zero. After further data exploration, these zero values were deemed erroneous and attributed to data quality deficiencies requiring correction.

Table 2.1: Frequency table for `LARGURA` with implausible values highlighted in orange

LARGURA [m]	0	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6.5	7	7.5
Count	70	1	54	38	45	22	54	5	5	1	1	1	1

Given the high proportion of missing values (70 out of 298 segments), the corresponding features could not be removed from the dataset. Instead, a new column `ROADWIDTH` was created, and new road width values were imputed for segments with missing data. An estimate for the missing values can be derived from the `REDE_DFCI` attribute, following the classification system defined in [Categorias de classificação da rede viária florestal](#) (Ministério da Agricultura 2014). According to these guidelines, **first-order roads** (*1.ª ordem*) should have a usable lane width of at least 6 m; **second-order roads** (*2.ª ordem*) should range between 4 m and 6 m; and **complementary roads** (*complementar*) can be narrower. Based on this classification:

- 67 segments (`REDE_DFCI` = 1) were assigned a width of 6 m;
- 1 segment (`REDE_DFCI` = 2) was assigned a width of 4 m;
- 2 segments (`REDE_DFCI` = 3) were assigned a width of 3 m.

This approach ensures a consistent and interpretable road width variable across the entire dataset, enabling more reliable analysis in subsequent modeling steps. The resulting dataset includes a complete `ROADWIDTH` column, as illustrated in fig. D.3.

2.1.4 Distinguishing Fire Roads from the Full Set of Roads by Width

To enable timber transport, roads must have a minimum width of 3 m. To accommodate two-way traffic for firefighting vehicles, a wider minimum width of 6 m is required, which calls for the

creation of a separate subset of wide roads. While fig. D.4 shows the entire network of *"timber roads"*, fig. D.5 displays only the wider *"fire roads"*.

Note that forest stands adjacent to roads wider than 6 meters are fully accessible for both timber extraction and firefighting activities, and can therefore be excluded from the scope of this study. This will be addressed in the following section.

2.1.5 Summary of Road Data for Further Usage

Summarizing the relevant results from this section, in the area under study, we have:

- a **timber road network**, consisting of 205 roads wider than at least 3 meters suitable for timber transport (see fig. D.4),
- a **wider road network**, subset of the above timber roads, consisting of 70 roads wider than at least 6 meters suitable for both timber transport and firefighting traffic (see fig. D.5).

2.2 Forest Management Units (Geospatial Data)

Following the preparation of the existing road network, we now turn our attention to the forest management units (stands) within the study area — equally essential elements in the problem under investigation. A forest management unit is a defined area of forest land that is managed as a unit. These units are central to the planning of forest road infrastructure.

The significance of forest stands in the modeling framework for this problem stems from both their access requirements and spatial boundaries. First, the characteristics of each stand determine its specific needs for road access. These needs are incorporated as constraints in the ILP model (see chapter 3, section 3.2.4.C), ensuring that any proposed road construction accounts for the stand-level factors of timber availability and fire resilience. Second, due to the fragmented ownership of stands across different entities, the design of future road infrastructure must respect stand boundaries to uphold property divisions and support coordinated management. In particular, this approach allows for a fair distribution of the area losses associated with road placement among different landowners, ensuring that no single entity bears a disproportionate impact on timber production or economic returns.

This section begins with a description of the dataset used (section 2.2.1) and the corrections necessary to make it usable (section 2.2.2), followed by an analysis of road accessibility of the forest stands (section 2.2.3) - because only inaccessible stands are part of the decision problem under study.³

2.2.1 Introduction to the Forest Management Units Dataset

The data on forest management units used in this study originates from a geospatial dataset provided in the form of a shapefile  `CasteloPaiva.shp`, made available by Susete Marques, PhD. According to its provider, the dataset was derived from **Pléiades** satellite imagery with a spatial resolution of 0.5 meters. Each forest management unit is represented as a polygon (vector data). The dataset is projected in **ETRS89 / Portugal TM06 (EPSG:3763)**⁴. It comprises **687 forest management units** that vary in shape and size (see fig. 2.2). Each unit (stand) is homogeneous with respect to species composition, slope, age and site index (soil quality). For the purposes of this study, the only attributes used — aside from the spatial geometry — are

³Nevertheless, all stands are relevant for the creation of the road network.

⁴Standard projected Coordinate Reference System (CRS) for mainland Portugal, for details see <https://epsg.io/3763>.

ID_UG, the unique identifier for each management unit, and Declive, representing the stand's average slope.⁵



Figure 2.2: Castelo da Paiva: Forest stands

2.2.2 Spatial Correction and Topology Cleaning of Stands Data in QGIS

During verification in QGIS using multiple baselayers (e.g., Google Maps, OpenStreetMap), a noticeable spatial misalignment in the dataset became evident. The geometries of the forest stands appeared systematically shifted relative to recognizable geographic features such as rivers and roads. In the absence of accompanying metadata and after unsuccessful reprojection attempts, the issue was ultimately resolved by **manually shifting** the entire layer. This

⁵According to the data provider, the polygons were designed to have relatively homogeneous slopes.

correction was particularly important, as subsequent analyses involve spatial relationships between forest stands and the road network, for which accurate alignment is essential.

Following this spatial realignment, additional topological inconsistencies were identified. Visual inspection and the QGIS [Topology Checker](#) tool revealed numerous small gaps and overlaps between adjacent polygons - errors that could compromise spatial operations such as intersection and buffering. Specifically, the tool flagged **165 overlaps** and **709 gaps**, a quantity impractical to address manually. To resolve these issues, the following steps were undertaken:

1. **Snapping:** The dataset was snapped to itself using a 1-meter tolerance to improve the alignment of adjacent polygon boundaries.
2. **Geometry Validation:** Invalid geometries were identified and repaired using QGIS's [Fix Geometry](#).
3. **Second Topology Check:** A second run of the [Topology Checker](#) reported **84 gaps** and **32 overlaps**.
4. **Manual Inspection and Cleaning:** Upon review, **65 gaps** were retained as legitimate features (e.g., clearings or administrative boundaries), while **19 gaps** and **all 32 overlaps** were manually corrected using the [Vertex Tool](#) to ensure topological consistency.
5. **Multipolygon Correction:** During the previous steps, **two multipolygon features** were inadvertently created. These were manually inspected and compared with the original geometries, then corrected back into simple polygons using the [Vertex Tool](#) to ensure consistency across the dataset and simplify further data processing.

These corrections were essential for preparing the data to be suitable for the spatial operations and optimization modeling conducted in later stages of this work - particularly for the creation of the network of potential roads along the stand boundaries (see section 2.6).

2.2.3 Classification of Forest Stands by Road Accessibility

In this work, a road is considered suitable for timber logging and transport operations if it has a minimum width of 3 meters, hereafter referred to as a "*timber road*".⁶ A stand that is adjacent to

⁶This definition does not account for legal ownership constraints, nor for other physical or environmental limitations that may affect actual usability. All roads in the provided dataset that are marked as operable and have a width of at least 3 meters are considered suitable for timber logging.

a timber road (i.e., one with a minimum width of 3 meters) is considered "*timber-accessible*".⁷ A stand that is adjacent to a wide road (i.e., one with a minimum width of 6 meters) is considered "*fully accessible*". The decision problem addressed in this study includes only those stands that are currently not timber-accessible, hereafter referred to as "*inaccessible*".⁸

A corresponding spatial analysis of the relationships between forest stands and the surrounding road network was performed using the `geometry.intersects` method from GeoPandas, which determines whether two geometric objects share any points or areas. To ensure spatial accuracy and address initial data imprecision, preprocessing steps included aligning the CRSs of the datasets and applying a 6-meter spatial buffer around the stands. In a first classification step, a set of **203 fully accessible stands** (see fig. D.6 in appendix D.2, shown in green) was identified.⁹

In a second step, the remaining 484 stands not accessible via wide roads (orange-hatched in fig. D.6) were further classified. Figure D.7 shows the resulting sets: There are **169 timber-accessible stands** (light green) and **315 inaccessible stands** (orange cross-hatched).

Only the 315 inaccessible stands (orange cross-hatched in fig. D.7 and solid orange in fig. D.8) are considered in the optimization. However, all stands remain relevant for the creation of the road network, as roads should follow stand boundaries.

2.2.4 Summary of Stand Accessibility Classification

The accessibility analysis yields the following classification:

- **315 inaccessible stands** (see fig. D.8)
→ These are included in the optimization model; their access needs are to be extracted from the harvesting schedule data (see section 2.3).
- **372 stands that are already accessible**
→ Excluded from the optimization, but relevant for geometry since stand boundaries constrain road placement.

⁷Since wider fire roads are a subset of timber roads, stands adjacent to them are also timber-accessible.

⁸Determining which narrow roads should be widened to improve firefighting access constitutes a separate planning problem and falls outside the scope of this study (discussed in chapter 4).

⁹Since both timber extraction and firefighting access to these stands are already supported by the existing road network, their access needs are considered fulfilled and must be excluded from the decision problem. This was done by filtering out the corresponding prescription data (see section 2.3). This exclusion reduces the required road construction, thereby lowering overall costs and simplifying the optimization by reducing its size and complexity.

2.3 Harvesting Schedules (Tabular Data)

This section outlines how the provided harvesting schedules - detailing when, where, and how much timber is to be removed - inform the road access requirements used in the optimization model. Starting from three pre-defined harvesting schedules (referred to as “solutions”),¹⁰ we determine, for each 10-year-period, which stands require road access and what type (timber road vs. wider fire road). Although this information is not explicitly included in the original datasets, it can be systematically inferred through a sequence of processing steps. After reformatting the stakeholder solution to match the structure of the other two (section 2.3.1), each solution was then merged with the full prescription dataset to incorporate the key attributes *Vremovido*, *rait0/rait5/rait10*, for each period (see section 2.3.2). Fire resistance values were assigned based on shrub cleaning periodicity (*ShPer*), which may occur every 5, or 10 years, or not at all (section 2.3.3). Special cases and anomalies in the data were identified and analyzed (section 2.3.4). Finally, the dataset was filtered to include only stands that are part of the decision problem, based on the results of the accessibility classification from section 2.2. Ultimately, the road access requirements for each stand and period were derived (section 2.3.5).

2.3.1 Introduction to the Harvesting Schedules Dataset

The harvesting schedules used in this study were developed by Susete Marques, PhD, Olha Nahorna, and Felipe Silva, based on data collected in the fall of 2019 and processed in 2020. These schedules represent the outcome of simulations of multiple forest management alternatives, using growth annulled simulators/yield-tables (e.g., SIMFLOR; for more details, see appendix B) developed in collaboration with stakeholders from the AFVS. Each management alternative is referred to as a “solution,” with every solution assigning a specific prescription to each forest stand.

The dataset includes **three alternative harvesting schedules**, each representing a distinct management strategy:

🌲 **MaxWood solution** — optimized for maximum wood production,

💧 **MaxRes solution** — optimized for maximum fire resistance,

¹⁰These were generated through a prior optimization process conducted independently of this study.

 **Stakeholder solution** — selected by stakeholders¹¹.

All three schedules cover the same set of forest management units and share a 50-year planning horizon, divided into five 10-year periods. The dataset was provided in the form of an Excel workbook  `Presc_dataset21_missingstakeholders_sol.xlsx` containing three worksheets, along with a separate log file:

 `Presc` — includes the full set of possible prescriptions (section 2.3.1.A);

 `MaxWood` — details the MaxWood solution (section 2.3.1.B);

 `MaxFireRES` — details the MaxRes solution (section 2.3.1.C);

 `Stakeholders.Solution.log` — logfile containing the stakeholder-selected solution (section 2.3.1.D).

Please note that the harvesting schedule data includes stands not only within [Paiva](#) (the study area for this work, introduced in section 1.4 and with corresponding stands detailed in section 2.2) but also from the neighboring ZIF [Entre Douro e Sousa](#). Since this Master's dissertation focuses exclusively on Paiva, stands outside the study area must be excluded. However, to ensure a comprehensive understanding of the dataset and maintain consistency during preprocessing, the harvesting schedule data were initially analyzed in full. Most preprocessing steps were applied to the complete dataset, with filtering of out-of-scope stands performed only at the final stage.

2.3.1.A Prescription Data

The prescription data  `Presc` is extensive, containing 41 attributes (columns) and 317,883 entries (rows), where each row corresponds to a possible prescription for a forest stand in a given period. Most of the columns include information irrelevant to the current problem, and a large portion of the rows represent prescriptions not selected in any solution. Therefore, only a subset of this data was used in this project. The attributes relevant to the problem under study are detailed in the grey box below and exemplified by the excerpt in Table 2.2 in the appendix.

¹¹According to the data provider, this selection was based on a Pareto front involving four optimization criteria (carbon, biodiversity, fire resistance and erosion), selected by the stakeholders in a 2021 meeting in Oporto

Remark. The reader may notice that some attribute names contain spaces, such as `Na sol MaxRES`. While this formatting is not ideal for programmatic handling and such spaces should ideally be removed, the names are shown here as-is to reflect the original dataset and to improve readability in the accompanying exemplary table extracts.

Attribute	Description
<code>UG</code>	Unique identifier for the forest stand
<code>presc</code>	Unique identifier for the prescription
<code>species</code>	Species prescribed for the stand
<code>area</code>	Area in hectares occupied by the species
<code>ug.Sp</code>	Control column for data consistency
<code>Control</code>	Control column for data consistency
<code>Na sol MaxRES</code>	Indicator whether the prescription is included in MaxRES solution
<code>Na sol MaxWood</code>	Indicator whether the prescription is included in MaxWood solution
<code>period</code>	Harvesting period
<code>vthin (ton)</code>	Volume of the stand thinned in tons
<code>vharv (ton)</code>	Volume of the stand harvested in tons
<code>Vremovido (ton)</code>	Total biomass removed (in tons)
<code>rait0</code>	Fire resistance indicator ¹ for no occurring shrub cleaning
<code>rait5</code>	Fire resistance indicator ¹ for shrub cleaning interval of 5 years
<code>rait10</code>	Fire resistance indicator ¹ for shrub cleaning interval of 10 years

Table 2.2: Illustration of relevant columns from the prescription data sheet

UG	ug.Sp	area	presc	Na sol MaxRES	Na sol MaxWood	period	species	vthin (ton)	vharv (ton)	Vremovido (ton)	rait0	rait5	rait10
1	1_Ec	42.6018	1	#N/A	#N/A	1	Ec	0	2999.2720	2999.2720	132.0655	166.147	136.3257
1	1_Ec	42.6018	1	#N/A	#N/A	2	Ec	0	5575.8278	5575.8278	140.5859	166.147	140.5859
...	...	...	...	...	...	...	...	...	...	...	...	...	...

Full table size: 317,883 rows across 41 columns.

2.3.1.B MaxWood Solution

The  MaxWood solution was provided in tabular format, comprising 1,512 entries across the seven columns described in the grey box below. This dataset, illustrated in Table 2.3, specifies the prescriptions selected under the MaxWood scenario — that is, the set of management actions aimed at maximizing wood production.

¹Deeper explanation of the `rait` values: They originate from a fire resistance indicator calculated in prior work (methods of Ferreira et al. 2015, adjusted and applied to Vale do Sousa by **marto** *addressing* 2017). Ranging from 0 to 1, the indicator was multiplied by the stand's area, producing the present numerical `rait` values in value range $\mathbb{R}^+$.

Attribute	Description
Decision_Variable_Name	Identifier of the decision variable, formed by concatenating Presc, UG, Sp, and ShPer
Control	Control attribute for data consistency (UG.Presc)
Presc	Unique identifier of the selected prescription
UG	Unique identifier of the forest stand
Sp	Prescribed species
ShPer	The prescribed periodicity of shrub cleaning (0, 5 or 10)
Sol	Indicator if prescription is included in the solution (Yes/No)

Table 2.3: Excerpt from MaxWood Solution sheet

Decision Variable Name	Control	Presc	UG	Sp	ShPer	Sol
Presc3502_Pa1_Ec_0	1.3502	3502	1	Ec	0	Yes
Presc19_Pa2_Ec_0	2.19	19	2	Ec	0	Yes
Presc21_Pa3_Ec_10	3.21	21	3	Ec	10	Yes
...	...	...	...	...	...	Yes

Full table length: 1512 rows

Remark. The length of the MaxWood solution table (1512 rows) does not correspond with the number of stands contained in it (1406 unique values in column UG), indicating the presence of stands that have two different prescriptions within one and the same harvesting plan. This will be further addressed in section 2.3.4.

2.3.1.C MaxRES Solution

Similarly, the sheet  MaxFireRES identifies the selected prescriptions for the MaxRes solution — that is, the prescriptions recommended to maximize the fire resistance of the forest. It was also provided in tabular format (1,512×7), as illustrated in Table 2.4. The columns are identical to those in the MaxWood Solution sheet; for a description, please refer to the grey box above.

Table 2.4: Excerpt from MaxRES Solution sheet

Decision Variable Name	Control	Presc	UG	Sp	ShPer	Sol
Presc1601_Pa1_Ec_5	1.1601	1601	1	Ec	5	Yes
...	...	...	...	...	...	Yes

Full table length: 1512 rows

Remark. As with the previous solution, the number of stands does not match the row count: while there are 1,512 rows, the solution contains prescriptions for 1,406 stands, of which 106 have two prescriptions instead of one. The reason for this discrepancy is explained in section 2.3.4.

2.3.1.D Stakeholder Solution

The stakeholder solution was provided in a different format — essentially as raw output in the form of a logfile generated directly by the optimizer. To make the data usable, the decision variable names and their corresponding solution values were extracted from the file and converted into a tabular format (see Table 2.5 below at the left). At this stage, the dataset contained only the two columns described in the grey box below. The stakeholder solution data contains 1,514 rows.

Attribute	Description
Variable Name	name of the decision variable, concatenating Presc, UG, Sp, ShPer
Solution Value	fraction of area of stand UG which is assigned to prescription presc

The data in the Variable Name column are not immediately usable, as they contain concatenated identifiers — specifically the prescription ID, forest stand ID, species, and shrub cleaning periodicity (Presc, UG, Sp, and ShPer). To enable further processing, these components had to be parsed and restructured into separate attributes, matching the format used in the MaxWood and MaxRes solutions. This was accomplished using regular expressions to extract the relevant elements: the numeric values following the prefixes “Presc” and “Pa” were assigned to the Presc and UG columns, respectively, and the shrub cleaning periodicity (ShPer) was also extracted. A species column was not created at this stage, as that information already exists in the complete prescription dataset, which would later be merged with each solution. An excerpt of the resulting structured data is shown in Table 2.6 at the right as an illustrative example.

During the transformation of the stakeholder solution, two management units were identified that contained multiple prescriptions in the same period, with fractional solution values between 0 and 1 (see Table 2.7). Consultation with the subject-matter expert indicated that the solution space was formulated using continuous decision variables bounded within $[0, 1]$, where each variable denotes the proportion of a stand’s area allocated to a specific prescription. Consequently, the fractional values for each management unit (UG) collectively sum to 1, representing full area coverage.

Table 2.5: Stakeholder solution (from logfile)

Variable Name	Solution Value
Presc503_Pa101_Pb_10	1.000000
Presc503_Pa1015_Pb_5	1.000000
Presc500602_Pa1016_Pb_0	1.000000
...	...

Full table length: 1514 rows

Table 2.6: Stakeholder solution (processed)

Presc	UG	ShPer	Solution Value
503	101	10	1.000000
503	1015	5	1.000000
500602	1016	0	1.000000
...	...	...	...

Full

table length: 1514 rows

Table 2.7: Fractional solution values in stakeholder solution

UG	Presc	ShPer	Solution Value
1253	36503	10	0.494295
1253	36	5	0.505705
1339	19700	0	0.567668
1339	19318	0	0.432332

Before delving deeper into the further processing, let us define the intended outcome:

2.3.1.E Objective of Preprocessing the Harvesting Schedule Data

The goal is to identify, for each period, the **set of stands that require road access** and determine the **appropriate road width**. This relies on two key stand-level attributes:

- **Timber removal volume** ($V_{removido}$) – Any stand scheduled for timber harvesting or thinning necessitates a path connected to the road network.
- **Fire resistance indicator** ($rait0/rait5/rait10$)^a – Stands with fire resistance below a predefined threshold need a wider road to ensure access for firefighting.

^aThe variable to be used is the one corresponding to the shrub cleaning periodicity prescribed.

2.3.2 Integrating Prescription Data into the Solutions

Each of the three solutions must be integrated with the prescription data to extract essential attributes: $V_{removido}$, $rait0$, $rait5$, $rait10$, and the corresponding harvesting period. This was accomplished by merging the relevant prescription data with each solution.

For the MaxRes and MaxWood solutions, the integration process began by filtering the prescription dataset to retain only those rows where the columns $Na_{sol\ MaxRES}$ or $Na_{sol\ MaxWood}$,

respectively, indicated inclusion in the solution. Then, `pandas.merge` was used with an outer join to combine the filtered prescription data with the corresponding solution, matching rows based on the `UG` and `Presc` columns. This outer join ensured that all rows from both tables were retained. The results of these merges are exemplarily illustrated in Table 2.8 and Table 2.9, with only the relevant columns shown.

Unlike the other solutions, the stakeholder solution lacks an explicit indicator in the prescription data identifying which prescriptions are included. As a result, no filtering was performed prior to merging. Instead, an outer join was applied directly, and any rows containing *NaN* values in the merged result were removed in a subsequent step. The resulting merged stakeholder solution is exemplarily illustrated in Table 2.10.

Table 2.8: MaxRes solution merged with prescription data

UG	Presc	species	period	Vremovido (ton)	rait0	rait5	rait10	ShPer
1	1601	Ec	1	2999.2720	110.7646	115.0248	85.2036	5
...	...	...	...	...	...	...	...	...

Length: 8212 rows

Table 2.9: MaxWood solution merged with prescription data

UG	Presc	species	period	Vremovido (ton)	rait0	rait5	rait10	ShPer
1	3502	Ec	1	3968.3560	136.3257	149.1062	119.285	0
...	...	...	...	...	...	...	...	...

Length: 7670 rows

Table 2.10: Stakeholder solution merged with prescription data

UG	Presc	species	period	Vremovido (ton)	rait0	rait5	rait10	ShPer
69	1	Ec	1	2999.2719	110.7646	115.0248	85.2036	5
...	...	...	...	...	...	...	...	...

Length: 8403 rows

The reader might wonder why, starting from 1,512 selected prescriptions in each solution (1,514 for stakeholder), the integration with the prescription data results in tables of completely different

lengths — none of which equal the product of the number of prescriptions times the number of periods:

$$\underbrace{1512}_{\text{\# prescriptions}} \times \underbrace{5}_{\text{\# periods}} = \underbrace{7560}_{\text{expected length of integrated solution/prescription data}}$$

This discrepancy arises because the harvesting schedule data includes special cases where, after a total harvest, a different species is planted. In these cases, two rows of data are generated for the period of change instead of one, which explains the increased number of rows. A detailed explanation of the impact of such special cases on the data is provided in section 2.3.4.

With these nuances accounted for, the integration process ensures that all necessary information is now in place, setting the stage for the next analytical step and allowing us to confidently assign the appropriate fire resistance values based on the harvesting schedules data.

2.3.3 Assigning the Appropriate Fire Resistance Value

Each prescription specifies a shrub cleaning periodicity (*ShPer*) — either 0, 5, or 10 years — which directly affects the stand’s fire resistance. Correspondingly, the dataset includes three fire resistance indicators: *rait0*, *rait5*, and *rait10*. Since decisions on road access requirements depend on a single fire resistance value per stand, only one of these indicators can be retained and used for each record.

To address this, a new column, *Rait*, was created. For each record, the appropriate fire resistance value was assigned by selecting the fire resistance indicator that corresponds to its specified *ShPer*, as illustrated in Table 2.11.

Table 2.11: Visualization of assigned fire resistance values (*Rait*) based on shrub cleaning periodicity (*ShPer*), selected values highlighted in yellow

<i>ShPer</i>	<i>rait0</i>	<i>rait5</i>	<i>rait10</i>	<i>Rait</i>
...	...	...	...	...
5	6.5650	7.8357	6.7768	7.8357
0	136.3257	149.1062	119.2850	136.3257
...	...	...	...	...

With all relevant attributes now available in the dataset for each stand and period, the data is nearly ready for use in the model. However, before moving on to formatting and storage considerations (see section 2.3.5), it is important to address a few special cases present in the data — namely, species conversions and mixed stands.

2.3.4 Notable Special Cases: Species Conversions and Mixed Stands

In preparing the dataset for model input, two noteworthy structural patterns emerged that require to be addressed: **species conversions** and **mixed stands**. Although they may appear similar at first glance — both involving more than one species in a stand — their data structure and implications are fundamentally different.

Each case is described in detail below, followed by an assessment of its impact on the size of the merged datasets.

Species Conversions

A species conversion occurs when, within a single prescription and stand, different species are listed for the same period.¹² This reflects a harvest-replant sequence within the 10-year period: the original species (e.g., *Ec*) is harvested and then replaced by a new species (e.g., *Ct*). These rows represent consecutive steps within the same prescription.

Table 2.12 illustrates an example: During period 1, prescription no. 1312 in stand no. 1 prescribes a change from *Ec* to *Ct*. Characteristically for such species conversions is that the "old" species shows a high *Vremovido* value, while the newly planted species has a *Vremovido* value of zero. Additionally, the space previously occupied by the "old" species is fully allocated to the "new" species (see *Area*).

Table 2.12: Example of a species conversion

UG	Presc	Species	Area	Period	Vremovido (ton)
1	1312	Ct	42.602	1	0.0
1	1312	Ec	42.602	1	2999.27
1	1312	Ct	42.602	2	0.0
1	1312	Ct	42.602	3	50.84

Mixed Stands

A mixed stand refers to a case where, in the same stand and period, *two different prescriptions* with distinct species are applied simultaneously. In contrast to species conversions, both

¹²It happens the first time a stand is harvested. Conversion can only happen once in the planning horizon, and there can be no conversion back into Eucalypt.

species are present together — each covering a different portion of the stand's area.

As illustrated in Table 2.13, during the first two periods, stand no. 1015 contains the species *Ec* along with the second species *Pb*, each applied via a different prescription (no. 3 and no. 503) on only a fraction of the total area.

Table 2.13: Example of a mixed stand

UG	Presc	Species	Area	Period
1015	3	Ec	2.934	1
1015	503	Pb	8.036	1
1015	3	Ec	2.934	2
1015	503	Pb	8.036	2

Impact on Row Counts

These structural variations explain the peculiarities mentioned previously: Each solution's prescriptions apply only to a total of 1,406 stands (see Table 2.3, Table 2.4 and the corresponding remarks below). Still, the solutions consist of 1,512 rows (stakeholder solution: 1,514) - which is caused by the presence of mixed stands: 106 (stakeholder solution: 108) stands in the solution have two prescription instead of one.

Secondly, the species conversions affect the number of rows in the merged solution datasets (Table 2.8, Table 2.9, Table 2.10). Each species conversion results in two entries per [UG, Period, Presc] combination, thus inflating row counts beyond the baseline of $1,512 \times 5 = 7,560$ (stakeholder solution: $1,514 \times 5 = 7,570$) expected for five periods per prescription. The numbers of species conversions occurrences in each solution dataset are as follows: 652 in the MaxRes solution, 110 in the MaxWood solution, and 833 in the Stakeholder solution - which explains the table lengths:

MaxRes solution: $7,560 + 652 = 8,212$ rows,

MaxWood solution: $7,560 + 110 = 7,670$ rows,

Stakeholder solution: $7,570 + 833 = 8,403$ rows.

Remark. We observe that the Stakeholder and MaxRes solutions — both prioritizing fire resistance — feature significantly more species conversions than the MaxWood solution, which

instead optimizes for timber yield. This difference arises from variations in growth rates among species. In particular, species conversions away from eucalyptus, which is currently dominant and grows rapidly, are rare in the MaxWood solution due to its emphasis on maximizing productivity. Notably, eucalyptus replanting is restricted under Portuguese regulations.¹³

2.3.5 Determining and Storing Access Requirements by Stand and Period

At this stage, preprocessing of the harvesting schedule data is essentially complete. Data pertaining to stands that fall outside the scope of the study can therefore be removed.

Filter for Relevant Data Only (Inaccessible Stands in Paiva) As outlined in section 2.2.1, the study area is only Castelo de Paiva. However, no separate filtering step was needed for that: As stated in section 2.2.4, of the 687 stands located in Castelo de Paiva, only the 315 that are classified as inaccessible are relevant for the road access analysis, while all accessible stands shall be excluded from the problem. Consequently, the preprocessed harvesting schedule data were filtered to retain only data on the 315 inaccessible stands in Paiva (see illustration in fig. D.8).

General Access Determination Rules Road access requirements for each stand and period were determined based on the following two key variables:

- **Vremovido:** Indicates the volume of timber removed during a given period. If the value is greater than zero, at least timber road access (i.e., the construction of at least a 3-meter road) is required.
- **Rait:** Indicates the level of fire resistance in a given period. If the value falls within the lowest 5% quantile¹⁴, wide firefighting road access (i.e., the construction of a 6-meter road) is required.

Remark. The *Rait* values used in this step are static and reflect the current infrastructure. Dynamically updating *Rait* in response to changes in the network over time could better capture fire resistance development of the stands and could be considered in future work.

¹³Following the devastating wildfires in 2017, the Portuguese government enacted legislation ([Decreto-Lei n.º 11/2019](#)) restricting eucalyptus planting, particularly its reintroduction in areas previously occupied by other species, with the aim to protect native forests and reduce fire risk.

¹⁴This threshold is illustrative and adjustable. Final selection should reflect expert and stakeholder judgment.

Handling Stands with Multiple Prescriptions In some cases, a stand may have more than one prescription per period due to special conditions such as species conversions or mixed stands (as discussed in section 2.3.4), leading to multiple $V_{removido}$ and R_{ait} values. These cases were handled as follows:

- $V_{removido}$: If multiple values exist for a given stand and period and any prescription of them has a value greater than zero, access is considered required.
- R_{ait} : If multiple values exist for a stand and period, the minimum R_{ait} value is used.¹⁵

For each harvesting schedule, a road access needs matrix was constructed. Each matrix identifies, for every stand and period, whether road access is required and of which type. The first column lists the stands, the next five columns indicate timber road access for periods $t = 1, \dots, 5$ via binary values (1 = access needed, 0 = not needed), and the last five columns indicate firefighting road access for the same periods.

2.3.6 Summary of Preprocessed Harvesting Schedules: Access Needs

The completed preprocessing of the harvesting schedule data resulted eventually in the wanted road access requirements by stand and period, stored in form of matrices (315x11).

Here presented exemplarily for the MaxRes solution:

UG	t_1^{timber}	t_2^{timber}	t_3^{timber}	t_4^{timber}	t_5^{timber}	t_1^{wide}	t_2^{wide}	t_3^{wide}	t_4^{wide}	t_5^{wide}
840	1	0	1	1	1	0	0	0	0	0
841	1	1	0	1	1	0	0	0	0	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1620	0	1	1	1	1	0	0	0	0	0

¹⁵The minimum was preferred, rather than the average or weighted average. This conservative, worst-case approach ensures that fire resistance is not overestimated, particularly in periods with species conversions or pending harvests, which could otherwise result in high economic losses in case of fire. In future work, alternative approaches could be explored.

2.4 Costs for Road Construction and Maintenance

The final pillar of information required to address the problem at hand concerns the costs associated with constructing and maintaining forest roads. As outlined before, the model is designed with the objective is to minimize these costs. Accordingly, a preparatory step involves quantifying the cost per kilometer for each of the following actions:

- Constructing a timber road,
- Constructing a wider fire road road,
- Maintaining an existing timber road,
- Maintaining an existing wider fire road,
- Upgrading (Widening) a timber harvesting path to a wider fire road.

This section begins by outlining the available cost data (section 2.4.1), followed by the assumptions used to fill data gaps (section 2.4.2), and concludes with the derived cost estimates applied in the optimization model (section 2.4.3).

2.4.1 Available Cost Data

The Portuguese Monitoring Commission for Forest Operations ([Comissão de Acompanhamento para as Operações Florestais](#) (CAOF)) provides a reference matrix detailing minimum and maximum costs for key forestry activities, including manual, mechanical, and mixed afforestation operations; reforestation; infrastructure development; and maintenance within forest territories. This [matriz de \(re\)arborização](#) (CAOF 2022) includes cost estimates for the construction and the maintenance of one kilometer of timber road, categorized by three slope classes.

The CAOF values refer to forest paths of standard width suitable for mechanized timber harvesting operations. However, this study also accounts for wider fire roads, assumed to be twice the width of standard timber roads. As not all actions considered in this work are explicitly covered in the cost listings of CAOF's [matriz de \(re\)arborização](#), several assumptions¹⁶ were necessary to estimate the remaining costs for the additional configurations.

¹⁶These assumptions were developed in consultation with the thesis supervisors who are considered subject matter experts.

2.4.2 Cost Estimation Assumptions

- (i) The cost of constructing a wider fire road is assumed to be 150% of the cost of constructing a standard timber road:

$$\text{CostC}_{\text{wide}} = 1.5 \cdot \text{CostC}_{\text{timber}} \quad (2.1)$$

- (ii) The cost of maintaining a wider fire road is assumed to be the same as maintaining a standard timber road:

$$\text{CostM}_{\text{wide}} = \text{CostM}_{\text{timber}} \quad (2.2)$$

- (iii) The cost of upgrading a standard timber road to a wider fire road is assumed to be 75% of the cost of constructing a standard timber road:

$$\text{CostsU} = 0.75 \times \text{CostC}_{\text{timber}} \quad (2.3)$$

- (iv) The cost of operating with a cross slope between 5% and 25% is approximated by averaging the minimum and maximum slope category costs:

$$\text{Cost}_{(5 < \text{slope} < 25\%)} = \frac{\text{Cost}_{(\text{slope} \leq 5\%)} + \text{Cost}_{(\text{slope} \geq 25\%)}}{2} \quad (2.4)$$

2.4.3 Result: Estimated Costs per Action per Road Kilometer

Table 2.14 presents the estimated costs in EUR per kilometer, based on the values provided by CAO and the assumptions outlined above. The color coding in the table corresponds to the colors used in the assumptions, indicating which assumption was applied to derive each value.

Using this data, we can determine the construction, maintenance, and upgrade costs for all potential road segments considered in this study, as detailed in section 2.6. Specifically, for each segment, once the appropriate slope category has been identified, its length is multiplied by the corresponding cost per kilometer to obtain the total cost. The obtained values are incorporated into the ILP model as input parameters. Given that the problem is multi-period, the fixed values shown in the Table 2.14 were discounted at a rate of 3.5% during implementation.

Table 2.14: Estimated Costs in EUR per action per kilometer; with highlighted entries color-coded to their respective assumptions.

Slope [%]	CostC _{timber}	CostM _{timber}	CostC _{wide}	CostM _{wide}	CostU
≤ 5	2147	1073.5	3220.5	1073.5	1610.25
$5 < \dots < 25$	4830.75	1878.625	7246.125	1878.625	3623.0625
≥ 25	7514.5	2683.75	11271.75	2683.75	5635.875

2.5 Segmentation of the Problem into Independent Subproblems

The problem was decomposed into smaller, more manageable subproblems through implementation of a graph-based spatial segmentation, as mentioned conceptually in section 3.3. Each inaccessible forest stand can be represented as a node $v \in V$ in an undirected graph $G = (V, E)$,¹⁷ with spatial adjacency relationships between the stands forming the "edges" $e \in E$ of the graph. Connected component analysis was used to partition this graph into geographically cohesive components (section 2.5.1). Each component corresponds to a spatial cluster of adjacent inaccessible stands that can be treated as an independent optimization subproblem. Around each of these inaccessible clusters, the ring of neighboring stands along which the inaccessible ones will be connected to the road segment were identified (section 2.5.2). To maintain consistency in the modeling, spatially adjacent components with potentially shared access roads were combined (section 2.5.3). This allows that road infrastructure will be shared, where geographically possible.

Remark. Note that natural geographical barriers, if existing, (e.g., rivers, streams, or steep terrain) should be additionally incorporated in a previous segmentation step. Such barriers may prevent connectivity between stands even if they are spatially adjacent, and therefore must be accounted for to ensure a realistic decomposition.

2.5.1 Identification of Clusters of Inaccessible Stands

To systematically identify clusters of inaccessible stands, a graph-based method using the Python library `NetworkX` was used. Each inaccessible stand in the dataset was represented

¹⁷This graph is not to be confused with the graph representing potential future roads, they are entirely different, just the same mathematical concept is applied.

as a node in an undirected graph, uniquely identified by its ID_UG attribute. An edge was created between two nodes if the stands they represented were spatially adjacent. Adjacency was determined by applying a 1-meter buffer to each stand polygon and checking for overlaps between these buffered geometries. Stands with overlapping buffers were considered neighbors and connected by an edge in the graph. This procedure resulted in a graph capturing the spatial relationships among inaccessible stands (see fig. 2.3). Connected components of the graph – subsets of nodes connected to each other but disconnected from other such subsets – were then computed. Each connected component corresponds to a contiguous cluster of inaccessible stands. Using this method, **51 clusters of inaccessible stands** were identified.

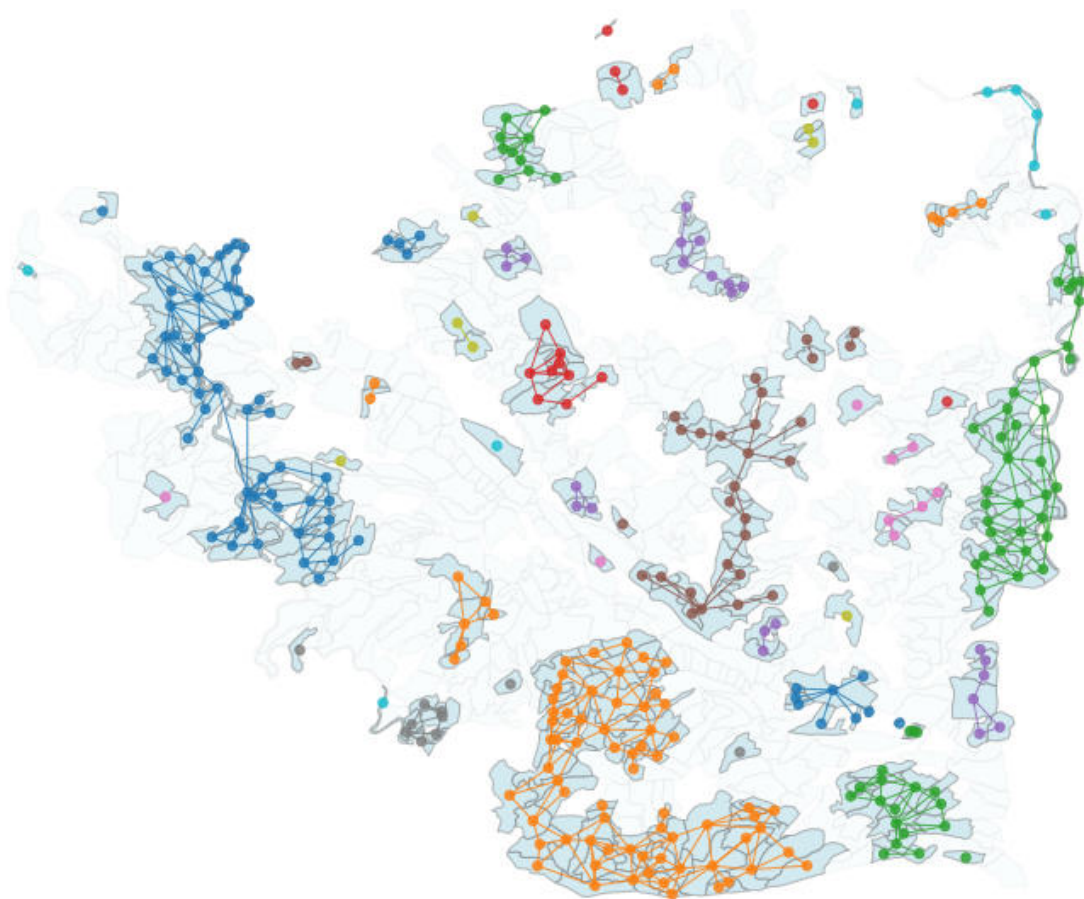


Figure 2.3: Inaccessible stands with their graph representation, forming 51 clusters

2.5.2 Merging Inaccessible Clusters with Surrounding Accessible Neighbors

The inaccessible patches had to be combined with their neighboring stands already connected to the road network, as new access roads would inevitably follow the boundaries of these accessible stands. To achieve this, each inaccessible cluster was dissolved into a single geometry, and a 1-meter buffer was applied to identify neighboring stands (their “ring of neighbors,” see fig. 2.4 in green). These neighbors form the immediate interface through which future access roads can connect to the existing infrastructure. After combining each inaccessible component with its neighboring stands, the boundaries of this newly created components were stored for further analysis.

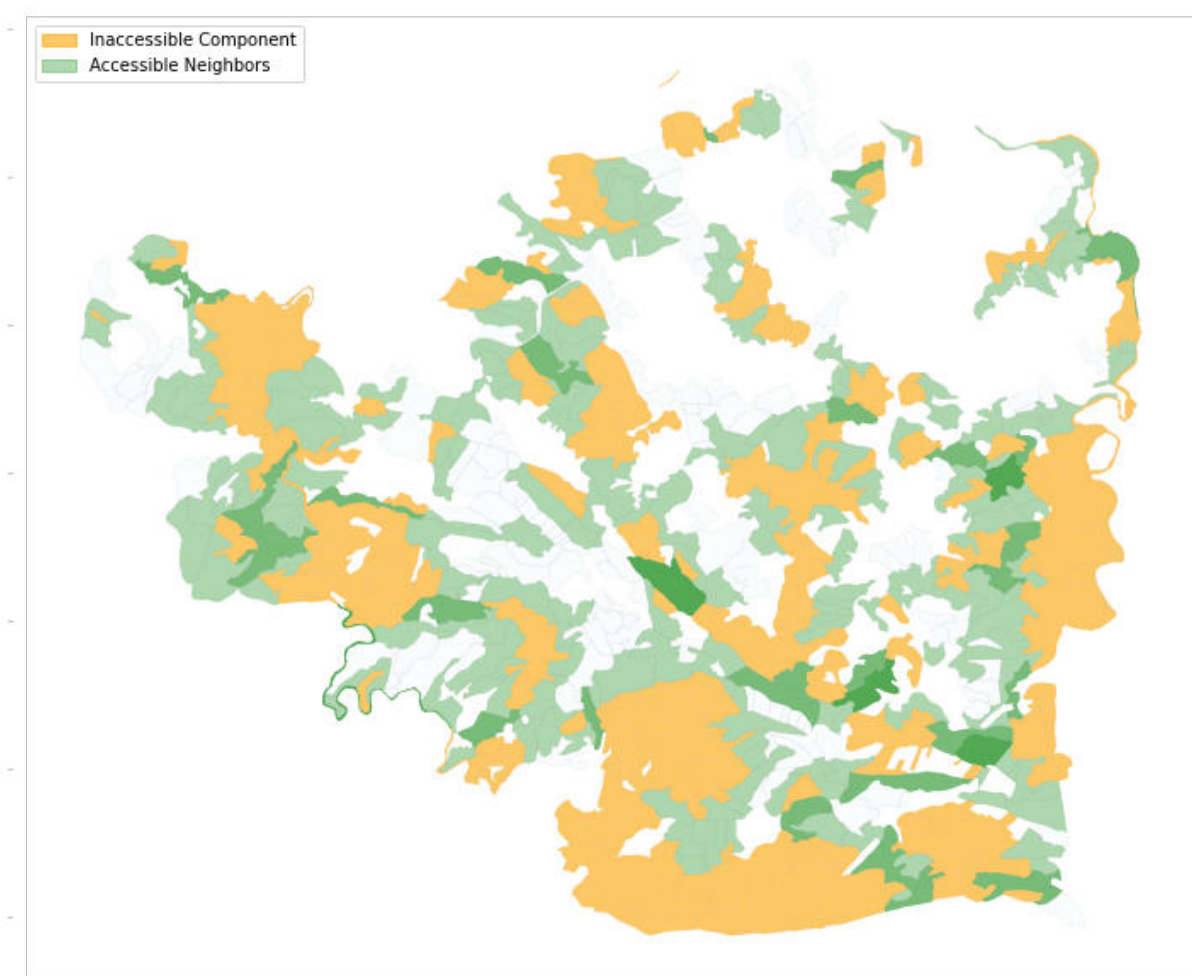


Figure 2.4: Inaccessible areas (yellow) surrounded by their neighboring already accessible stands (green). Darker shades of green reflect that an accessible stand is neighboring several inaccessible clusters.

As indicated by the darker green tones in fig. 2.4, several accessible stands are adjacent to more than one inaccessible component, which requires particular attention.

2.5.3 Handling Overlaps Between Rings of Accessible Neighboring Stands

Overlaps occur when an accessible stand is adjacent to more than one cluster of inaccessible stands, thereby belonging to multiple of the components created in section 2.5.2. To avoid redundant infrastructure planning, such overlapping components must be merged whenever access roads to one cluster could also serve another. In contrast, if two clusters share an accessible stand but lie on opposite sides of the existing road network – so that their future access roads cannot realistically overlap – their planning can remain independent, since access to one does not affect access to the other. Similarly, if two inaccessible clusters touch along their boundaries but connecting them would require a significantly longer detour than linking each directly to the network, they are treated as independent “opposite-side” cases.

Applying this rule, overlapping components were merged into joint units for access planning. This procedure yielded a decomposition into **35 independent components**, which are presented in appendix D.

2.5.4 Result: Independent Subproblems for Road Network Planning

Figures D.10 to D.43 illustrate the segmentation of the study area into 35 independent components. Each segment consists of at least one of the components created in section 2.5.2 – inaccessible stands combined with their accessible neighboring stands – along whose boundaries the inaccessible areas are to be connected to the pre-existing road network. The resulting segments provide the basis for the subsequent optimization in CPLEX, with each segment treated as a separate decision problem whose solutions will later be integrated.

Remark. One component (comp 17, see fig. D.21) can not be processed within the current study using the chosen approach, as it is not adjacent to any existing roads and has no neighboring stands whose boundaries could guide the modeling of future access. For road construction decisions in this area, stakeholders need to coordinate with the owners of surrounding lands through which future connections to the road network could be established.

2.6 Creation of the Network of Potential Road Segments

Next, for each area segment ("component") determined in section 2.5.4, the graph structure representing the network of potential forest road segments (mathematically introduced in chapter 3, definition 3.6) was created. What are the key requirements on the design of that road network?

- Each road segment must align with the borders between forest stands to ensure that the impact of road construction is distributed among different forest stands and their respective owners.
- The network must guarantee that every forest stand has at least one viable connection to the existing public road infrastructure.¹⁸

To construct the graph for this network, each component determined in section 2.5.1 was processed to extract stand boundaries and first generate an initial graph representing the road network via nodes and edges (section 2.6.1), create exit nodes (section 2.6.2), assign respective costs (section 2.6.3), and then simplify the graph by retaining only the necessary nodes to improve computational efficiency.

In the following sections the key steps from the workflow are outlined. Additionally, data storage, visualization, and verification steps were performed which will not be explicitly detailed here, but can be reviewed in the accompanying notebooks. Throughout the construction and modification of the graph, special attention was given to the integrity of the spatial and cost-related attributes. At each major step — such as the addition of exit nodes, the splitting of edges, and the introduction of centroid-to-boundary connections — cumulative metrics (total edge length and aggregated costs) were tracked. This continuous monitoring served both as a validation mechanism to detect unintended topological distortions and as a verification tool to ensure that cost manipulations were correctly applied.

2.6.1 Extracting Nodes and Edges from Stand Boundaries to Create a Road Network Graph

To construct the road network graph, nodes and edges were extracted directly from the boundaries of forest stands. The graph nodes were identified, accessed and stored in form of the

¹⁸Without this, the model would not be able to generate feasible solutions, as certain stands would remain inaccessible regardless of the selected road segments.

coordinate pairs they represent. The process starts by parsing the polygonal geometries within the `GeoDataFrame` of stands of the component. Each polygon's exterior boundary is analyzed to derive graph vertices (nodes) and the linear segments between them (edges), along with associated attributes such as edge length and slope. It iterates through each feature's geometry and checks whether it is a `Polygon`.¹⁹ Each exterior coordinate is snapped to a fixed spatial grid to improve precision and ensure consistency across adjacent stands. These snapped coordinates form the graph's nodes. Edges are constructed by connecting each consecutive pair of exterior points. For each edge, two main attributes are calculated:

- **Edge length**, computed as the Euclidean distance between the two vertices.
- **Slope**, inherited from the "Declive" attribute of the associated stand polygon.²⁰

This returns five key elements: A list of all unique boundary vertices (nodes), a list of edges (pairs of nodes), a list of dictionaries storing edge attributes (length, slope, etc.), a list of exit points, where stand boundaries intersect the existing road network, and a mapping of nodes to the stands they belong to. These are then used to construct an undirected graph using the `networkx` library, where nodes represent snapped boundary coordinates, and edges represent the physical path segments along stand boundaries. An illustration of one resulting graph component (component # 1) is shown in fig. D.44 in the appendix.

Remark. It is critical for this step that the polygon geometries in the shapefile are complete and not corrupted. This requirement enforced the extensive preprocessing steps described in section 2.2.2, whose completion ensured valid, clean geometries such that the boundary extraction process was eventually working without weird things happening.

2.6.2 Creating Exit Nodes

To integrate potential exit points into the network model, a series of spatial procedures were applied to ensure correct mapping to the graph topology. Each candidate exit point is associated with the closest graph node based on Euclidean distance. If the nearest node lies within

¹⁹In this study's dataset, all geometries were simple `Polygons`. In case the data includes `MultiPolygons`, it must be ensured that only the outer boundaries are used. Interior "holes" are enclosed within a single stand and do not connect to neighboring stands or the existing road network, making them irrelevant for graph construction.

²⁰Although this per-polygon slope approximation may seem coarse, the data provider indicated that the stands were designed to be relatively homogeneous with respect to slope. To avoid excessively increasing the scope of this master's dissertation, which is already quite extensive, improving the approximation using additional data sources was left for future research (see also the discussion in chapter 4).

a 10-meter threshold, the exit point is merged with that node; otherwise, it is added as a new node. If the exit point does not coincide with an existing node, the graph is modified to introduce a new vertex at the closest edge by splitting it into two segments connected via the new exit node. The new edges retain key attributes like slope and are flagged with a `has_exit` attribute for downstream use. See fig. D.45 as an example.

2.6.3 Assigning Cost Attributes to Network Edges

The costs for each road segment in each period were calculated using the base costs per action performed on road kilometer as estimated in section 2.4 and their individual length and slope. The respective costs were assigned as attribute to the edges. Based on the distinction made by CAO, three slope categories were distinguished:

- **Flat Terrain** ($\leq 5^\circ$)
- **Moderate Slope** (between 5 and 25°)
- **Steep Terrain** ($\geq 25^\circ$)

The function created to assign construction, maintenance, and upgrade costs to each edge in the road network graph G based on its slope and length iterates over all edges. Each edge is classified into one of three slope categories, and the to build and maintain for both widths, and upgrade of the road are computed multiplying the predefined base cost for each action performed on a kilometer of road (see Table 2.14) with the length of each edge. The resulting costs are stored as edge's attributes.

Remark. To account for the time value of money, all cost parameters were discounted at a rate of 3.5%. The discounting was applied during implementation of the model, see section 3.5.

2.6.4 Eliminating Irrelevant Nodes and Merging Edges

As shown in fig. D.44, some edges in the graph created are quite short, resulting in unnecessarily fragmented road segments. There is no justification to keep this over-fragmentation, it occurs purely due to the shape of the geometries. To simplify the network and improve efficiency (model strengthening), all nodes that are not exit nodes or road junctions (degree > 2) were removed and their edges merged accordingly, which reduces the number of decision

variables.²¹ The resulting graph with all the created additional information was then stored to be used in the next step.

Remark. Please note that the order of the processing steps matters: the exit nodes and the length of each road segment must be determined using the geographically accurate data before the graph can be abstracted like that.

The function `merge_edges()` iteratively simplifies the graph G by merging adjacent edges while preserving essential structural nodes: The algorithm examines each edge and attempts to remove intermediate nodes that are no intersections ($\text{degree} > 2$) and neither the closest exit point to an intersection. To ensure topological consistency, nodes with a degree greater than two (representing road crossings) and nodes marked as exits are maintained. When an irrelevant node is identified, the function connects its neighboring nodes directly, effectively replacing two consecutive edges with a single, aggregated edge.

For each successful merge, the function:

- Determines the nearest neighboring edge to the candidate node and combines the two corresponding edge segments.
- Computes the new edge attributes by summing relevant parameters such as `edgelenlength`, `Build5m`, `Maintain5m`, `Build10m`, `Maintain10m`, and `Upgrade`, while setting auxiliary attributes (e.g., `slope`, `has_source`, and `has_exit`) accordingly.
- Records removed intermediate nodes in a dedicated attribute (`removed_nodes`) to maintain merge traceability.
- Deletes the original edges and the merged intermediate node, then inserts the newly created edge connecting the remaining endpoints.

This process repeats iteratively until no additional merges occur. The result is a topologically consistent but simplified representation of the original network, where redundant short segments are collapsed into longer, continuous links. While preserving the connectivity and key structural features of the road system, this reduction in graph complexity supports more efficient network analysis.

After merging edges, the integrity of the network was verified by visually comparing pre- and post-merged graphs, and especially the relevant information (like, road intersections and exit

²¹This is actually part of the model strengthening and could be explained in section 3.3, but to better understand the whole process it is explained here.

nodes are maintained). Furthermore, key attributes, such as network length, and total costs, were checked to ensure that no inconsistencies were introduced.

Remark. Please note that after this abstraction, the edges in the graph no longer represent the stands' borders with geographical accuracy. However, this was the exact goal and is not a concern, as the essential information (edge length for verification purposes and cost attributes for optimization) remain preserved. This aligns with the fundamental principle of modeling: simplifying the representation while retaining the necessary elements.

2.6.5 Adding Stand-internal Edges

The centroid was calculated for each stand polygon and added as a node in the graph. From this centroid, edges were created to connect it with the polygon's boundary vertices. Although these edges do not represent actual physical paths, they were intentionally added because the network model later requires these internal connections to formulate all necessary constraints. Since these stand-internal edges are not meant to be constructed in reality and therefore do not contribute to the total cost of building or maintaining the road network, each of them was assigned a cost value of zero.

Remark. Assigning zero cost to certain edges obviously affects the optimization model and can lead to illogical selections – for instance, the repeated construction of zero-cost road segments (instead of maintenance in the following period), or choosing to construct the same segment in the same period for both possible road widths, or simultaneously maintaining, upgrading, and constructing it. Such outcomes are unrealistic, as only one of these actions can occur for a given segment. While this kind of behavior does not generally pose a problem – since the stand-internal edges are not intended to be constructed anyway – it makes it considerably more difficult to verify the solution against the expected logic or behavior. Additional constraints were introduced (see section 3.2.4.F) to prevent such undesired behavior.

2.6.6 Summary: Network Evolution Across Processing Steps & Characteristics

The evolution of the network in terms of nodes, edges, and associated attributes at each processing step is summarized in Table 2.15, providing an overview of the network characteristics after each processing step. From here on, all analyses were performed using the fully preprocessed network (row 5 of Table 2.15).

Table 2.15: Overview of network graph characteristics after each processing step.

Step	Description	#Nodes	#Edges	#Sources	#Exits	Length [km]	Costs [€]
0	Initial network	16,729	17,299	0	0	759.19	0
1	+ exit nodes	17,160	17,730	0	736	759.19	0
2	+ costs	17,160	17,730	0	736	759.19	3,263,460
3	simplified network	1,530	2,100	0	448	759.19	3,263,460
4	+ source nodes	1,844	3,735	314	448	759.19	3,263,460

The difference between the results of step 3 and step 4 show clearly how the network simplification procedure of only retaining structural relevant nodes reduces network complexity while preserving the key attributes edge length and cost. Eliminating structurally irrelevant nodes reduced the number of edges by about 88%. The contribution of each component to the totals is shown in table 2.16. The table highlights that some components are considerably larger than others, implying that solving the optimization problem for these components is substantially more computationally demanding than for the smaller ones.

Table 2.16: Component network graph statistics sorted by number of edges (descending).

Component	Nodes	Edges	Sources	Exits	Length [km]	Costs [€]
comp_5_39_merged	287	682	68	41	105.33	462,512
comp_2_44_merged	286	635	54	53	117.58	474,323
comp_14_15_48_merged	210	447	37	48	77.02	350,483
comp_20_25_28_33_merged	192	373	28	54	74.71	320,116
comp_4_11_12_13_51_merged	145	301	28	28	56.05	252,471
comp_9	68	118	6	26	27.70	117,909
comp_10	58	111	10	16	22.20	105,666
comp_31_34_merged	54	94	6	17	22.78	96,528
comp_16	42	93	9	6	22.34	87,587
comp_23	45	83	6	12	17.53	74,510
comp_21_40_merged	41	73	5	11	17.33	66,871
comp_37	33	73	7	5	12.46	45,147
comp_24	35	73	8	6	14.09	68,007
comp_3_43_merged	30	57	5	8	12.79	68,682
comp_49	31	56	4	10	15.64	71,829
comp_6	31	56	4	10	12.42	47,861
comp_32	28	40	1	15	16.90	69,536
comp_22	22	40	3	7	11.76	58,140
comp_7	20	31	2	9	7.51	42,298
comp_27	19	31	2	7	7.94	34,915
comp_19_41_merged	18	30	3	6	7.00	25,353

Continued on next page

Table 2.16 – continued from previous page

Component	Nodes	Edges	Sources	Exits	Length [km]	Costs [€]
comp_36_50_merged	19	28	2	7	17.32	68,520
comp_8	15	26	2	5	5.57	21,654
comp_45	16	24	1	7	7.82	32,933
comp_29	13	22	2	4	4.78	17,320
comp_38	14	21	1	6	6.59	23,874
comp_26	11	20	2	3	4.65	26,203
comp_42	13	19	1	7	8.10	31,803
comp_18	9	17	2	2	3.41	15,963
comp_30	8	14	1	2	5.16	18,696
comp_46	9	13	1	3	3.24	11,743
comp_35	8	13	1	2	3.84	13,918
comp_1	8	12	1	3	7.06	25,586
comp_47	6	9	1	2	2.57	14,503
comp_17	4	6	1	0	0.79	4,439
Totals	1,844	3,735	314	448	759.19	3,263,460

3

Optimization Model and Solution Approach

Contents

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THIS chapter presents and explains the model developed to address the problem outlined in the introduction and the necessary fundamentals. A single-objective ILP approach is adopted, allowing the usage of an efficient solver to find the minimum cost solution to the problem defined. In section 3.1, fundamental definitions and basic terminology are introduced. In section 3.2, the model is formulated, to cover all aspects of the real life problem which are relevant to the optimization. Section 3.3 briefly outlines the model-strengthening methods used to enhance convergence toward a viable solution, while section 3.4 provides a concise overview of the formulated model.

The model was then implemented and solved using Python and the CPLEX Python API (section 3.5): Each of the subproblems was solved independently, and their solutions were subsequently aggregated to form a complete road network for the entire study area, with the total cost obtained by summing the results across all components. This was performed for each of the three different harvesting schedules, but due to page limitations in this Master's dissertation, only the results of the stakeholder-based solution are presented in this document. The results for all three schedules are available in the accompanying [GitHub repository](#).

3.1 Basic Definitions & Terminology

The network of road segments that can be selected for construction, maintenance and upgrade will be modeled as a graph $G = (V, E)$, where V is a set of vertices, and E is a set of edges connecting adjacent vertices. Throughout this chapter, we will work with the following sets and indices:

Sets

T	set of harvesting periods t ,
S	set of forest stands,
$B(s)$	the <i>Boundary</i> of a stand $s \in S$,
V_{source}	set of <i>Source Nodes</i> , representing the forest stands,
V_{exit}	set of <i>Exit Nodes</i> , representing connection points to the existing road network,
V_{transit}	set of <i>Transit Nodes</i> , representing all other nodes (neither exit nor source),
V	set of all Vertices/Nodes , $V := V_{\text{source}} \cup V_{\text{exit}} \cup V_{\text{transit}}$,
E	set of <i>Edges</i> $\{u, v\}$, $u, v \in V$, representing road segments,
A_{fwd}	set of <i>Forward arcs</i> (u, v) , $u, v \in V$, starting at u and ending at v ,
A_{bwd}	set of <i>Backward arcs</i> (v, u) , $u, v \in V$, starting at v and ending at u ,
A	set of all Arcs (u, v) and (v, u) , $A := A_{\text{fwd}} \cup A_{\text{bwd}}$,
$N(v)$	Neighborhood of vertex $v \in V$ (nodes adjacent to v),
$B(s)$	<i>Boundary</i> of stand s (nodes on the stand's geographical border).

Indices

w	the <i>road width</i> (either <i>timber</i> or <i>wide</i>)
t	a <i>harvesting period</i> (from 1 to 5)
s	a forest stand (identified via its associated source node)
v	a <i>node</i> in the road network
u	a <i>node</i> in the road network
$e := \{u, v\}$	an <i>edge</i> , denoting the road segment connecting the nodes u and v
$a := (u, v)$	an <i>arc</i> (directed edge), denoting the road segment from start u to end v

Before defining the decision variables, objective function and constraints, we must establish the framework for accurately representing the network of forest road segments that can be constructed. In this context, we differentiate between various types of vertices (nodes):

Definition 3.1 (Source Node). A *source node* (*root*) is a point located at the center of a forest stand. Each forest stand has exactly one source node, and each source node uniquely identifies the forest stand surrounding it.

Definition 3.2 (Exit Node). An *exit node* (*sink*) is a point that connects a forest road segment to the public road infrastructure.

Definition 3.3 (Transit Node). A *transit node* is a point that is neither a source node nor an exit node.

By definition, each node can be only one option of the three above. With all node types defined, we can represent the network of potential roads segments formally.

Definition 3.4 (Vertices/Nodes). Let the set of vertices

$$V := V_{\text{source}} \dot{\cup} V_{\text{exit}} \dot{\cup} V_{\text{transit}} = \left\{ (x, y) \text{ coordinate pairs} \right\},$$

be the disjoint union of all source nodes (V_{source}), exit nodes (V_{exit}), and transit nodes (V_{transit}).

Definition 3.5 (Edges). Let

$$E := \left\{ \{u, v\} \mid u \in V, v \in V \text{ and } u \neq v \right\}$$

be the set of *edges*, linking the vertices, that represent undirected road segments.

Definition 3.6 (Network of potential road segments). With the sets V and E as defined above, the graph

$$G = (V, E)$$

represents the *network of potential road segments*.

To formulate certain constraints, it is necessary to incorporate information about the direction of *flow* along each edge. For this purpose, we define a set of directed edges, referred to as *arcs*. While an edge simply denotes a connection between two nodes without implying direction, an arc explicitly specifies the direction by indicating a *start* node and an *end* node. This distinction allows us to represent the movement of entities along a segment in a directed manner.

Definition 3.7 (Arcs). Given the set of vertices V , as defined in definition 3.6, we define the set of **arcs** (directed edges) as the set of ordered pairs

$$A := \{(u, v) \in V \times V \mid u \neq v\},$$

where each pair (u, v) represents a directed connection in a road segment (arc) from node u to node v .

We additionally define a digraph $D = (V, A)$ that captures the directionality of movement. For each undirected edge representing a potential road segment, the digraph contains two directed arcs (one in each direction), rather than a single undirected edge. Moreover, the reader may have noticed that the defined road network includes segments connecting the interior of a stand s to points on its boundary $B(s)$. However, we do not intend to construct such segments within the stands. Why, then, are they included in the model?

These segments serve a conceptual purpose: they are *virtual* road segments, introduced to facilitate the formulation of certain constraints that ensure the existence of required access paths. These virtual segments are assigned zero cost and are, within the current work, not intended to be physically constructed, maintained, or upgraded.

While the solution to the linear model may formally select some of these segments for construction, maintenance, or upgrade, this selection is purely auxiliary and does not correspond to real-world implementation. These virtual segments primarily support the representation of real-world requirements.

Definition 3.8 (Directed network of potential road segments). The digraph

$$D = (V, A)$$

represents the *directed network of potential road segments*.

The set of arcs A can be partitioned into two equally sized subsets representing the two possible directions along each undirected edge. Specifically, if $\{u, v\} \in E$, then $(u, v) \in A_{\text{fwd}}$ and $(v, u) \in A_{\text{bwd}}$ represent the forward- and backward-directed arcs, respectively.

Definition 3.9 (Forward- and backward-directed arcs). Let A be the set of directed arcs corresponding to an undirected edge set E . We define two subsets of A as follows:

$$A_{\text{fwd}} = \{(u, v) \in A \mid \{u, v\} \in E\},$$

$$A_{\text{bwd}} = \{(v, u) \in A \mid \{u, v\} \in E\},$$

where A_{fwd} contains the forward-directed arcs and A_{bwd} contains the corresponding backward-directed arcs. These subsets satisfy:

$$A_{\text{fwd}} \cap A_{\text{bwd}} = \emptyset, \quad A_{\text{fwd}} \cup A_{\text{bwd}} = A.$$

Let us now introduce the concept of a *neighborhood*:

Definition 3.10 (Neighborhood). Let $G = (V, E)$ be a graph and let $v \in V$ be a vertex. The **neighborhood** of v , denoted $N(v)$, is the set of all vertices adjacent to v :

$$N(v) = \{u \in V \mid (v, u) \in E\}.$$

We refer to the neighborhood $N(s)$ of a source node $s \in V_{\text{source}}$ as the *boundary* $B(s)$. It contains all the nodes situated along the geographical border of stand s :

Definition 3.11 (Boundary). The **boundary** $B(s)$ is the neighborhood of a source node $s \in V_{\text{source}}$, representing the nodes located along the geographical border of the forest stand s .

Let us further agree on some basic terminology:

Definition 3.12 (Road Existence). A road segment is considered to *exist* in period t at width w if it undergoes one of the following actions during that period:

- construction at width w ,
- maintenance at width w ,
- upgrade to width w .

3.2 Model Formulation

In this section, we define the decision variables representing the choices to be made (see section 3.2.1) and the necessary parameters (see section 3.2.2), before formulating the objective function (see section 3.2.3). The objective function is subject to several constraints, which are explained in section 3.2.4.

While developing the cost-minimization objective function is relatively straightforward, the more complex challenge lies in formulating appropriate constraints that ensure the model accurately reflects real-world requirements. For the scenario under study, these constraints must, on the one hand, guarantee the existence of roads that facilitate timber harvesting and transport during each period where necessary, and on the other hand, ensure fire truck accessibility to stands with low wildfire resistance.

3.2.1 Decision Variables

A variety of decision variables is necessary to accurately represent the problem. Each additional distinction from the "real world" increases the number of variables in the model, as will be demonstrated below. We begin by defining the binary decision variables.

3.2.1.A Binary Decision Variables

For each road segment, binary decisions must be made regarding five possible actions: construction at timber or wide width¹, maintenance at timber or wide width², and upgrade³:

¹Construct the road? (Y/N)

²Maintain the existing road? (Y/N)

³Upgrade the existing timber road to a wide road? (Y/N)

Definition 3.13 (Decision variables, binary). For $e \in E$, $w \in \{\text{timber}, \text{wide}\}$, and $t \in \{1, 2, 3, 4, 5\}$, define:

$$C_{e,t}^w := \begin{cases} 1 & \text{if edge } e \text{ is **selected for construction** at width } w \text{ in period } t, \\ 0 & \text{otherwise;} \end{cases}$$

$$M_{e,t}^w := \begin{cases} 1 & \text{if edge } e \text{ is **selected for maintenance** at width } w \text{ in period } t, \\ 0 & \text{otherwise;} \end{cases}$$

$$U_{e,t} := \begin{cases} 1 & \text{if edge } e \text{ is **selected for upgrade** to wide width in period } t, \\ 0 & \text{otherwise.} \end{cases}$$

Considering a harvesting schedule spanning multiple time periods, these decisions must be made iteratively for each period – leading to five decisions per possible action, per road segment. Additionally, the construction and maintenance decisions must be made for both possible road widths, which increases the number of corresponding decision variables. In summary, given five time periods, we have

$$\underbrace{5 \cdot 2}_{\text{construction}} + \underbrace{5 \cdot 2}_{\text{maintenance}} + \underbrace{5}_{\text{upgrade}} = 25 \quad (3.1)$$

decisions to take per road segment e , leading to a total of $25 \times |E|$ binary decision variables, represented by $C_{e,t}^w$, $M_{e,t}^w$, and $U_{e,t}$ – see Table C.2.

3.2.1.B Integer Decision Variables

To model access feasibility, we introduce auxiliary integer decision variables, commonly referred to as *flow variables*. While such variables in network optimization typically represent the movement of goods or traffic along arcs, here they only indicate whether an arc will be used for traffic and for which type of traffic (see constraints in section 3.2.4.B). Specifically, the *timber flow* and *wide flow variables* on each directed arc a , listed in Table C.1, count the number of forest stands whose access path uses that arc during a given time period.

Definition 3.14 (Auxiliary variables, integer). For $a \in A$, $w \in \{\text{timber}, \text{wide}\}$ and $t \in \{1, 2, 3, 4, 5\}$:

$$\begin{aligned} flow_{a,t}^{\text{timber}} &:= \text{number of stands whose timber traffic traverses arc } a \text{ in period } t; \\ flow_{a,t}^{\text{firefighting}} &:= \text{number of stands whose fire vehicle traffic traverses arc } a \text{ in period } t. \end{aligned}$$

These variables allow us to ensure connectivity: Via the flow variables, we can create constraints (as described in section 3.2.4.C) that encode stand-level access requirements by limiting the outgoing flow from a Source Node s to either 0 (stand does not need access) or 1 (needs access) via the access needs parameters (see definition in definition 3.16), and then ensure that each stand requiring access gets linked to an exit point via a valid sequence of road segments selected for construction (see constraints in section 3.2.4.E).

Since flow may occur in both directions — forward (fwd) and backward (bwd) — and since each edge yields two directed arcs, we have 20 flow variables per undirected edge e : two types of flow (timber and wide), two directions per flow type, across five time periods. This results in a total of:

$$\underbrace{5 \cdot 2}_{\text{timber (fwd + bwd)}} + \underbrace{5 \cdot 2}_{\text{wide (fwd + bwd)}} = 20 \cdot |E| = 10 \cdot |A| \quad (3.2)$$

integer variables, as illustrated in Table C.1.

Remark. While the flow variables refer to directed arcs a , with $A_{\text{fwd}} \ni (u, v) \neq (v, u) \in A_{\text{bwd}}$, they are by definition direction-sensitive, so it is $flow_{(u,v),t} \neq flow_{(v,u),t}$. In contrast, the binary decision variables refer to undirected edges e , where $\{u, v\} = \{v, u\}$, so it is $C_{\{u,v\},t}^w = C_{\{v,u\},t}^w$; $M_{\{u,v\},t}^w = M_{\{v,u\},t}^w$; and $U_{\{u,v\},t} = U_{\{v,u\},t}$.

3.2.2 Parameters

To represent the costs associated with the construction, maintenance, and upgrade of each road segment, we use the parameters $CostC_{e,t}^w, CostM_{e,t}^w, CostU_{e,t}^w$.

Definition 3.15 (Parameters to represent costs).

$CostC_{e,t}^w :=$ Cost of constructing road segment e at width w in period t ,

$CostM_{e,t}^w :=$ Cost of maintaining road segment e at width w in period t ,

$CostU_{e,t} :=$ Cost of upgrading road segment e to wider width in period t .

Secondly, we introduce the parameters $need_road_{s,t}^{timber}$ and $need_road_{s,t}^{wide}$ to represent the varying access requirements of forest stands across time periods. If a stand s has timber to harvest in period t — that is, it requires an access path — we set $need_road_{s,t}^{timber} = 1$. Similarly, if stand s has low fire resistance and requires a wide access path in period t , we set $need_road_{s,t}^{wide} = 1$.

Definition 3.16 (Parameters to represent stand access needs).

$$need_road_{s,t}^{timber} := \begin{cases} 1 & \text{if stand } s \text{ has timber to harvest during period } t, \\ 0 & \text{otherwise.} \end{cases}$$

$$need_road_{s,t}^{wide} := \begin{cases} 1 & \text{if the fire resistance in } s \text{ is below the defined limit} \\ 0 & \text{otherwise.} \end{cases}$$

3.2.3 Objective Function

To identify the minimum cost configuration among all feasible road networks, we define the objective function Z to represent the total costs of road construction, maintenance, and upgrades across all road segments, time periods, and road widths. The goal is to determine values for the decision variables such that the total cost Z is minimized.

Definition 3.17 (Objective Function). Minimize

$$Z(C_{e,t}^w, M_{e,t}^w, U_{e,t}) = \sum_{t \in T} \left(\sum_{w \in W} \left(\sum_{e \in E} (C_{e,t}^w \cdot CostC_{e,t}^w + M_{e,t}^w \cdot CostM_{e,t}^w + U_{e,t} \cdot CostU_{e,t}) \right) \right)$$

3.2.4 Constraints

The objective function is subject to a set of constraints that reflect the real-world requirements of the problem. These include logical relationships between road construction, maintenance, and upgrades, as well as the necessity of ensuring access to all forest stands. Since we distinguish between timber and wide roads, each requirement is formulated through two corresponding constraints. As previously explained, fire vehicle access requires wide roads, while timber transport can be handled by narrower ones. Because fire access roads can also accommodate timber transport, the two constraints derived from each real-world requirement are not necessarily identical in form.

3.2.4.A Maintenance or Upgrade Only When Road Already Exists

Any maintenance or upgrade action must build on an existing road. Specifically, constraint (3.3) ensures that a road segment can only be maintained at timber width or upgraded to wide width in period t if, in the previous period $t-1$, it was either constructed or maintained at timber width:

$$\forall t \in T, \forall e \in E : M_{e,t}^{\text{timber}} + U_{e,t} \leq C_{e,t-1}^{\text{timber}} + M_{e,t-1}^{\text{timber}} \quad (3.3)$$

Similarly, constraint (3.4) ensures that a road segment can only be maintained at wide width in period t if, in the previous period $t-1$, it was either constructed or maintained at wide width, or upgraded from timber to wide width:

$$\forall t \in T, \forall e \in E : M_{e,t}^{\text{wide}} \leq C_{e,t-1}^{\text{wide}} + M_{e,t-1}^{\text{wide}} + U_{e,t-1} \quad (3.4)$$

The defined set T contains the harvesting periods 1 through 5. In constraints (3.3) and (3.4), the time index t appears together with $t-1$. To ensure that these constraints are well-defined in the first period ($t = 1$), we **initialize all variables indexed with $t = 0$ to zero**, representing the non-existence of the road segments previous to the first period.

Road Maintenance and Upgrade Constraints

$\forall t \in T \quad \forall e \in E:$

$$M_{e,t}^{\text{timber}} + U_{e,t} \leq C_{e,t-1}^{\text{timber}} + M_{e,t-1}^{\text{timber}} \quad (3.3)$$

$$M_{e,t}^{\text{wide}} \leq C_{e,t-1}^{\text{wide}} + M_{e,t-1}^{\text{wide}} + U_{e,t-1} \quad (3.4)$$

with Initialization for $t = 0$:

$$C_{e,0}^w = 0, \quad M_{e,0}^w = 0, \quad U_{e,0} = 0 \quad \forall e \in E, \forall w \in W$$

3.2.4.B Flow only on Existing Roads

The flow of fire suppression forces or timber through a road segment is only permitted if two conditions are met:

1. The road segment exists.
2. The road segment meets the minimum required width.

Accordingly, we must establish dependencies between the relevant decision variables. This model focuses on whether *any* flow occurs from a given source. This introduces a binary decision framework: either timber flow (firetruck flow, respectively) from a specific stand s occurs (flow = 1), or it does not occur (flow = 0).

Consequently, for each road segment $e = \{u, v\} \in E$ and each period $t \in T$, the total timber flow (fire truck flow, respectively) in both directions, $flow_{(u,v),t} + flow_{(v,u),t}$, is bounded by the number of stands requiring road access (wide-road access, respectively) during that period. If, in period t , the road segment e is to be used for timber harvest/transport, then it must exist (be constructed, maintained or upgraded) during that period. Conversely, if no such timber road exists between u and v in period t , then the timber flow along that segment must be zero, see (3.5):

$\forall t \in T \quad \forall e = \{u, v\} \in E$

$$\underbrace{flow_{(u,v),t}^{\text{timber}} + flow_{(v,u),t}^{\text{timber}}}_{\in \mathbb{N}_0} \leq \underbrace{\left(\sum_{s \in S} \overset{\text{\# of stands needing timber road}}{need_road_{s,t}^{\text{timber}}} \right)}_{\in \mathbb{N}_0} \cdot \underbrace{\left(\overset{\text{Existence of timber or wide road segment } e}{C_{e,t}^{\text{timber}} + M_{e,t}^{\text{timber}} + C_{e,t}^{\text{wide}} + M_{e,t}^{\text{wide}} + U_{e,t}} \right)}_{\in \{0,1\}} \quad (3.5)$$

where e denotes the edge $\{u, v\}$, while $(u, v) \in A_{\text{fwd}}$ and $(v, u) \in A_{\text{bwd}}$ denote the corresponding directed arcs. This constraint (3.5) is formulated as an inequality because not every stand requiring firetruck access must use the same road segment. The number of stands with low fire resistance may exceed the actual firetruck flow over any given segment, as access routes can differ across the network.

Similarly, if, in period t , the road segment between u and v shall enable firetruck flow, the road segment e must exist at wide width during that period. And if in period t no road segment between u and v exists, the firetruck flow between u and v must be zero during that period, enforced by (3.5).

$$\forall t \in T, \quad \forall e = \{u, v\} \in E :$$

$$flow_{(u,v),t}^{\text{firefighting}} + flow_{(v,u),t}^{\text{firefighting}} \leq \overbrace{\left(\sum_{s \in S} need_road_{s,t}^{\text{wide}} \right)}^{\text{\# of stands requiring wide-road access}} \cdot \overbrace{\left(C_{e,t}^{\text{wide}} + M_{e,t}^{\text{wide}} + U_{e,t} \right)}^{\text{existence of wide road segment } e} \quad (3.6)$$

Let us summarize:

Maximum Capacity Constraints (Flow only on existing roads)

$$\forall t \in T \quad \forall e = \{u, v\} \in E$$

$$flow_{(u,v),t}^{\text{timber}} + flow_{(v,u),t}^{\text{timber}} \leq \left(\sum_{s \in S} need_road_{s,t}^{\text{timber}} \right) \left(C_{e,t}^{\text{timber}} + M_{e,t}^{\text{timber}} + C_{e,t}^{\text{wide}} + M_{e,t}^{\text{wide}} + U_{e,t} \right) \quad (3.5)$$

$$flow_{(u,v),t}^{\text{firefighting}} + flow_{(v,u),t}^{\text{firefighting}} \leq \left(\sum_{s \in S} need_road_{s,t}^{\text{wide}} \right) \left(C_{e,t}^{\text{wide}} + M_{e,t}^{\text{wide}} + U_{e,t} \right) \quad (3.6)$$

3.2.4.C Access Points for Stands That Need Road Access

Now we benefit from the imaginary road segments. Each stand's access road (and its timber/firetruck flow) must begin somewhere — specifically, at a point along the boundary of that stand. Thus, we must enforce the existence of a "stand access point": During period t , for each stand s requiring an access road, at least one imaginary outgoing arc from inside the stand to a node in its boundary $B(s)$ must have flow. This boundary node then becomes the entrance to stand s , because non-zero flow already enforces the existence of the corresponding road segment in that period (see 3.2.4.B; thanks to (3.5) and (3.6), it suffices to enforce flow from stands when road existence is needed).

If stand s has timber to harvest in period t ($need_road_{s,t}^{timber} = 1$), then one of the outgoing arcs from s must carry a flow of 1 during that period. If s has no timber to harvest, then all the outgoing timber flow from s (on all segments) must be zero, as represented via equation (3.7):

$$\forall t \in T \quad \forall s \in V_{source} : \quad \overset{\text{outgoing timber flow from } s}{\sum_{v \in B(s)} flow_{(s,v),t}^{timber}} = need_road_{s,t}^{timber} \quad (3.7)$$

Similarly, if stand s has low fire resistance in period t ($need_road_{s,t}^{wide} = 1$), we ensure that some "firetruck flow" exits s through one of its boundary arcs. If the stand's fire resistance is not particularly low, then no such flow is permitted; represented by equation (3.8).

$$\forall t \in T \quad \forall s \in V_{source} : \quad \overset{\text{outgoing firetruck flow from } s}{\sum_{v \in B(s)} flow_{(s,v),t}^{firefighting}} = need_road_{s,t}^{wide} \quad (3.8)$$

Remark. In contrast to the general flow variables, which allow integer values, the outgoing flow on the stand-internal auxiliary road segments is binary due to constraints (3.7) and (3.8): The flow from a source node to a boundary node can only take values of either 0 or 1.

Constraints ensuring that stands needing access have an access point

$$\forall t \in T, \quad \forall s \in V_{source} :$$

$$\sum_{v \in B(s)} flow_{(s,v),t}^{timber} = need_road_{s,t}^{timber} \quad (3.7)$$

$$\sum_{v \in B(s)} flow_{(s,v),t}^{firefighting} = need_road_{s,t}^{wide} \quad (3.8)$$

3.2.4.D Avoiding In-flow into Stands

Would any forest owner be pleased if another stand's access road cut right through their own stand? Clearly not. However, to build access paths for other stands, the model would "prefer" to select the imaginary road segments located inside the other stands, since they have zero cost assigned – "minimum cost". We forbid the selection any of the imaginary road segments for construction, except when the stand containing that segment has outgoing flow — in which case the access point is genuinely needed.⁴

⁴As previously mentioned, the imaginary segment from the access point in the boundary into the stand will not actually be built, which is why the assigned costs are zero. They are purely theoretical and very useful to represent our real world constraints in mathematical terms (as seen in section 3.2.4.C).

What we must ensure is that the model cannot incorporate such segments into the access path of another stand. To prevent these stand-crossing access roads, we require that the total incoming flow into each Source Node s be zero (3.9):

Constraints forbidding in-flow into stands

$$\forall t \in T \quad \forall s \in V_{\text{source}} \quad \sum_{u \in B(s)} \left(\overset{\text{total incoming flow to stands}}{flow_{(u,s),t}^{\text{firefighting}} + flow_{(u,s),t}^{\text{timber}}} \right) = 0 \quad (3.9)$$

3.2.4.E Flow Conservation

To ensure that each access path forms a complete and uninterrupted connection to the public road network, the model must preserve flow continuity at all internal points. In other words, the total *incoming flow* to each Transit Node must equal the total *outgoing flow*.

Specifically, for every transit node $v \in V_{\text{transit}}$, the total incoming firetruck flow must match the total outgoing firetruck flow. This guarantees that a complete route from a stand access point to an exit point in the public road system is possible, see equation (3.10). Similarly, the total incoming timber flow to each transit node must equal the total outgoing timber flow (3.11):

$$\forall t \in T \quad \forall v \in V_{\text{transit}} \quad \sum_{u \in N(v)} \overset{\text{incoming "firetruck flow" to } v}{flow_{(u,v),t}^{\text{firefighting}}} = \sum_{u \in N(v)} \overset{\text{outgoing "firetruck flow" from } v}{flow_{(v,u),t}^{\text{firefighting}}} \quad (3.10)$$

$$\forall t \in T \quad \forall v \in V_{\text{transit}} \quad \sum_{u \in N(v)} \overset{\text{incoming "timber flow" to } v}{flow_{(u,v),t}^{\text{timber}}} = \sum_{u \in N(v)} \overset{\text{outgoing "timber flow" from } v}{flow_{(v,u),t}^{\text{timber}}} \quad (3.11)$$

These constraints ensure that any road built — initiated by an outgoing flow from a stand access point via (3.6) and (3.5) — will ultimately connect to the public road network. This prevents access roads from ending in dead-ends and guarantees that each constructed path forms a valid and continuous connection.

Remark (Transit nodes versus Exit nodes). The effectiveness of the above constraints relies on the clear separation between transit nodes and exit nodes. Since flow conservation applies exclusively to transit nodes, the model is naturally incentivized to route flows to exit nodes in the most cost-effective manner. At exit nodes, flow conservation is not enforced, allowing paths to terminate there without incurring additional costs.

Flow conservation

$$\forall t \in T \quad \forall v \in V_{\text{transit}}$$

$$\sum_{u \in N(v)} flow_{(u,v),t}^{\text{firefighting}} = \sum_{u \in N(v)} flow_{(v,u),t}^{\text{firefighting}} \quad (3.10)$$

$$\sum_{u \in N(v)} flow_{(u,v),t}^{\text{timber}} = \sum_{u \in N(v)} flow_{(v,u),t}^{\text{timber}} \quad (3.11)$$

3.2.4.F Additional Constraints introduced due to Zero-Cost Edges

While implementing the model, two additional constraints were introduced due to the presence of imaginary edges with costs assigned as zero. In models where all edges have positive costs assigned, the objective function of cost minimization naturally prevents invalid selections, such as double construction or overlapping usage, without the need for additional constraints. However, in our problem, there are imaginary edges with costs set to zero. For these edges, the usual cost-minimization logic does not automatically prevent undesirable configurations. To ensure the solution remains consistent, verifiable, and reasonable, we therefore introduce two additional constraints. The first additional constraint enforces that at most one type of action variable can be active at a given time step:

$$\forall t \in T \quad \forall e \in E : \quad C_{e,t}^{\text{timber}} + C_{e,t}^{\text{wide}} + M_{e,t}^{\text{timber}} + M_{e,t}^{\text{wide}} + U_{e,t} \leq 1 \quad (3.12)$$

Now let us extend this logic to consecutive time steps, preventing the "double" construction of a road segment that already exists:

$$\forall t \in T \quad \forall e \in E : \quad C_{e,t-1}^{\text{timber}} + C_{e,t-1}^{\text{wide}} + M_{e,t-1}^{\text{timber}} + M_{e,t-1}^{\text{wide}} + U_{e,t-1} + C_{e,t}^{\text{timber}} + C_{e,t}^{\text{wide}} \leq 1 \quad (3.13)$$

Note that, since (3.12) is logically implied by (3.13) for $t \in \{1, 2, 3, 4\}$, constraint (3.12) is actually needed only for $t = 5$, rather than for all $t \in T$.

Although these constraints may seem redundant in a purely positive-cost model, they ensure that the solution remains consistent, reasonable, and verifiable in the presence of zero-cost imaginary edges. Additionally, explicitly formulating such constraints helps reduce the feasible solution space, guiding the solver more efficiently through the branch-and-bound process.

3.3 Model Strengthening

The high number of decision variables (see section 3.2.1.A) underscores the need for model strengthening to reduce computational effort. With the implementation already described in chapter 2, the problem was decomposed into smaller, logically distinct subproblems by geographically separating areas whose decisions are unlikely to influence each other. Each subproblem corresponds to a geographically cohesive cluster of inaccessible forest stands, separated by already-accessible areas. Since road network decisions in one cluster do not affect others, the subproblems can be solved independently without compromising global optimality. This segmentation reduces the number of decision variables and constraints per instance, leading to faster solution times, while ensuring realistic network designs that respect practical road-building constraints and enabling parallel or sequential optimization.

In addition to geographical decomposition, further model strengthening was achieved by simplifying the underlying graph representation. Only the minimal set of nodes – necessary to represent the road segments along the boundaries of each inaccessible component and its neighboring accessible stands until reaching an "exit point" (connection to pre-existing road network) – was retained, reducing computational complexity without affecting the logical structure of the optimization problem. This approach further enhances solving efficiency by shrinking the number of decision variables, and thus, possible solution combinations.

The explicit constraints from section 3.2.4.F can also be seen as a form of model strengthening. While they were primarily required to maintain solution consistency and verifiability, they also reduce and prune the feasible solution space, guiding the solver more efficiently.

Remark. While the current model relies on the problem-specific strategies described above to reduce the feasible solution space and improve solver performance, additional general mixed-integer programming strengthening techniques could be explored in future work. For instance, symmetry-breaking constraints, variable fixing, or cutting planes derived from LP relaxations were not applied here. Incorporating such techniques could potentially further enhance computational efficiency, particularly for larger instances or more complex network structures, and may provide additional guidance to the solver in navigating the solution space.

3.4 Model Summary

Sets

S	Set of inaccessible forest stands
$T := \{1, 2, 3, 4, 5\}$	Harvesting periods
$W := \{\text{timber}, \text{wide}\}$	Road width options
V_{source}	set of <i>Source Nodes</i> , representing the forest stands,
V_{exit}	set of <i>Exit Nodes</i> , representing connection points to the existing road network,
V_{transit}	set of <i>Transit Nodes</i> , representing all other nodes (neither exit nor source),
V	set of all <i>Vertices/Nodes</i> , $V := V_{\text{source}} \dot{\cup} V_{\text{exit}} \dot{\cup} V_{\text{transit}}$,
E	set of undirected <i>Edges</i> $\{\{u, v\} u \in V, v \in V\}$ (road segments)
A	set of all directed <i>Arcs</i> (u, v) and (v, u)
$N(v)$	<i>Neighborhood</i> of vertex $v \in V$ (nodes adjacent to v)
$B(s)$	nodes that are part of the <i>Boundary</i> of stand s

Decision Variables

$C_{e,t}^w, M_{e,t}^w, U_{e,t}$	$\in \{0, 1\}$	Decision variables (road construction, maintenance, upgrade)
$flow_{a,t}^{\text{timber}}, flow_{a,t}^{\text{firefighting}}$	$\in \mathbb{N}_0$	Auxiliary variables ("timber flow", "firefighting flow")

Parameters

$CostC_{e,t}^w, CostM_{e,t}^w, CostU_{e,t}$	$\in \mathbb{R}_{\geq 0}$	Cost parameters for road construction, maintenance, upgrade
$need_road_{s,t}^{\text{timber}}, need_road_{s,t}^{\text{wide}}$	$\in \{0, 1\}$	Parameters for access requirements per stand s and period t

Objective function

Minimize

$$Z(C_{e,t}^w, M_{e,t}^w, U_{e,t}^t) = \sum_{t \in T} \left(\sum_{w \in W} \left(\sum_{e \in V} (C_{e,t}^w \cdot CostC_{e,t}^w + M_{e,t}^w \cdot CostM_{e,t}^w + U_{e,t}^t \cdot CostU_{e,t}) \right) \right)$$

subject to

Constraints

- Maintenance or Upgrade Only When Road Already Exists

$\forall t \in T, \forall e \in E :$

$$M_{e,t}^{\text{timber}} + U_{e,t} \leq C_{e,t-1}^{\text{timber}} + M_{e,t-1}^{\text{timber}} \quad (3.3)$$

$$M_{e,t}^{\text{wide}} \leq C_{e,t-1}^{\text{wide}} + M_{e,t-1}^{\text{wide}} + U_{e,t-1} \quad (3.4)$$

- Flow only on Existing Roads

$$\forall t \in T \quad \forall e = \{u, v\} \in E :$$

$$flow_{(u,v),t}^{timber} + flow_{(v,u),t}^{timber} \leq \left(\sum_{s \in S} need_road_{s,t}^{timber} \right) (C_{e,t}^{timber} + M_{e,t}^{timber} + C_{e,t}^{wide} + M_{e,t}^{wide} + U_{e,t}) \quad (3.5)$$

$$flow_{(u,v),t}^{firefighting} + flow_{(v,u),t}^{firefighting} \leq \left(\sum_{s \in S} need_road_{s,t}^{wide} \right) (C_{e,t}^{wide} + M_{e,t}^{wide} + U_{e,t}) \quad (3.6)$$

- Access Points for Stands That Need Road Access

$$\forall t \in T \quad \forall w \in W \quad \forall s \in V_{source} :$$

$$\sum_{v \in B(s)} flow_{(s,v),t}^{timber} = need_road_{s,t}^{timber} \quad (3.7)$$

$$\sum_{v \in B(s)} flow_{(s,v),t}^{firefighting} = need_road_{s,t}^{wide} \quad (3.8)$$

- Avoiding In-flow into Stands

$$\forall t \in T \quad \forall s \in V_{source} : \quad \sum_{u \in B(s)} (flow_{(u,s),t}^{firefighting} + flow_{(u,s),t}^{timber}) = 0 \quad (3.9)$$

- Flow Conservation

$$\forall t \in T \quad \forall v \in V_{transit} :$$

$$\sum_{u \in N(v)} flow_{(u,v),t}^{firefighting} = \sum_{u \in N(v)} flow_{(v,u),t}^{firefighting} \quad (3.10)$$

$$\sum_{u \in N(v)} flow_{(u,v),t}^{timber} = \sum_{u \in N(v)} flow_{(v,u),t}^{timber} \quad (3.11)$$

- Additional Constraints introduced due to Zero-Cost Edges

$$\forall t \in T \quad \forall e \in E :$$

$$C_{e,5}^{timber} + C_{e,5}^{wide} + M_{e,5}^{timber} + M_{e,5}^{wide} + U_{e,5} \leq 1 \quad (3.12)$$

$$C_{e,t-1}^{timber} + C_{e,t-1}^{wide} + M_{e,t-1}^{timber} + M_{e,t-1}^{wide} + U_{e,t-1} + C_{e,t}^{timber} + C_{e,t}^{wide} \leq 1 \quad (3.13)$$

Initialization

$$\forall e \in E \quad \forall w \in W : \quad C_{e,0}^w = 0, \quad M_{e,0}^w = 0, \quad U_{e,0} = 0$$

3.5 Model Implementation and Solution using CPLEX Python API

The model was implemented and solved using Python and the Python API of CPLEX, provided through the [CPLEX](https://pypi.org/project/CPLEX/) package⁵. This interface enables direct interaction with the IBM® ILOG® CPLEX® Optimization Studio (CPLEX) solver engine, which must be installed locally. The theoretical formulation, summarized in Section 3.4, was translated into a computational representation compatible with CPLEX.

The implementation follows a modular workflow that iterates over each harvesting schedule and all spatial components identified during preprocessing (see section 2.5.4 and figures in appendix D). For each combination of harvesting schedule and component, the script loads the corresponding graph data (nodes, edges, and attributes), classifies node types (source, transit, exit), and prepares edge attributes reflecting operational and economic parameters. Prior to model construction, automated verification steps ensure data consistency – such as matching node identifiers, validating bidirectional arc pairs, and confirming that all required constraints can be instantiated.

Decision variables were defined for road construction, maintenance, and upgrading actions across all planning periods, as well as for the flow of timber and wide-access needs between sources and exits. Constraints were programmatically added to enforce flow conservation, logical dependencies between activities (e.g., a road must exist before it can be maintained), and non-overlapping construction periods. Cost parameters were discounted at a rate of 3.5% to account for the time value of money, applied to the midpoint of each 10-year period (i.e., years 5, 15, 25, 35, and 45) according to

$$\text{Cost}(t) = \frac{\text{Cost}}{(1 + 0.035)^{(10t-5)}}.$$

Solver configurations were applied uniformly across all components. When optimality was not reached within the given thresholds, the solver was rerun using a warm start from the previous solution with an extended time limit. Solver logs, model files (LP, MPS, SAV), and summaries of the solution status were exported for each component to ensure reproducibility and transparency. The modular structure allows independent solution of components, which could be executed sequentially or in parallel to enhance computational efficiency.

Finally, component-level solutions were aggregated per harvesting schedule and combined to obtain the full model results for each of the three optimization scenarios, forming the basis for the analysis presented in Chapter 4.

⁵<https://pypi.org/project/CPLEX/>

4

Results and Discussion

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This chapter presents and discusses the results of the five-period, dual-purpose forest road network optimization model under three alternative harvesting schedule scenarios for the Paiva study area. Owing to the size and complexity of the computed solutions, the chapter cannot report the complete, spatially explicit road network designs for all scenarios and periods. Instead, results are examined at an aggregated level using solver performance indicators, summary metrics, and representative visualizations in appendix D.5, which illustrate the structure and behavior of the computed solutions. The full set of computational outputs, including detailed network-level results, is made available in the associated [GitHub repository](#). Methodological limitations and practical considerations related to data quality and model adoption are discussed alongside the reported results.

4.1 Understanding Model Results

The model was run for three alternative harvesting schedule scenarios (MaxRes, MaxWood, Stakeholder solution), which determine the access requirements for forest stands in each planning period and thereby drive the demand for construction, maintenance, or upgrade of forest road segments that was “fed” into the model. All other model components – including the road network graph, cost parameters, and mathematical formulation – are held constant, so differences in results only reflect variations in access needs between the harvesting schedules, based on scheduled timber extraction and the fire resistance of each stand in each period.

For each scenario, the optimization model computes a set of road segments to meet these access requirements while enforcing connectivity to the public road network through auxiliary flow variables. The selected segments include both timber-access and firetruck-access roads, with appropriate road width assigned according to the model rules. Representative spatial visualizations of selected solutions are provided in the appendix in fig. D.48 and fig. D.49, together with the underlying access requirements shown in table D.1.

4.2 Core Metrics and Computational Performance

This section summarizes the computational performance of the optimization model for each harvesting schedule scenario. Fundamental characteristics of the full road network for the Paiva study area were presented earlier in chapter 2 (see Table 2.15). The network comprises 1,844 nodes (314 source nodes and 448 exit nodes) and 3,735 edges, with a total length of 759.19 km and total estimated costs of €3,263,460.

All computations were conducted on a laptop equipped with an Intel® Core™ i5-7300U processor (2.6 GHz, 2 cores / 4 logical processors) and 8 GB of RAM. The CPLEX solver was configured with an optimality gap target of 0.01% and a per-component runtime limit of ten minutes for the initial runs. Components that did not reach this threshold within the allocated time were re-solved using a warm-start strategy with an extended runtime limit of four hours.

Table 4.1 reports global solver performance metrics for all three harvesting schedules under the ten-minute runtime limit. The Global Upper Bound (UB) column reports the sum of the best feasible solution costs across all components, while Global Lower Bound (LB) shows the sum of the solver’s theoretical lower bounds (both in EUR, rounded to the nearest integer). The Aggregated MIP Gap is calculated as

$$\text{Aggregated MIP Gap (\%)} = 100 \times \frac{\text{Global UB} - \text{Global LB}}{\text{Global UB}},$$

and Total Runtime [s] indicates the sum of solver runtimes across all components.

Table 4.1: Global solver performance metrics for all three harvesting schedules, runtime limit = 10 min.

Harvesting Schedule	Global UB [€]	Global LB [€]	Agg. MIP Gap	Total Runtime [s]
MaxRes	518,290	353,958	31.71%	3,801
MaxWood	568,570	377,566	33.59%	3,661
Stakeholder	563,466	384,749	31.72%	3,821

Component-level results for the three harvesting schedule scenarios are presented in Tables 4.2, C.3, and C.4. These results reveal a consistent pattern: smaller, less complex components were typically solved quickly and to optimality, while a few large components accounted for the majority of residual MIP gaps. Extending the per-component runtime limit to four hours yielded only modest improvements, as shown in Tables 4.3, 4.4, and 4.5.

The rather modest improvements with extended runtime suggest that the corresponding integer linear programming instances are structurally challenging for branch-and-bound methods. This behavior is consistent with observations in the literature¹, which note that weak linear relaxations in Mixed Integer Programming (MIP) formulations can cause solvers to spend substantial time exploring nodes that contribute little to closing the optimality gap. Despite these challenges, feasible solutions were consistently found across all scenarios.

¹See, for example, Vielma [2015](#)

Table 4.2: Solver performance for the *MaxRes* harvesting schedule, runtime limit 10 min. Components are sorted by decreasing MIP gap.

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]
comp_2_44_merged	90,385	39,654	56.13%	600.2
comp_5_39_merged	122,926	60,194	51.03%	600.29
comp_4_11_12_13_51_merged	57,845	36,601	36.73%	600.15
comp_20_25_28_33_merged	53,611	37,810	29.47%	600.1
comp_14_15_48_merged	54,267	40,447	25.47%	600.13
comp_24	15,611	15,610	0.01%	64.87
comp_16	16,245	16,244	0.01%	479.29
comp_31_34_merged	10,438	10,438	-	30.4
comp_1	1,842	1,842	-	0.1
comp_10	14,651	14,651	-	111.72
comp_18	2,280	2,280	-	0.1
comp_19_41_merged	7,114	7,114	-	1.28
comp_21_40_merged	8,564	8,564	-	14.01
comp_22	1,568	1,568	-	0.51
comp_23	8,532	8,532	-	10.06
comp_26	2,936	2,936	-	0.11
comp_27	1,765	1,765	-	0.11
comp_29	2,769	2,769	-	0.53
comp_30	429	429	-	0.1
comp_32	124	124	-	0.22
comp_35	1,496	1,496	-	0.14
comp_36_50_merged	2,375	2,375	-	0.31
comp_37	8,160	8,160	-	26.97
comp_38	88	88	-	0.07
comp_3_43_merged	10,209	10,209	-	9.92
comp_42	462	462	-	0.09
comp_45	688	688	-	0.18
comp_46	208	208	-	0.14
comp_47	2,473	2,473	-	0.23
comp_49	782	782	-	0.11
comp_6	8,518	8,518	-	0.95
comp_7	184	184	-	0.04
comp_8	1,247	1,247	-	0.08
comp_9	7,499	7,499	-	48.38

Table 4.3: Solver performance for extended run (4h, stakeholder harvesting schedule).

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]	#Variables	#Nonzero
comp_2_44_merged	89,962	56,041	37.71%	14,409	31,750	2,164
comp_5_39_merged	133,450	86,660	35.06%	14,405	34,100	2,803
comp_4_11_12_13_51_merged	59,077	41,818	29.21%	14,408	15,050	1,244
comp_20_25_28_33_merged	54,230	40,883	24.61%	14,408	18,650	1,162
comp_14_15_48_merged	55,659	44,693	19.70%	14,408	22,350	1,441

Table 4.4: Solver performance for extended run (4h, MaxWood harvesting schedule).

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]	# Variables	# Nonzero
comp_2_44_merged	91,623	55,810	39.09%	14,408	31,750	2,151
comp_5_39_merged	135,962	85,333	37.24%	14,407	34,100	2,932
comp_4_11_12_13_51_merged	56,841	40,342	29.03%	14,408	15,050	1,189
comp_20_25_28_33_merged	54,123	41,449	23.42%	14,401	18,650	1,167
comp_14_15_48_merged	55,659	44,713	19.67%	14,405	22,350	1,441

Table 4.5: Solver performance for extended run (4h, MaxRes harvesting schedule).

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]	# Variables	# Nonzero
comp_2_44_merged	84,644	55,438	34.50%	14,405	31,750	2,039
comp_5_39_merged	118,930	81,909	31.13%	14,401	34,100	2,488
comp_4_11_12_13_51_merged	53,151	39,075	26.48%	14,400	15,050	1,082
comp_20_25_28_33_merged	52,884	40,897	22.67%	14,401	18,650	1,056
comp_14_15_48_merged	53,097	43,192	18.65%	14,406	22,350	1,255

4.3 Model Scope and Assumptions

The model was developed for Vale do Sousa, Northern Portugal, where forests are primarily managed for timber harvesting. Its scope is therefore limited to timber-oriented forests and does not directly apply to national parks, forests with other management objectives, or areas lacking pre-existing road networks.

In this study, only inaccessible stands were targeted for new road construction. Limiting the model to these stands improves tractability and simplifies the optimization problem. Additional

motivations include: uncertain road width data² for existing roads (see section 2.1.3); prioritization of new access over upgrading existing roads to address urgent connectivity gaps; and practical considerations regarding ownership, as many existing roads are publicly owned whereas new roads on private land allow more straightforward implementation.

In addition, the forest stands considered are not fully contiguous, as some areas are under different ownership. When inaccessible stands were separated from the existing road network by areas outside the dataset, no road segments could be proposed across those lands. One such case occurred for Component 17 (fig. D.21). These limitations define the boundaries of the model's applicability and should be considered when interpreting the results.

4.4 Limitations and Data Considerations

The results of this study should be interpreted with caution due to inherent limitations in the input data. The forest stands layer required manual alignment during preprocessing, as the features did not align logically with roads and rivers. Consequently, all subsequent analyses carry a comparable level of uncertainty. The dataset was originally collected for other purposes, and granularity limitations remain. Key sources of potential inaccuracies include:

- Geospatial inaccuracies in stand or road data that may affect connectivity, neighborhood analyses, and component segmentation.
- Incorrect road width values that could lead to infeasible connections or bottle necks when newly built firetruck-accessible roads connect to pre-existing roads that are smaller than their reported width.
- Imprecise cost estimates that may affect solution optimality.
- Hydrographic data gaps: minor streams were assumed seasonal and non-obstructive; only major rivers were treated as boundaries.
- Coarse slope data, which may not fully capture local variations affecting road construction costs.

²Road widths were imputed where data was missing or implausible: segments with REDE_DFCI = 1 were assigned 6 m, REDE_DFCI = 2 assigned 4 m, and REDE_DFCI = 3 assigned 3 m.

4.5 Methodological Reflections

The problem addressed is a deterministic, multi-period, dual-purpose forest road network optimization. ILP provides a robust and interpretable framework capable of producing globally optimal or bounded solutions, making it particularly suitable for infrastructure planning with linear constraints. The CPLEX solver efficiently handled large-scale instances while providing verifiable optimality certificates, and integration through the Python API enabled reproducible, automated computations.

MILP was preferred over heuristic or AI-based methods because it guarantees constraint satisfaction and solution quality, whereas metaheuristic approaches trade optimality guarantees for exploratory search efficiency. The chosen methodology balances interpretability, solution verifiability, and computational feasibility, even on modest hardware.

Taken together, these considerations define both the capabilities and limitations of the proposed forest road network optimization model. While the methodology ensures interpretable, verifiable, and computationally feasible solutions, the applicability of the results is constrained by data quality, assumptions regarding existing infrastructure, and land ownership boundaries. Understanding these constraints is essential for interpreting the scenario-specific model results presented in this chapter and highlights avenues for refinement. As discussed in the following chapter, future work could expand the model's scope, improve input data fidelity, and explore alternative computational approaches to enhance both realism and decision-making support in timber-oriented forest management.

5

Conclusions and Future Work

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5.1 Key Contributions

This master's dissertation developed an ILP model to solve a multi-period, minimum cost forest road network problem that accounts for both timber harvesting and firetruck accessibility. The model is applicable to forest areas where road network development is required for timber extraction, while ground-based firefighting access to vulnerable stands is considered desirable. The resulting road network designs provide actionable guidance for forest managers, offering solutions that can be implemented in practice to achieve both economic efficiency of timber road development and improved wildfire preparedness, thereby bridging the gap between theoretical optimization and real-world application.

The model was exemplarily applied to the ZIF Paiva case study, a shared forest management area representative of fire-prone Mediterranean forests in Northern Portugal. Following extensive data preparation, the model produced feasible road network designs for each scenario, determining which segments should be constructed, maintained, or upgraded at which road width during each period. The performed geographical decomposition and graph simplification reduced computational complexity, enabling the solver to produce feasible solutions on modest hardware. Although global optimality could not always be proven within the set time limits (only for 30 out of 35 components), the reported optimality gaps provide transparent information on solution quality, allowing stakeholders to interpret and apply the results confidently.

Beyond the optimization, further actionable contributions of this work are the graph representations created for each inaccessible component, capturing the network of potential road segments. These graphs reflect the practical planning principle of boundary-following roads rather than roads crossing the interior of forest stands, which is particularly advantageous in shared management areas with stands under different ownership. As such, they provide a realistic and transferable representation of feasible road network alternatives. These graph structures may be reused in future projects within the study area and can be enriched with additional information, such as edge weights representing costs, environmental impacts, or operational constraints, depending on the specific problem being addressed.

Equally important is the documented and reproducible methodology for generating these network graphs, which is implemented in Jupyter notebooks and made available in the associated [GitHub repository](#). Except for necessary manual corrections of corrupted stand-boundary polygons, all data processing steps are reproducible in an automated way. The resulting processed datasets could constitute helpful resources for both operational planning and future research on the case study, and the workflow can serve as a practical reference for similar forest infrastructure planning problems beyond the present case study.

In summary, the contributions of this work are threefold:

1. **Modeling:** A multi-period, dual-purpose forest road network optimization model, integrating timber extraction and wildfire preparedness goals into road network planning.
2. **Computational Implementation:** A tractable ILP formulation solved with CPLEX, with verifiable, bounded solutions for the case study ZIF Paiva.
3. **Data and Workflow:** A comprehensive, reproducible workflow for preparing the necessary forest and road data and, in particular, generating the road network graphs, providing a methodological reference and practical tool for forest managers.

5.2 Future Work and Recommendations

Building on the limitations identified in chapter 4, several avenues for future work emerge, which can be broadly grouped into model scope extensions, data improvements, and methodological enhancements.

5.2.1 Potential Extensions to Model Scope

- **Incorporating pre-existing roads:** Roads that currently exist but do not meet firetruck-accessibility standards could be included, enabling the model to evaluate whether upgrades are more cost-effective than constructing new roads. Coordination with public authorities may be necessary for such interventions.
- **Multi-ownership integration:** Inaccessible stands currently separated by areas outside the target ownership could be addressed through collaboration with adjacent landowners, improved data sharing, or by enlarging the ZIF Paiva boundaries. This would enable the model to generate network solutions spanning multiple ownership domains and allow components such as Component 17 in this study – previously excluded due to ownership constraints and missing boundary information on the area between the component and the nearest public road – to be incorporated in future applications.
- **Additional operational and ecological considerations:** The model could be extended to include multi-purpose road segments, such as extra-wide roads that also function as fuelbreaks, depending on location-specific fuelbreak suitability criteria. In addition, the

intra-stand connectivity edges could be assigned explicit weights and constraints, allowing them to represent operational, ecological, or social considerations, such as sensitive habitats, terrain-related limitations, or access restrictions.

5.2.2 Data Improvements

High-quality input data is essential for reliable optimization. Future work would profit from:

- **Road data validation:** Validating and, if necessary, updating road widths and operability information to reflect real-world accessibility.
- **Hydrographic improvements:** Incorporating data on all streams in the area, including seasonal behavior and depth, to improve graph decomposition and connectivity analysis.
- **High-resolution terrain data:** Using fine-grained digital elevation models (DEM) to capture local slope variations and refining construction cost estimates and solution realism.

5.2.3 Methodological Enhancements

Future research could explore the following methodological enhancements:

- **Alternative Mixed Integer Linear Programming (MILP) formulations:** Adjusting the problem formulation to improve solver efficiency and reduce reliance on integer variables.
- **Hybrid and AI-based approaches:** Combining MILP with heuristics or machine learning to address stochastic or dynamic scenarios (e.g., wildfire spread, variable timber yields, or temporary road blockages). MILP outputs can serve as benchmarks or training data for such methods.
- **Computational scalability:** Investigating approaches that enhance performance for larger networks, multi-period horizons, or more complex constraint sets.

5.2.4 Economic and Strategic Considerations

Future work should also explore the trade-off between upfront investment in dual-use road networks and long-term savings from avoided wildfire damage. While low-intensity fires play an ecological role, the increasing frequency and severity of extreme events highlights the need

for cost-effective, resilient infrastructure planning. Stakeholder engagement and investment decisions remain critical factors alongside computational optimization.

5.3 Closing Remarks

This master's dissertation demonstrates how multi-period, dual-purpose forest road network planning can integrate timber harvesting efficiency with wildfire preparedness. The ILP framework, coupled with a reproducible Python workflow, provides actionable guidance for both operational planning and strategic decision-making. While data limitations and simplifications impose some constraints, the framework delivers clear, interpretable, and verifiable results.

The proposed methodology lays a foundation for increasingly resilient, cost-effective, and wildfire aware road networks in timber forests. By extending the model scope, improving input data, and exploring hybrid computational approaches, subsequent research can further enhance the practical and methodological impact of forest road network planning in fire-prone landscapes. Ultimately, however, the effectiveness of any planning framework depends on its adoption in practice: even the most rigorous models and compelling results remain of limited value if scientific insights are not translated into policy decisions and concrete implementation by stakeholders and decision-makers.

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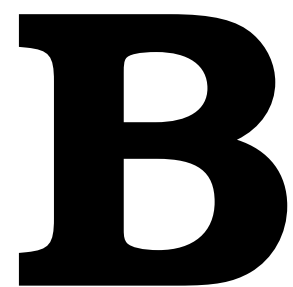


The FIRE-RES project

FIRE-RES is a four-year European project (2021–2025) dedicated to developing integrated forest management strategies to address the challenges posed by Extreme Wildfire Events (EWEs). It aims to harmonize wildfire emergency response with landscape-level planning and evaluate the economic feasibility of resilience-enhancing measures. The project's overarching goal is to develop tools, processes, and methodologies that promote integrated approaches to managing EWEs. It further advocates for proactive governance to improve societal resilience, raise risk awareness, and support effective communication—ultimately fostering fire-resilient landscapes and communities.

FIRE-RES is implemented through eleven Living Labs, which serve as open innovation ecosystems located in fire-prone regions across Europe and beyond. These include: Bulgaria, the Canary Islands, Catalonia and Galicia (Spain), Chile, Germany/The Netherlands, Greece, Norway/Sweden, Nouvelle-Aquitaine (France), Portugal, and Sardinia (Italy). Measures and approaches developed within these contexts are intended for broader replication and scaling.

For more information, see: <https://fire-res.eu>.



Forest Growth Projections

To provide context for the forest growth projections underlying the harvesting schedules the present work used and was built upon, the following description, coming from Marques et al. (2024), is reproduced from that source:

"Forest growth was assessed via specific empirical growth and yield models over a planning horizon of 100 years, allowing us to estimate the evolution of wood production volumes originated by thinnings, and accounting with different assortments of biomass and carbons. Each prescription is a schedule of silvicultural activities (e.g., planting, thinning, regeneration, fertilization, or harvesting), which when implemented on a stand is expected to achieve the desired outcomes. Usually, a myriad of different prescriptions is technically or biologically possible for each stand and were then simulated using species-specific empirical growth and yield models or yield tables. The *Globulus 3.0* and the *Pinaster* models, both implemented in the `standsSIM-MD` module included in the `SIMFLOR` platform (Lisbon, Portugal), were

used for *E. globulus* and *P. pinaster*, respectively. The *SUBER v5.0*, implemented in the same platform, was used for the growth simulation of *Q. suber*. The *Sim-Galiza* simulator, developed for *Q. robur* stands in Galicia (Spain), was also used, considering the similar soil and climate conditions from that region and our case study area. For *C. sativa*, yield tables were used. We obtained, from riparian stand national databases, structural parameters over the planning horizon.”

(Adapted from Marques et al. (2024))

C

Large Tables

Flow type	Flow direction	Period	Decision Variable
timber	backwards	1	$flow_{(v,u),1}^{timber}$
timber	backwards	2	$flow_{(v,u),2}^{timber}$
timber	backwards	3	$flow_{(v,u),3}^{timber}$
timber	backwards	4	$flow_{(v,u),4}^{timber}$
timber	backwards	5	$flow_{(v,u),5}^{timber}$
timber	forwards	1	$flow_{(u,v),1}^{timber}$
timber	forwards	2	$flow_{(u,v),2}^{timber}$
timber	forwards	3	$flow_{(u,v),3}^{timber}$
timber	forwards	4	$flow_{(u,v),4}^{timber}$
timber	forwards	5	$flow_{(u,v),5}^{timber}$
wide	backwards	1	$flow_{(v,u),1}^{wide}$
wide	backwards	2	$flow_{(v,u),2}^{wide}$
wide	backwards	3	$flow_{(v,u),3}^{wide}$
wide	backwards	4	$flow_{(v,u),4}^{wide}$
wide	backwards	5	$flow_{(v,u),5}^{wide}$
wide	forwards	1	$flow_{(u,v),1}^{wide}$
wide	forwards	2	$flow_{(u,v),2}^{wide}$
wide	forwards	3	$flow_{(u,v),3}^{wide}$
wide	forwards	4	$flow_{(u,v),4}^{wide}$
wide	forwards	5	$flow_{(u,v),5}^{wide}$

Table C.1: Flow variables

Action	Road Width	Period t	Decision Variable
Construction	timber	1	$C_{e,1}^{timber}$
Construction	timber	2	$C_{e,2}^{timber}$
Construction	timber	3	$C_{e,3}^{timber}$
Construction	timber	4	$C_{e,4}^{timber}$
Construction	timber	5	$C_{e,5}^{timber}$
Construction	wide	1	$C_{e,1}^{wide}$
Construction	wide	2	$C_{e,2}^{wide}$
Construction	wide	3	$C_{e,3}^{wide}$
Construction	wide	4	$C_{e,4}^{wide}$
Construction	wide	5	$C_{e,5}^{wide}$
Maintenance	timber	1	$M_{e,1}^{timber}$
Maintenance	timber	2	$M_{e,2}^{timber}$
Maintenance	timber	3	$M_{e,3}^{timber}$
Maintenance	timber	4	$M_{e,4}^{timber}$
Maintenance	timber	5	$M_{e,5}^{timber}$
Maintenance	wide	1	$M_{e,1}^{wide}$
Maintenance	wide	2	$M_{e,2}^{wide}$
Maintenance	wide	3	$M_{e,3}^{wide}$
Maintenance	wide	4	$M_{e,4}^{wide}$
Maintenance	wide	5	$M_{e,5}^{wide}$
Upgrade	wide	1	$U_{e,1}$
Upgrade	wide	2	$U_{e,2}$
Upgrade	wide	3	$U_{e,3}$
Upgrade	wide	4	$U_{e,4}$
Upgrade	wide	5	$U_{e,5}$

Table C.2: Binary Decision Variables

Table C.3: Solver performance for *Stakeholder* harvesting schedule, runtime limit 10min, sorted by largest MIP gap.

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]
comp_5_39_merged	143,907	66,818	53.57%	600.13
comp_2_44_merged	93,977	43,677	53.52%	600.08
comp_4_11_12_13_51_merged	61,371	39,604	35.47%	600.07
comp_20_25_28_33_merged	55,376	39,841	28.05%	600.09
comp_14_15_48_merged	56,144	42,124	24.97%	600.20
comp_16	16,527	16,526	0.01%	208.16
comp_19_41_merged	7,631	7,630	0.01%	2.43
comp_23	9,122	9,121	0.01%	27.69
comp_24	16,227	16,226	0.01%	105.13
comp_37	9,092	9,091	0.01%	27.91
comp_1	2,139	2,139	-	0.06
comp_10	18,742	18,742	-	355.18
comp_18	2,280	2,280	-	0.11
comp_21_40_merged	9,283	9,283	-	30.65
comp_22	1,568	1,568	-	0.63
comp_26	2,936	2,936	-	0.15
comp_27	1,765	1,765	-	0.25
comp_29	2,993	2,993	-	0.46
comp_30	503	503	-	0.07
comp_31_34_merged	12,055	12,055	-	27.32
comp_32	145	145	-	0.09
comp_35	1,755	1,755	-	0.05
comp_36_50_merged	2,489	2,489	-	0.10
comp_38	88	88	-	0.07
comp_3_43_merged	11,638	11,638	-	3.85
comp_42	462	462	-	0.08
comp_45	688	688	-	0.17
comp_46	321	321	-	0.07
comp_47	2,473	2,473	-	0.05
comp_49	2,046	2,046	-	0.07
comp_6	8,623	8,623	-	0.39
comp_7	184	184	-	0.05
comp_8	1,388	1,388	-	0.08
comp_9	7,529	7,529	-	29.20

Table C.4: Solver performance for the *MaxWood* harvesting schedule, runtime limit 10 min. Components are sorted by decreasing MIP gap.

Component Name	UB [€]	LB [€]	MIP Gap	Runtime [s]
comp_2_44_merged	95,787	40,677	57.53%	600.05
comp_5_39_merged	147,879	64,055	56.68%	600.25
comp_4_11_12_13_51_merged	62,389	39,604	36.52%	600.25
comp_20_25_28_33_merged	54,207	38,947	28.15%	600.07
comp_14_15_48_merged	56,144	42,124	24.97%	600.2
comp_16	16,527	16,526	0.01%	206.39
comp_37	9,092	9,091	0.01%	21.89
comp_24	16,227	16,226	0.01%	36.69
comp_23	9,122	9,121	0.01%	12.65
comp_10	18,742	18,742	-	226.38
comp_3_43_merged	11,638	11,638	-	3.31
comp_31_34_merged	12,055	12,055	-	28.7
comp_21_40_merged	9,283	9,283	-	17.1
comp_1	2,139	2,139	-	0.07
comp_18	2,280	2,280	-	0.18
comp_19_41_merged	7,114	7,114	-	1.23
comp_22	1,568	1,568	-	0.22
comp_26	2,936	2,936	-	0.06
comp_27	1,765	1,765	-	0.08
comp_29	2,993	2,993	-	0.2
comp_30	503	503	-	0.05
comp_32	145	145	-	0.1
comp_35	1,755	1,755	-	0.04
comp_36_50_merged	2,489	2,489	-	0.09
comp_38	88	88	-	0.05
comp_42	462	462	-	0.06
comp_45	688	688	-	0.23
comp_46	321	321	-	0.06
comp_47	2,473	2,473	-	0.09
comp_49	2,046	2,046	-	0.15
comp_6	8,623	8,623	-	1.95
comp_7	184	184	-	0.1
comp_8	1,378	1,378	-	0.28
comp_9	7,529	7,529	-	102.14

D

Visualizations

D.1 Road Data Preprocessing Results

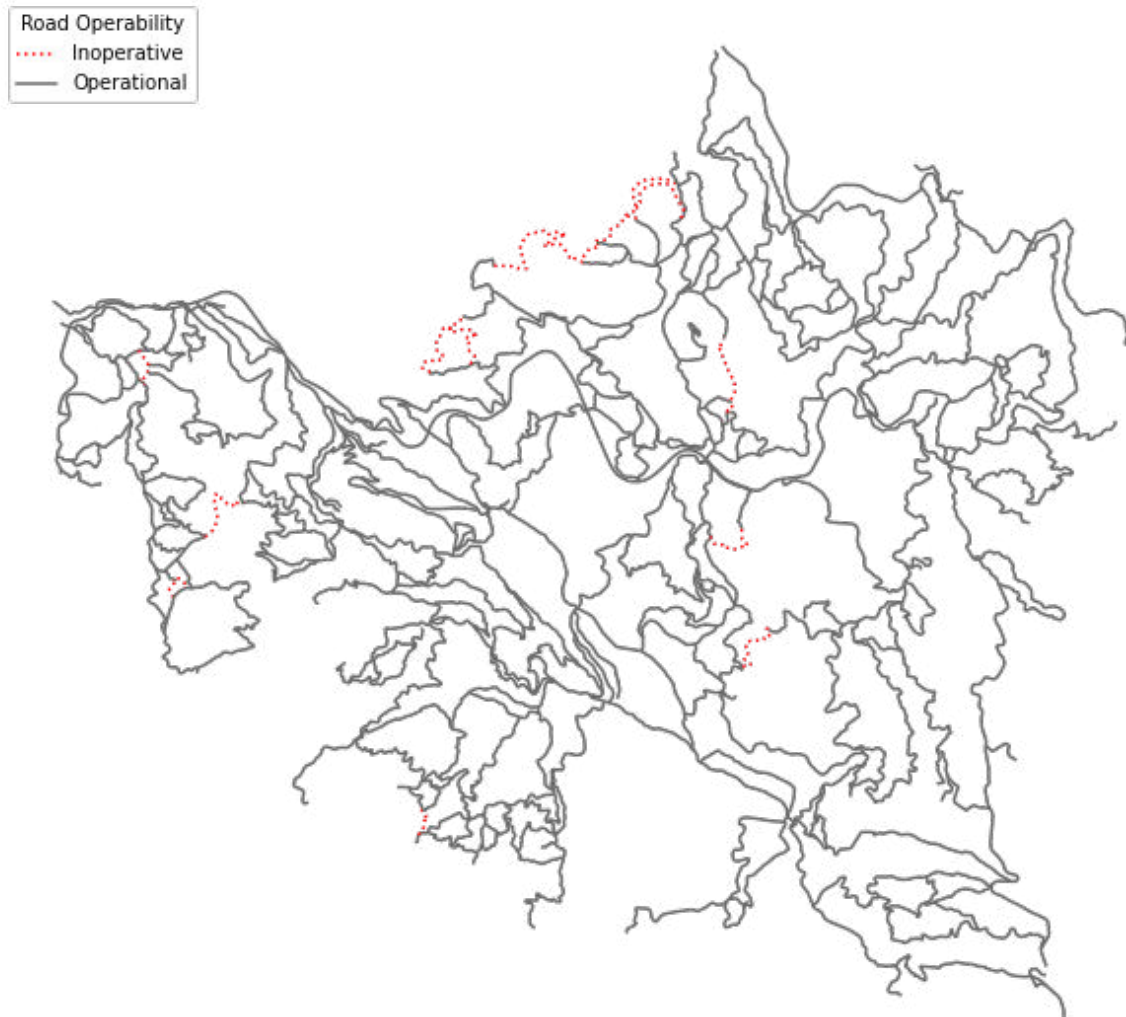


Figure D.1: Operability status of the road network in Paiva; inoperative roads shown as dotted red lines.



Figure D.2: Visualization of implausible road width information on operable road segments, segments with road width of zero shown as dashed lines in orange

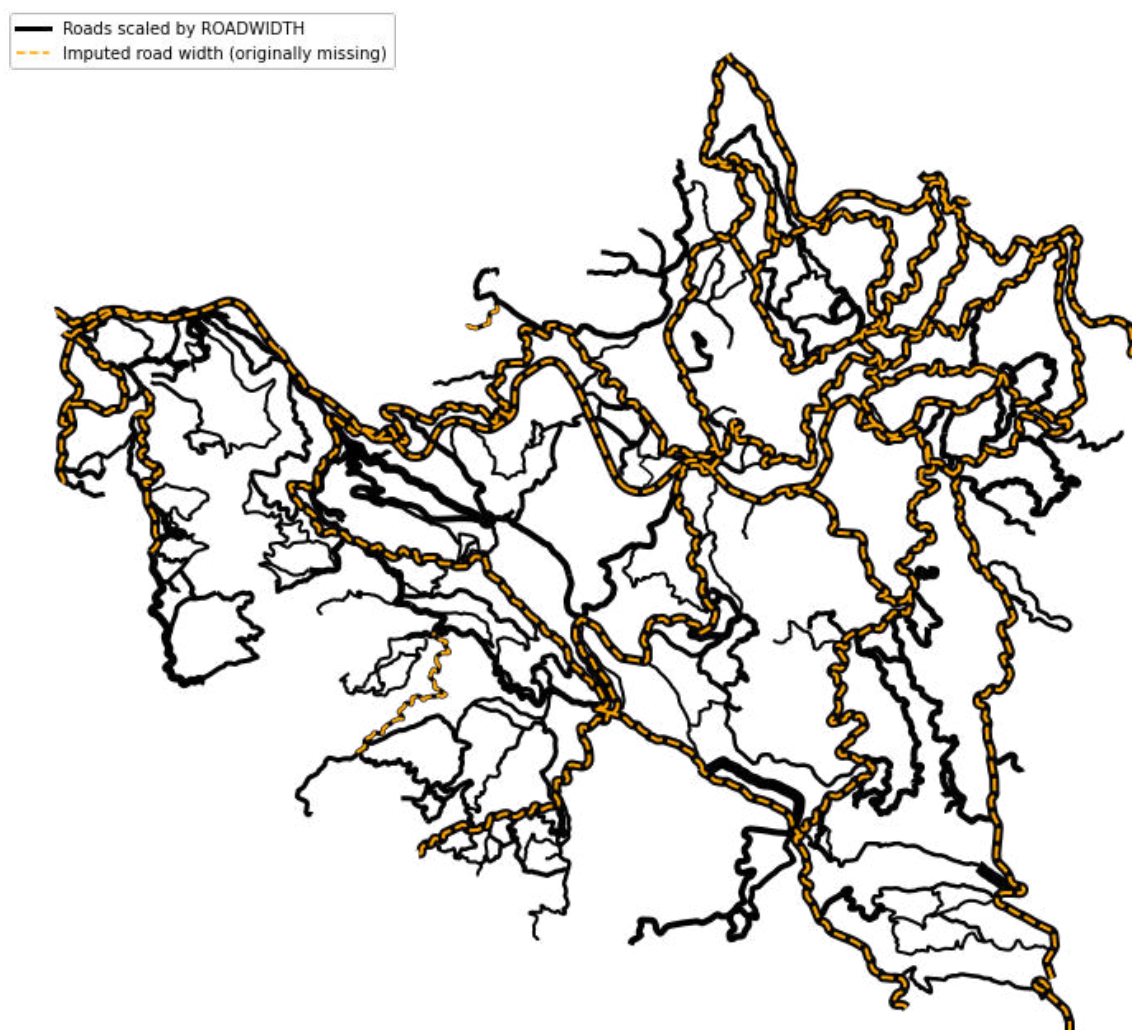


Figure D.3: Operable roads displayed with line widths proportional to road widths; imputed road width values highlighted as dashed lines in orange

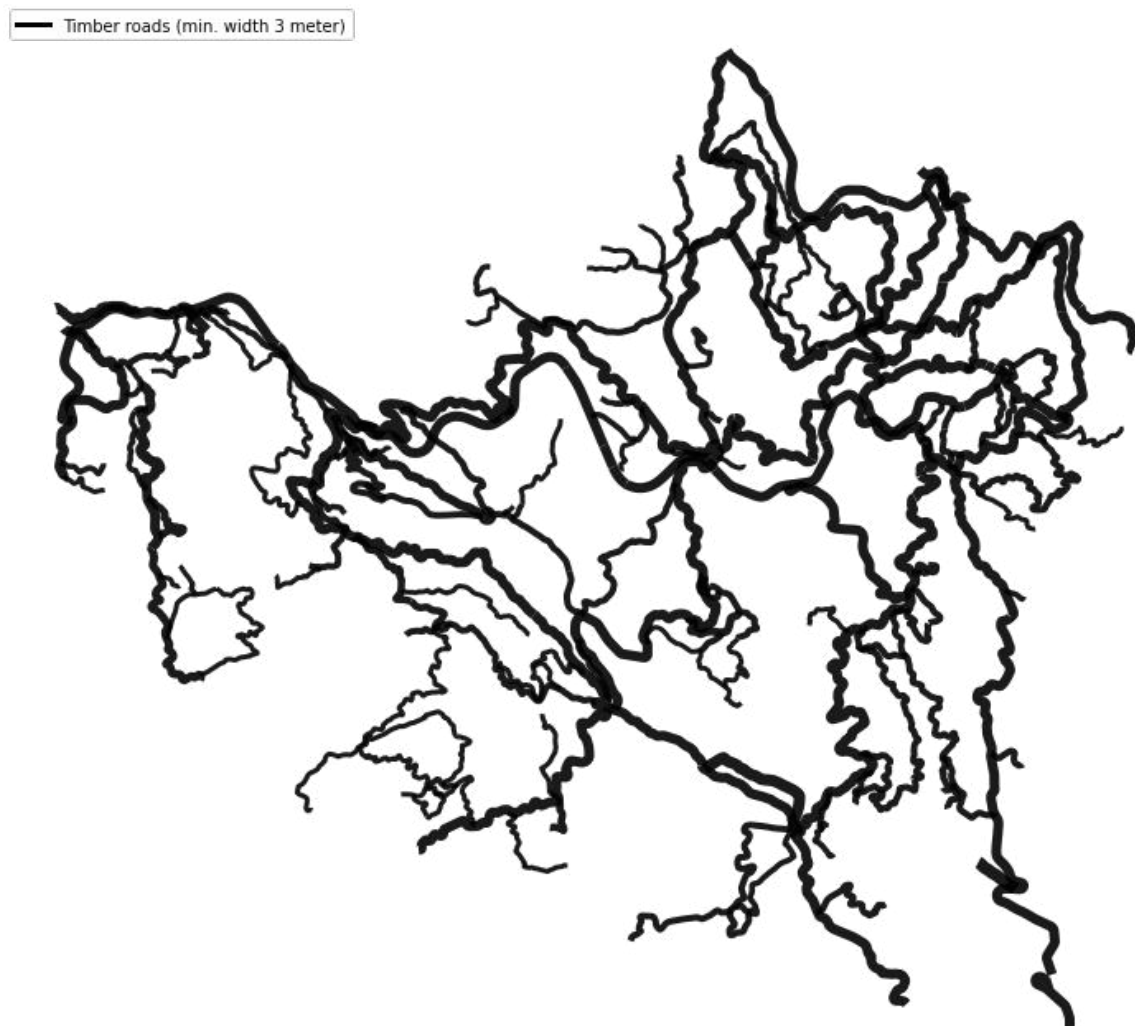


Figure D.4: Network of operable roads with widths greater than 3 m ("timber roads"), visualized with line width proportional to assigned road width values and colored by road type.

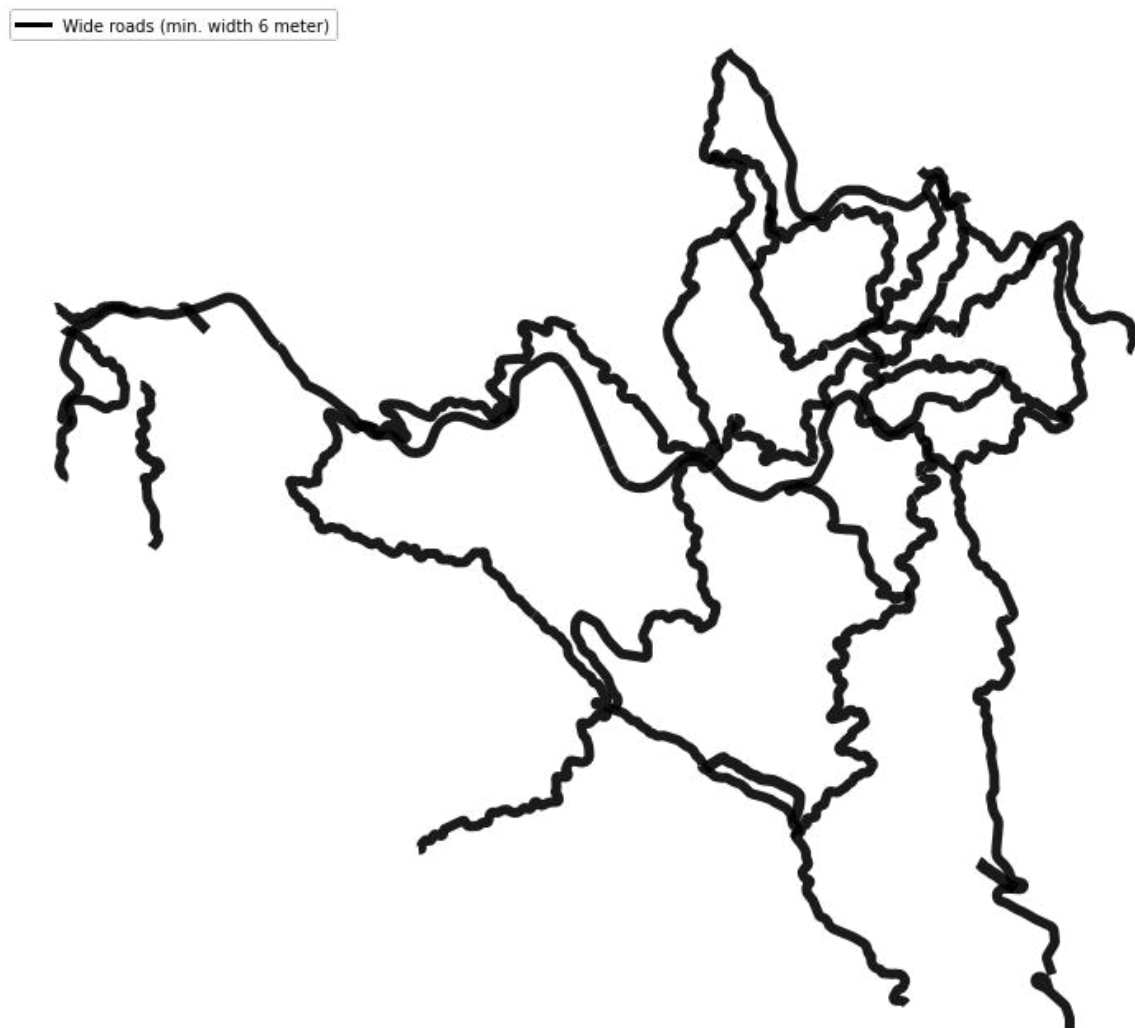


Figure D.5: Network of operable roads with widths greater than 6 m ("fire roads"), visualized with line width proportional to road width values.

D.2 Forest Stand Road Accessibility Analysis Results

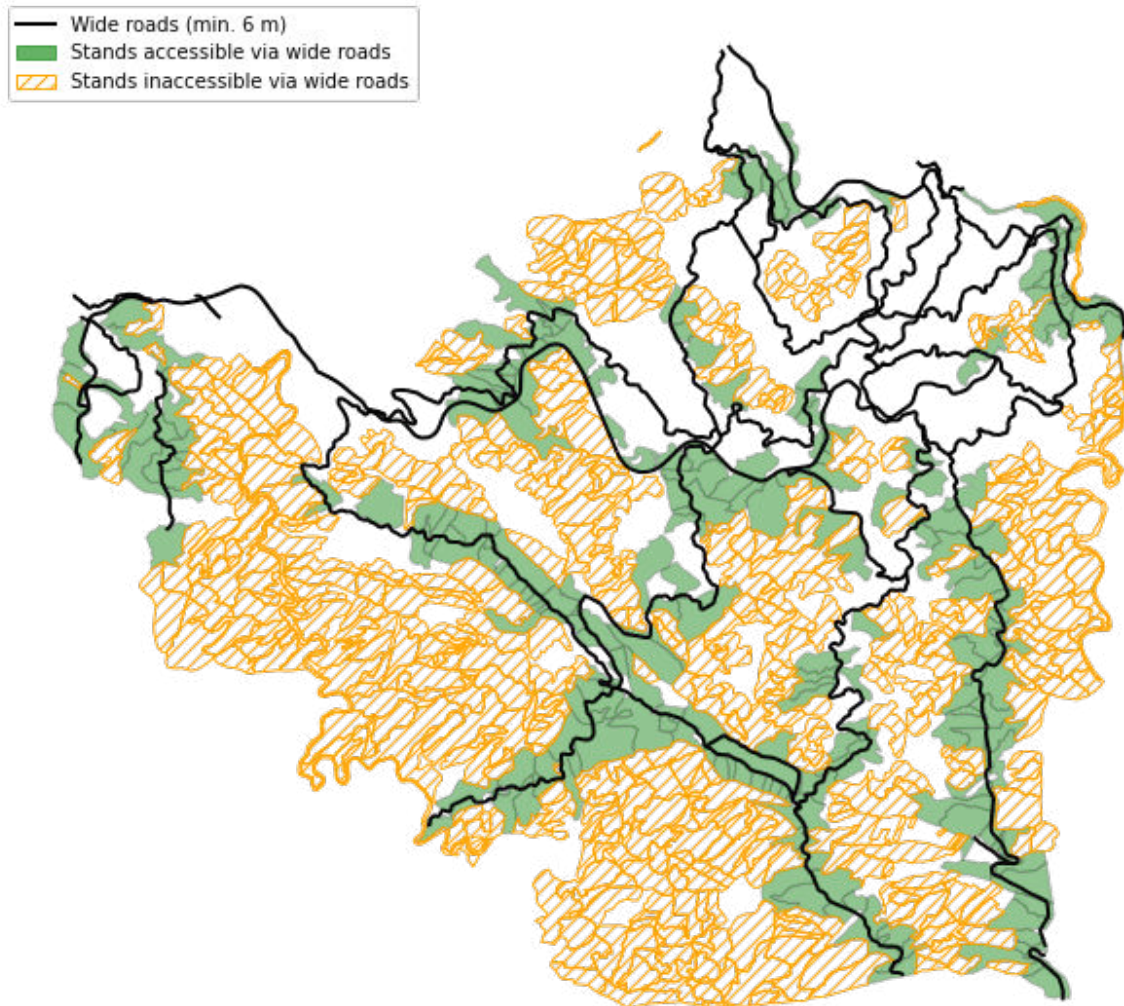


Figure D.6: Stand accessibility via the existing network of wide roads

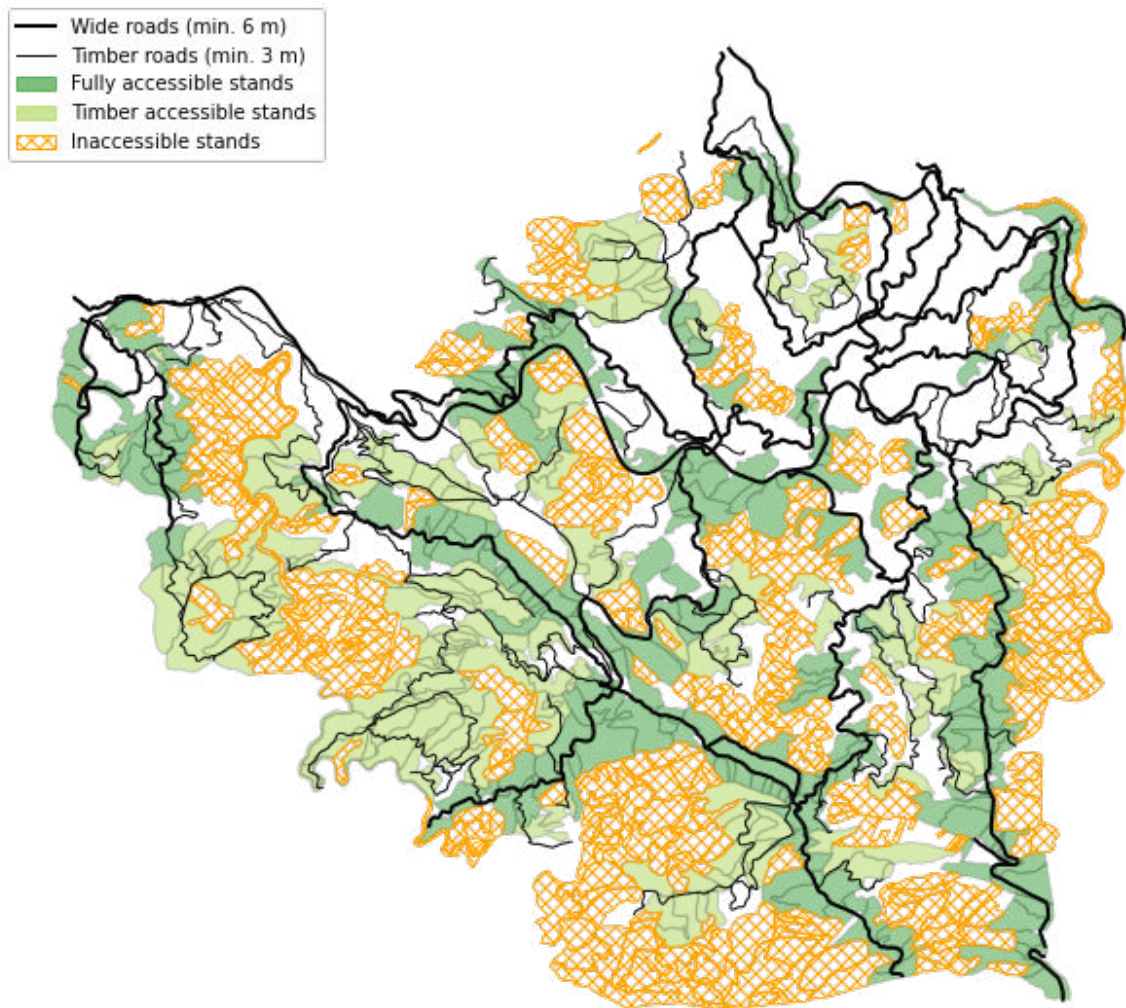


Figure D.7: Stand accessibility via the existing timber road network

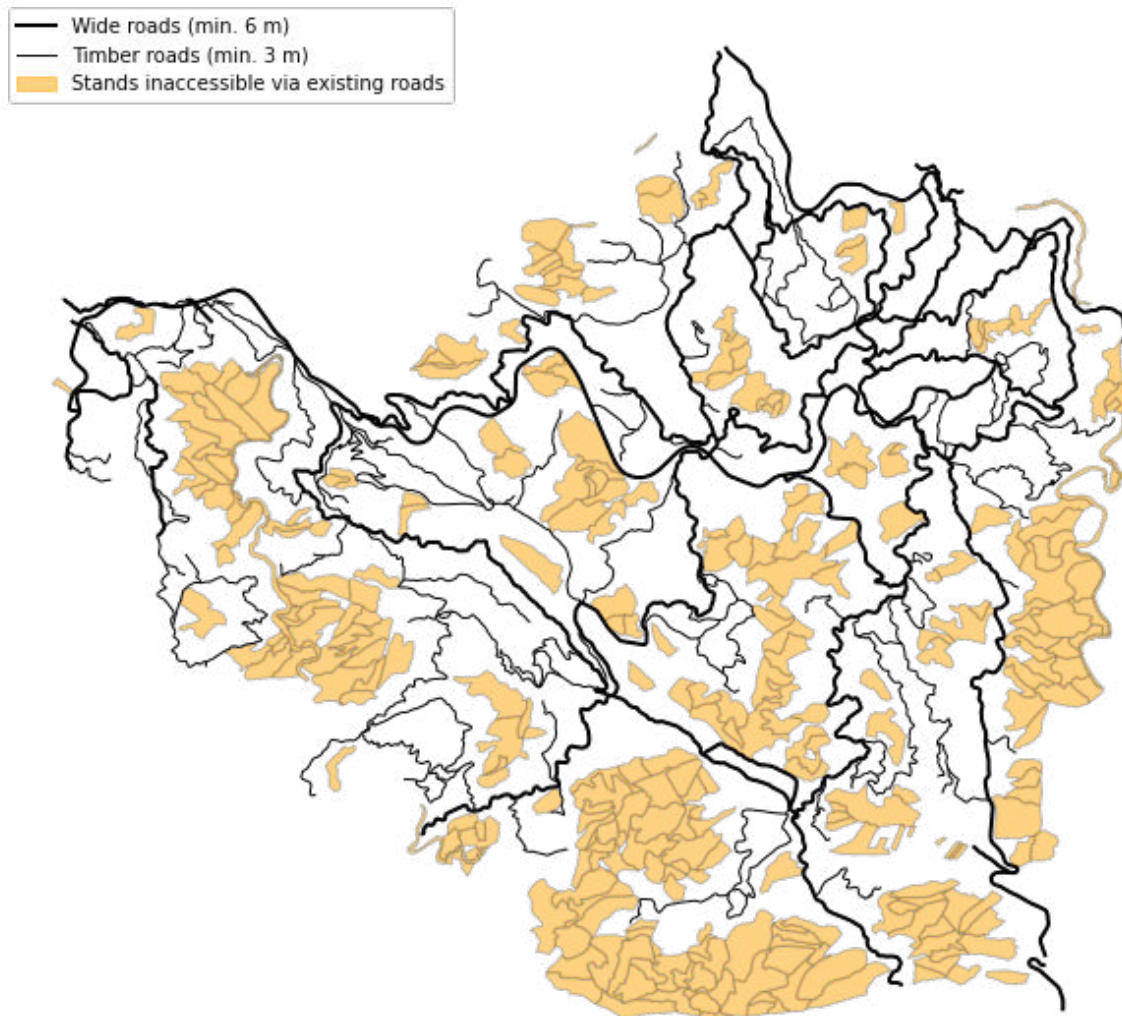


Figure D.8: Sets of stands whose access needs are considered in the decision problem

D.3 Segmentation Results

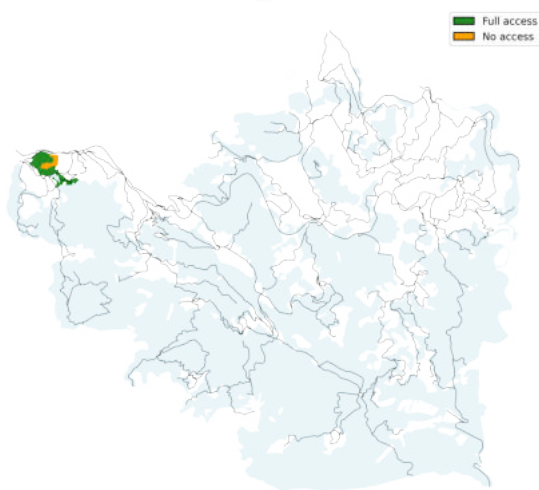


Figure D.9: Component 1

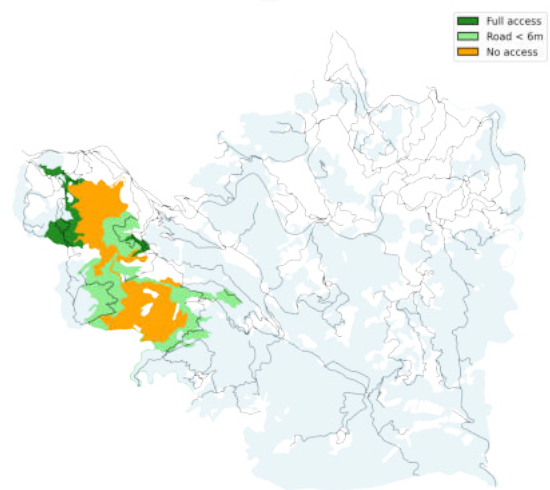


Figure D.10: Component 2.44 merged

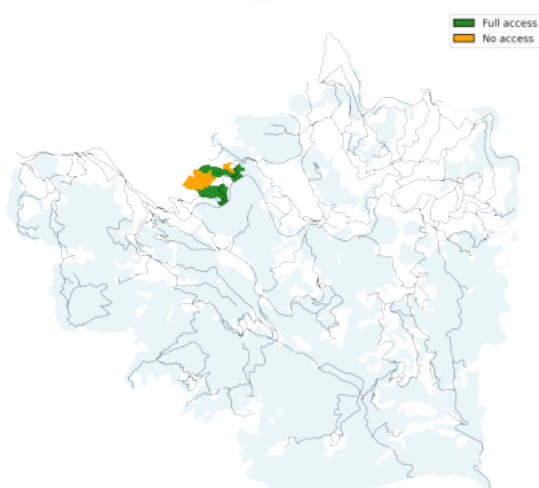


Figure D.11: Component 3.43 merged

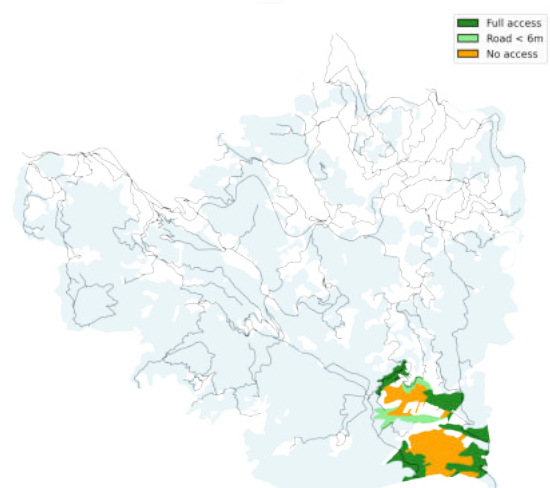


Figure D.12: Component 4.11_12_13_51 merged

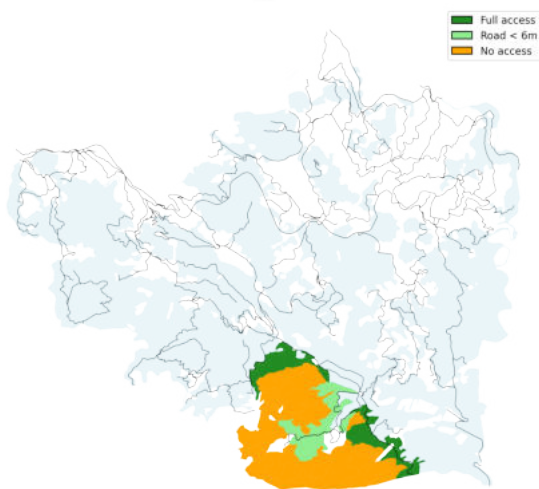


Figure D.13: Component 5_39 merged

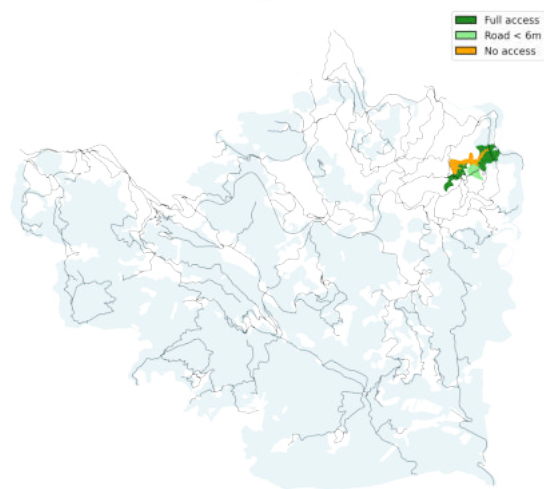


Figure D.14: Component 6



Figure D.15: Component 7

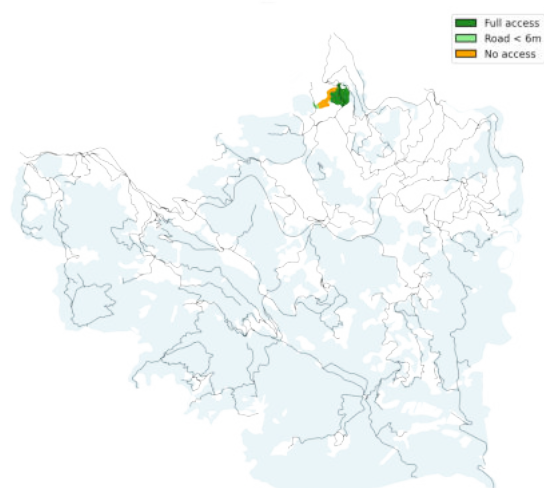


Figure D.16: Component 8

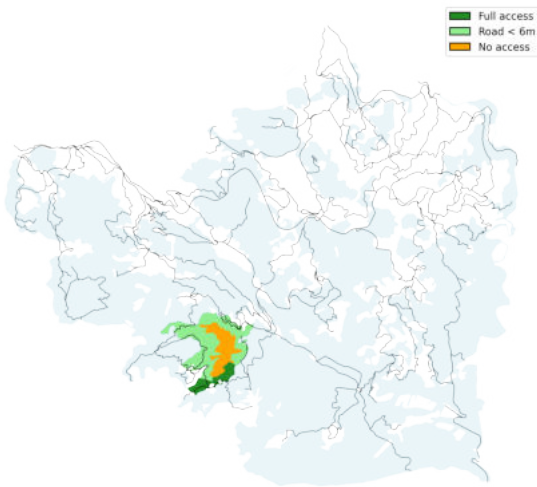


Figure D.17: Component 9

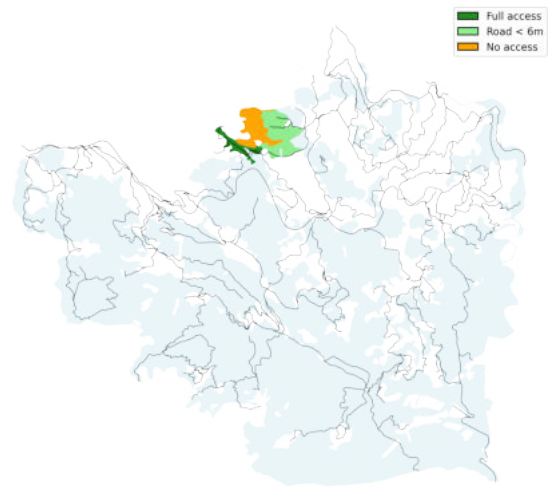


Figure D.18: Component 10

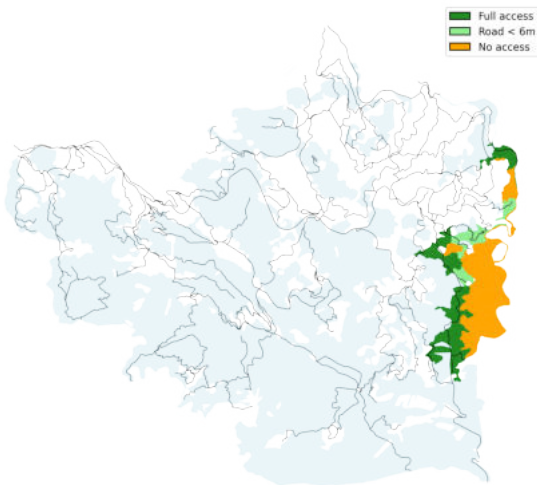


Figure D.19: Component 14_15_48 merged

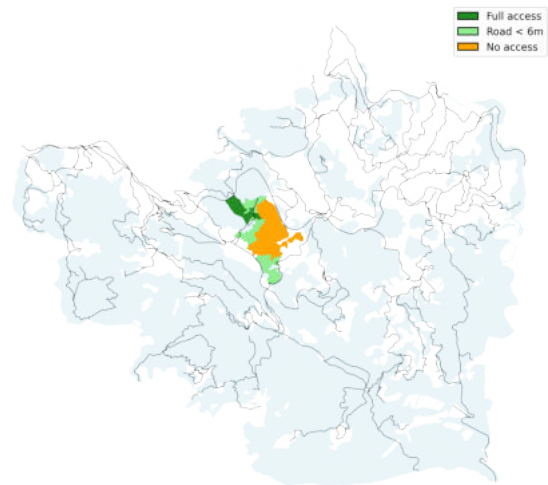


Figure D.20: Component 16

Remark. Component 17, the tiny yellow spot on the very top of fig. D.21, could not be addressed in the current project, as the area between this inaccessible component and the existing road network is under different ownership.

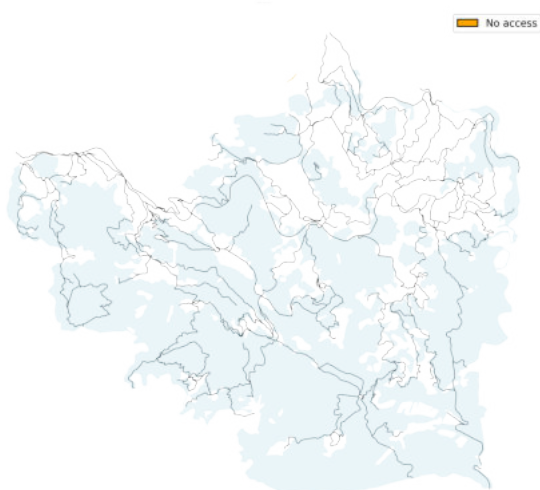


Figure D.21: Component 17

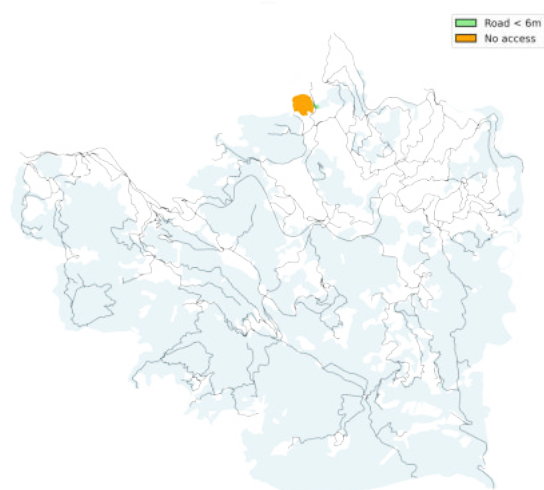


Figure D.22: Component 18

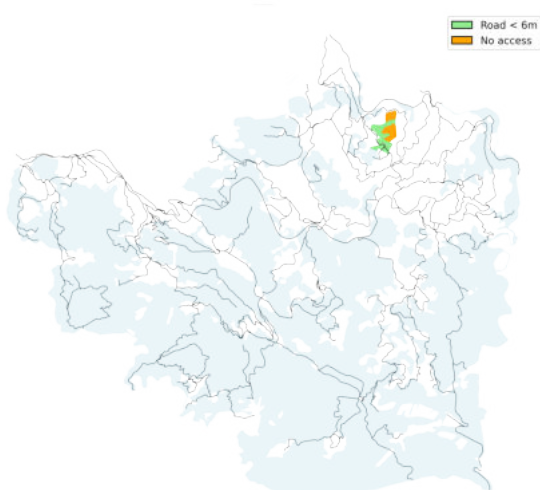


Figure D.23: Component 19_41 merged

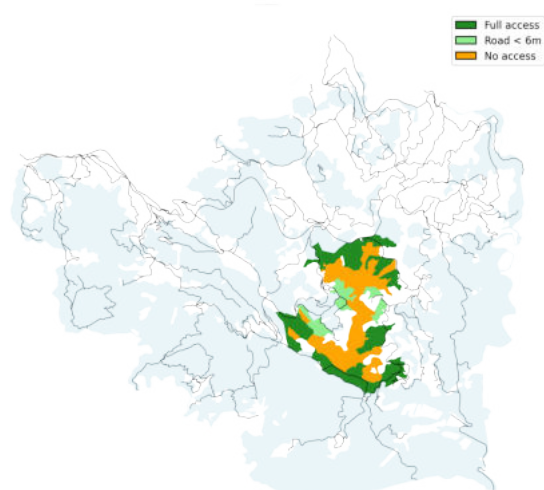


Figure D.24: Component 20_25_28_33 merged

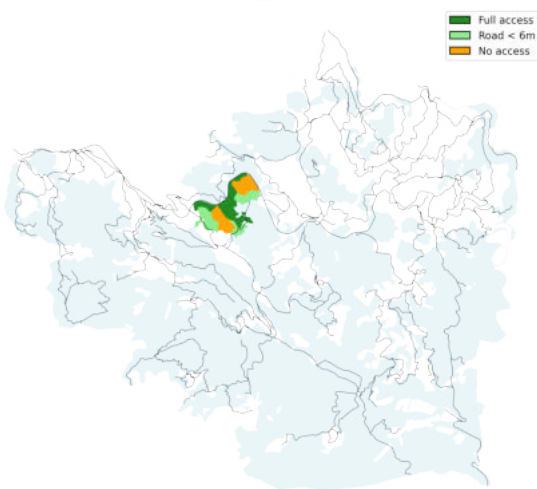


Figure D.25: Component 21_40 merged

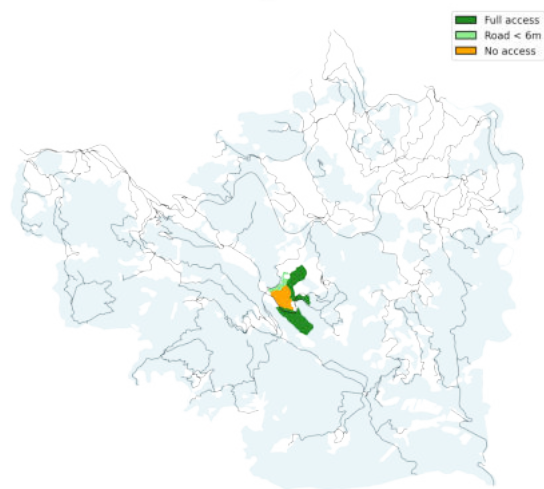


Figure D.26: Component 22

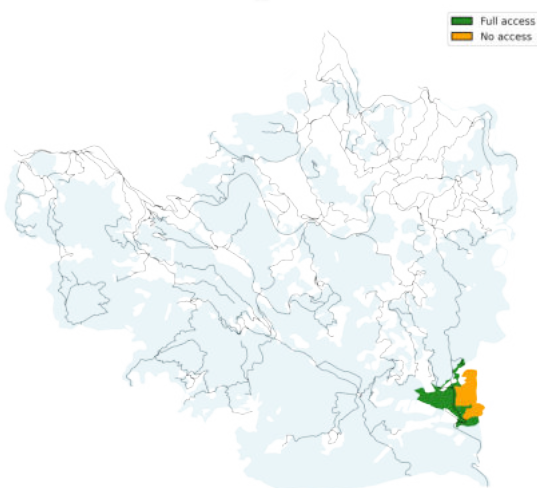


Figure D.27: Component 23

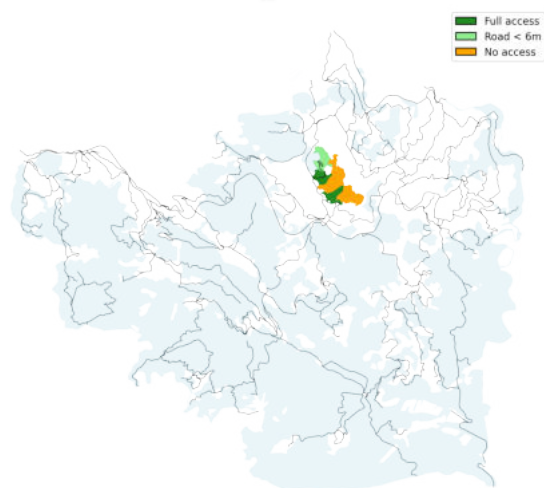


Figure D.28: Component 24

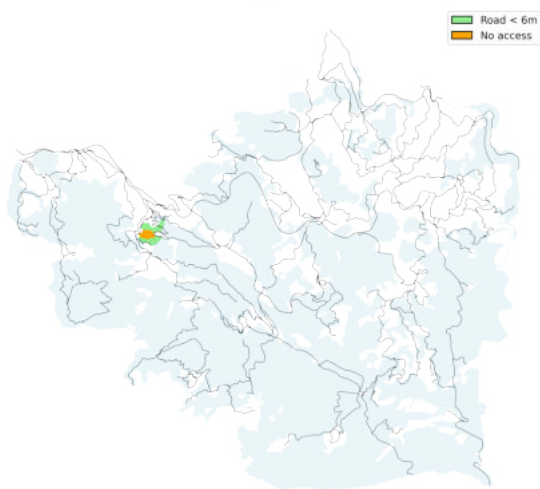


Figure D.29: Component 26

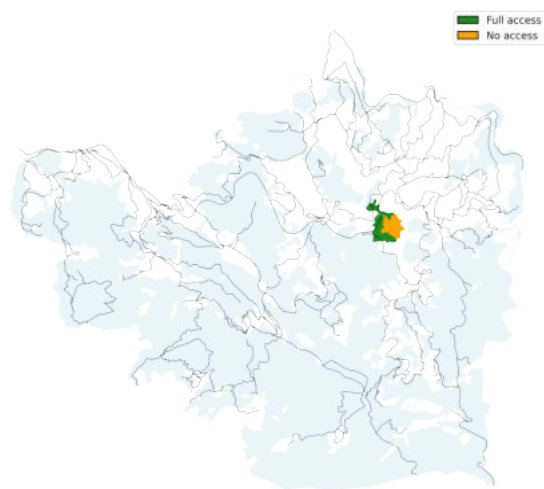


Figure D.30: Component 27

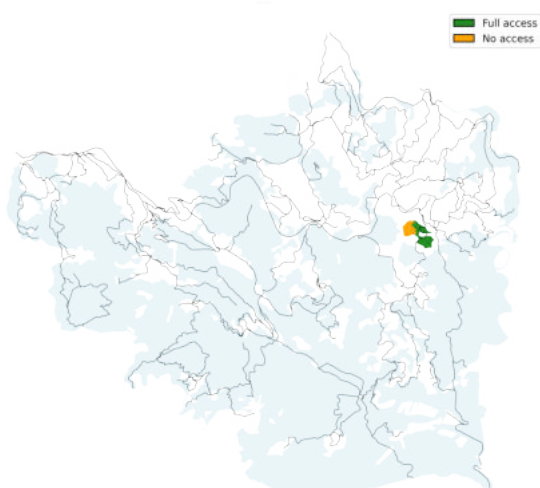


Figure D.31: Component 29

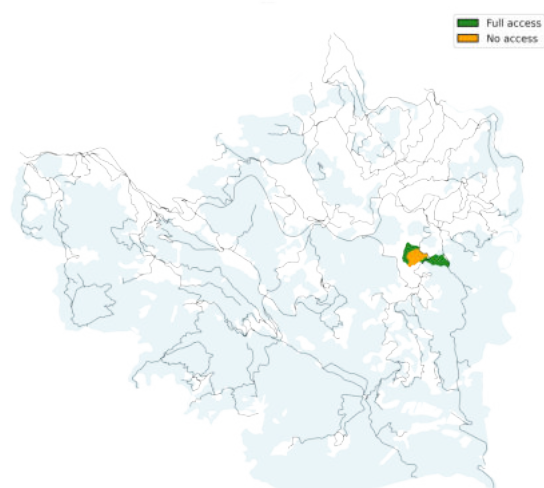


Figure D.32: Component 30

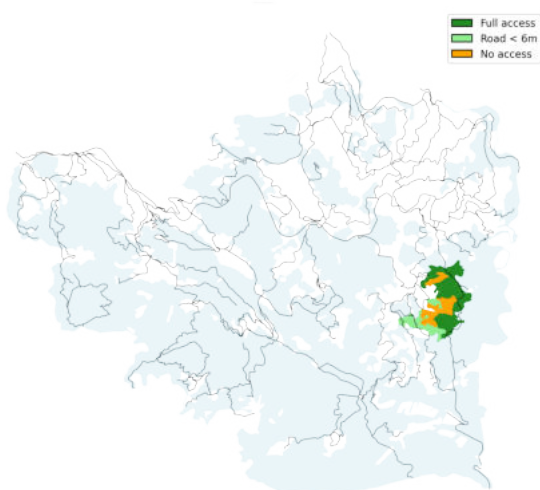


Figure D.33: Component 31_34 merged

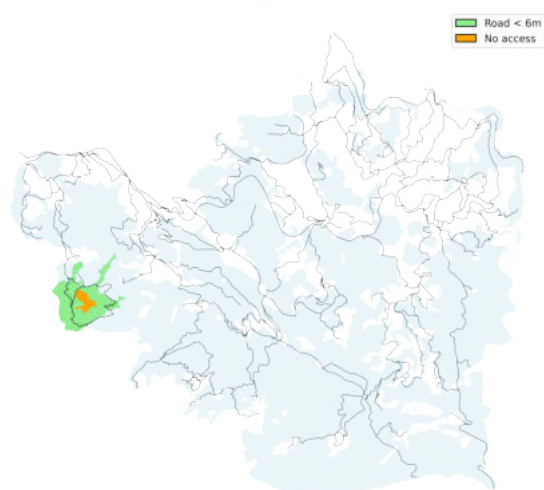


Figure D.34: Component 32

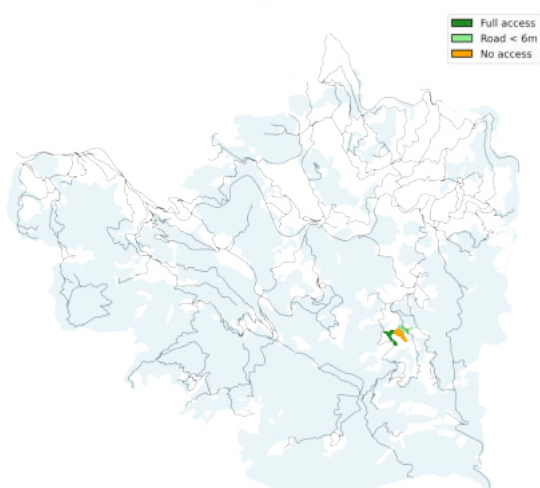


Figure D.35: Component 35

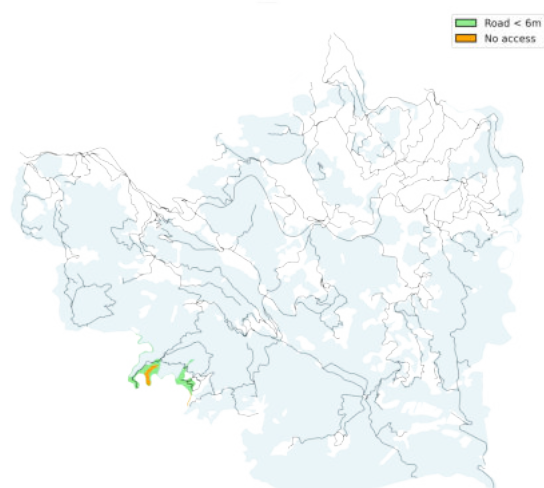


Figure D.36: Component 36_50 merged

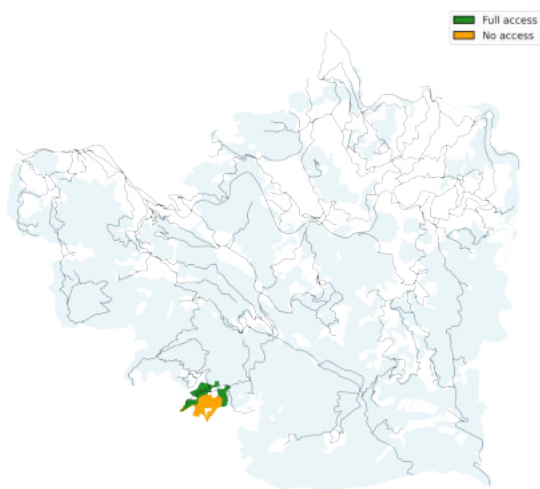


Figure D.37: Component 37

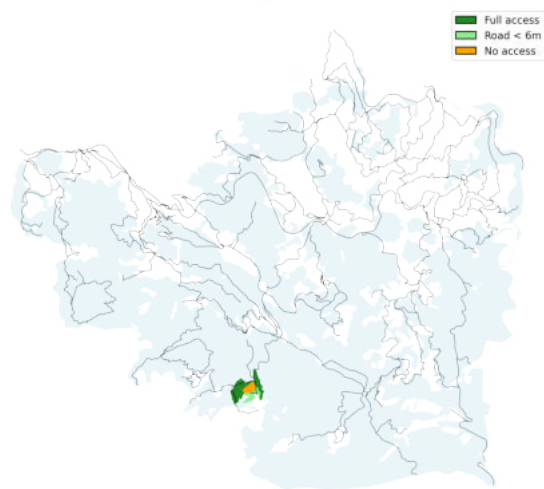


Figure D.38: Component 38

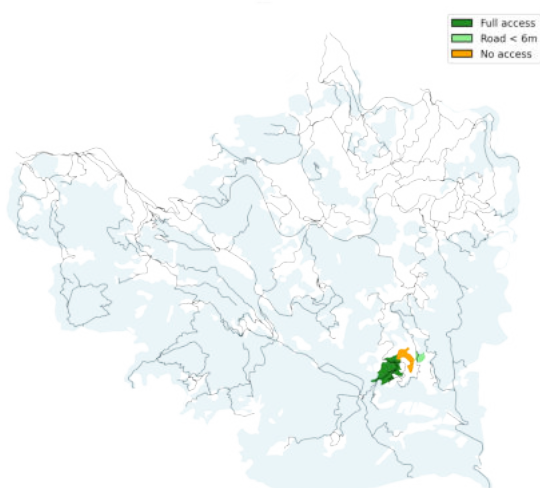


Figure D.39: Component 42

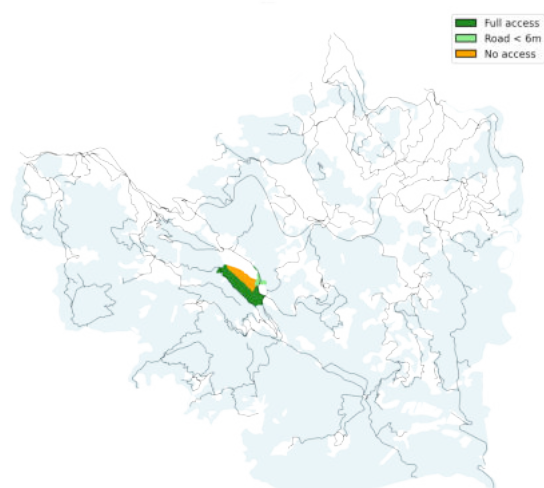


Figure D.40: Component 45

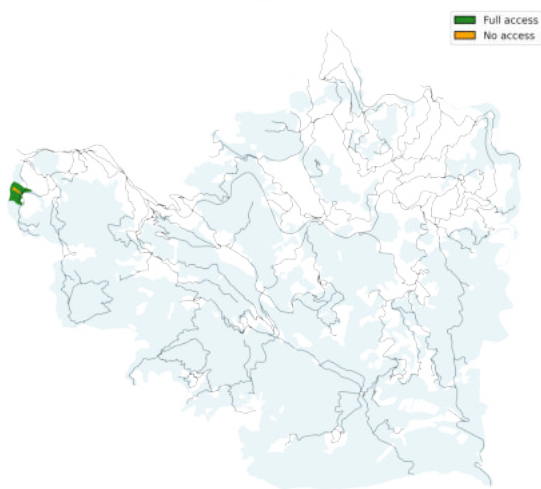


Figure D.41: Component 46

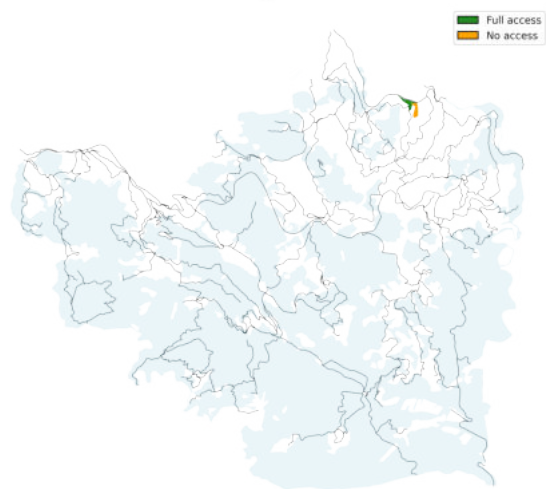


Figure D.42: Component 47

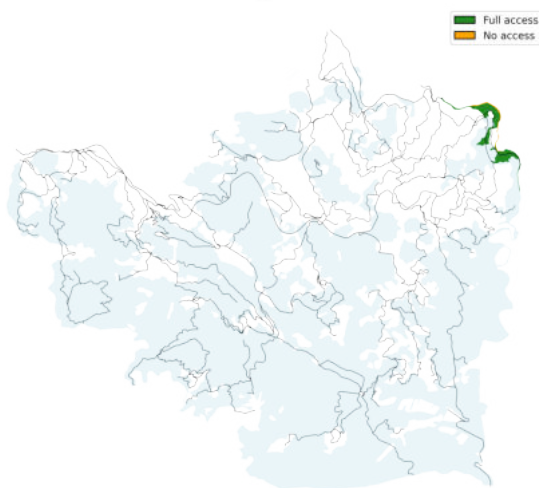


Figure D.43: Component 49

D.4 Road Network Graph Development - Example: Component 10

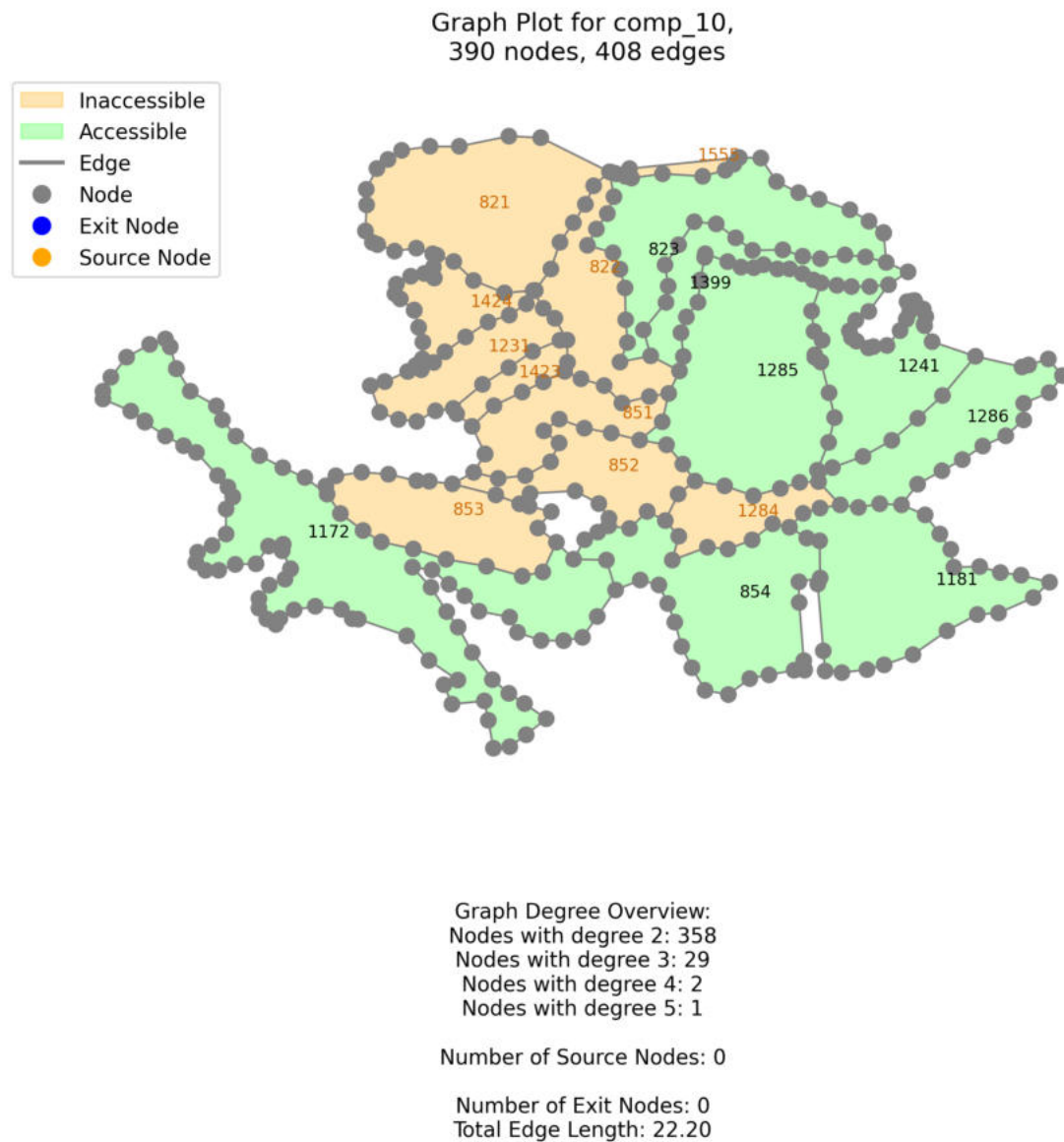


Figure D.44: Road network graph for component 10

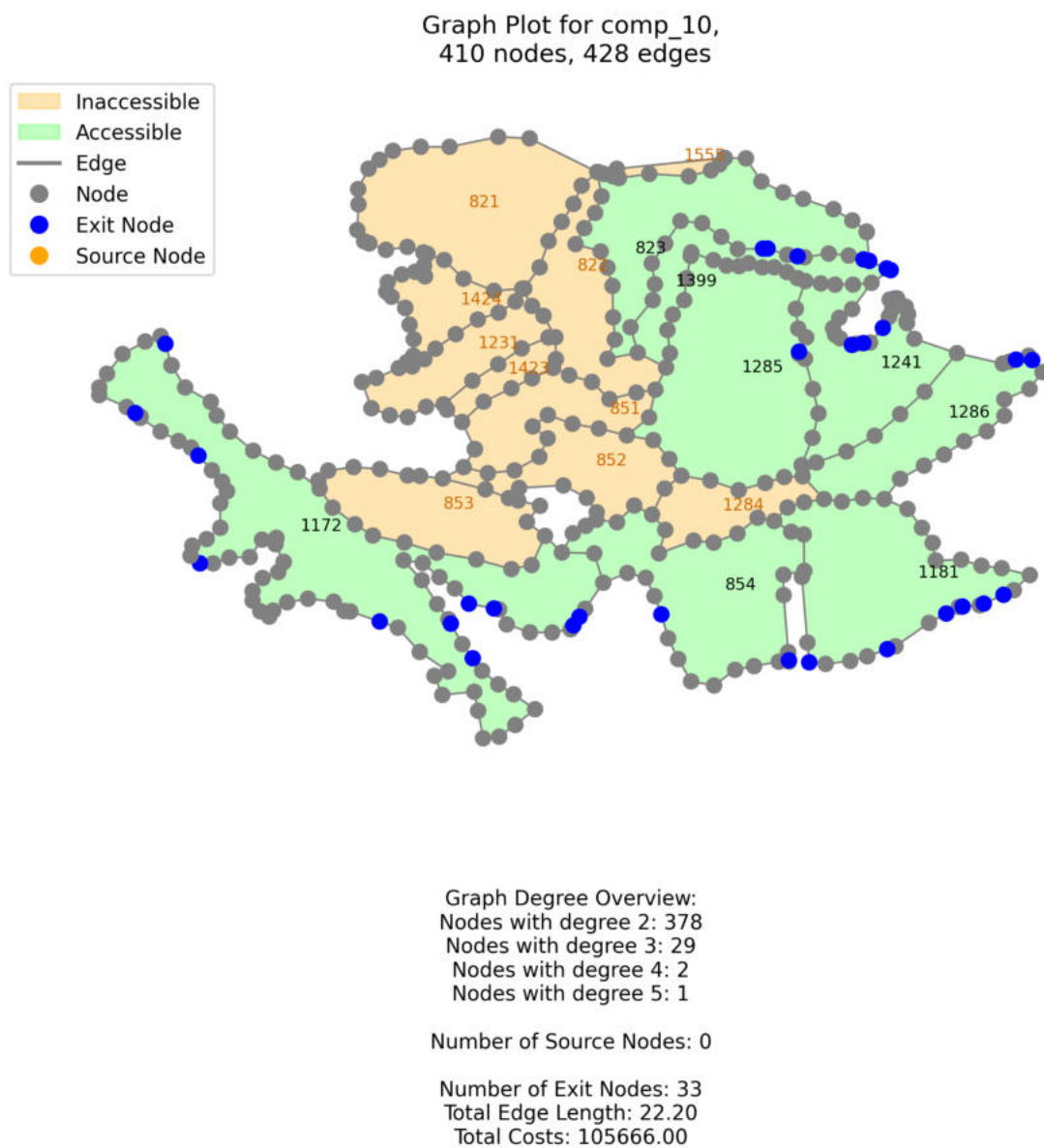


Figure D.45: Road network graph for component 10 with exit nodes

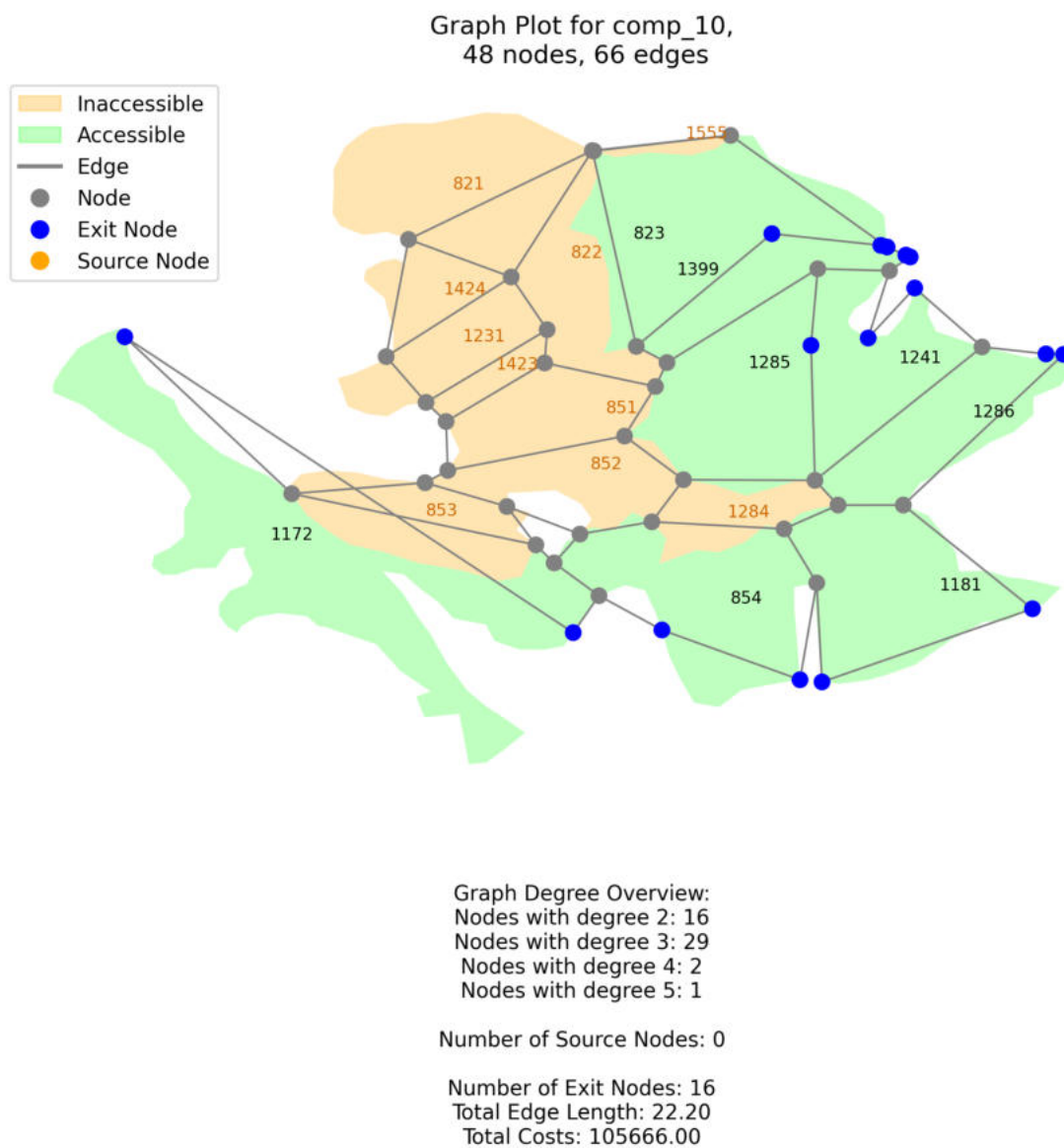


Figure D.46: Abstracted road network graph of component 10

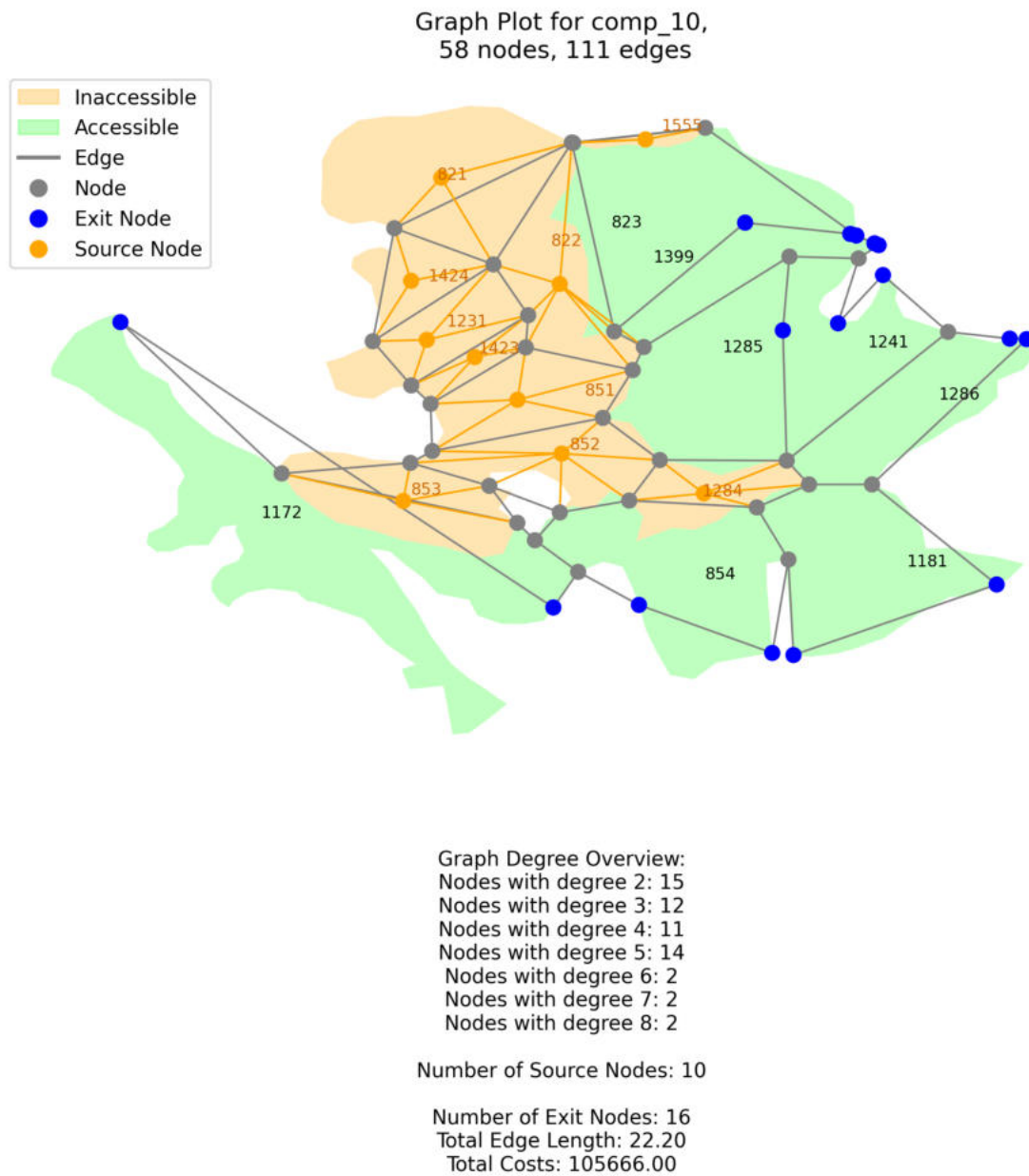


Figure D.47: Abstracted road network graph of component 1 with source nodes and their edges

D.5 Visualized Solution - Example: Component 10

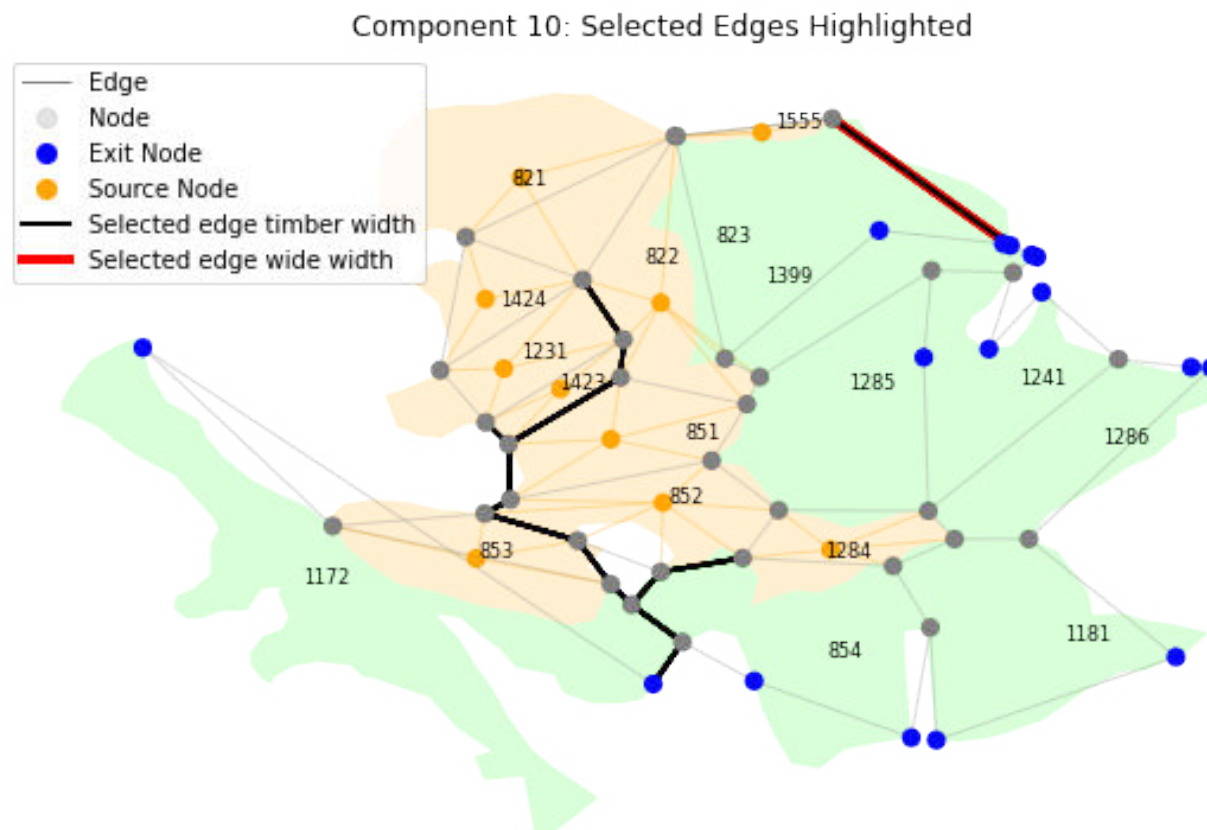


Figure D.48: Visualization of solver-selected edges over all periods

Figure D.49: Selected edges for construction (C), maintenance (M), and upgrade (U), and the flow, for both road widths, component 10

ID_UG	timber_1	timber_2	timber_3	timber_4	timber_5	wide_1	wide_2	wide_3	wide_4	wide_5
1423	1	1	1	1	1	0	0	0	0	0
851	1	0	0	1	1	0	0	0	0	0
821	0	1	1	0	0	0	0	0	0	0
822	0	1	0	1	0	0	0	0	0	0
852	1	1	1	1	1	0	0	0	0	0
853	1	0	0	0	0	0	0	0	0	0
1231	1	0	0	1	1	0	0	0	0	0
1284	1	1	1	1	1	0	0	0	0	0
1424	1	1	1	1	0	0	0	0	0	0
1555	0	0	0	0	0	0	0	0	1	1

Table D.1: Access needs matrix for component 10 under MaxRes harvesting schedule.

Table D.1 shows the access needs for component 10 under the MaxRes harvesting schedule. Each row corresponds to a stand (ID_UG), and each column indicates whether the stand requires road access of a given type and period, which is the foundation for the model selected road network visualized in fig. D.49. Columns named `timber_y` indicate whether timber-width access is needed in period y (1 = yes, 0 = no), while columns named `wide_y` indicate whether wide road access is needed in period y (1 = yes, 0 = no).

